

POSITIVE SOLUTIONS OF THE EXPLICIT FORMULA FOR $\zeta(s)$: NEAR-CRITICALITY AND UNIQUENESS

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ABSTRACT. We study the positive solutions of Weil’s explicit formula with the archimedean data of the Riemann zeta function: pairs of positive measures, one on the zero side and one on the prime side beyond $\log 2/2\pi$. By linear-programming duality such pairs exist if and only if a positivity statement involving neither zeros nor primes holds; its slack κ^* gives the optimal degree-one explicit-formula conductor bound $q_{\min} = e^{-2\pi\kappa^*}$. We conjecture that ζ ’s own pair is the only solution; this does not imply the Riemann hypothesis, but makes it equivalent to existence. We prove the conjecture for pairs carried by the zeros of ζ plus finitely many points and by the Bondarenko–Radchenko–Seip nodes, using positivity of the prime measure only on one residue class of nodes, and of the zero measure only at the origin. Certificates of the form $\Xi^2(H + \varepsilon e^{-\pi t^2})$, with H a polynomial in t^2 , vanish at every zero of ζ and show unconditionally that $\kappa^* \leq 1.1508391 \cdot 10^{-906}$. Uniqueness would follow from asymptotic properties of a family of such functions, defined by Hermite interpolation at the prime powers.

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1. INTRODUCTION

1.1. The linear problem behind explicit-formula conductor bounds. Weil’s explicit formula [81] (in the form of Guinand [38] for ζ) is a linear identity between two measures: one on the zeros of ζ , one on the prime powers. For even test functions F that are analytic and decay

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in a strip, and with $\widehat{F}(\xi) = \int F(t)e^{-2\pi i \xi t} dt$, it reads
(1.1)

$$F\left(\frac{i}{2}\right) + F\left(-\frac{i}{2}\right) + \frac{1}{2\pi} \int_{\mathbb{R}} F(t) \left(\operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right) dt = \sum_{\rho} F\left(\frac{\rho - \frac{1}{2}}{i}\right) + \frac{1}{\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \widehat{F}\left(\frac{\log n}{2\pi}\right).$$

The left side, which we denote by $\mathcal{A}(F)$, contains only the archimedean data of ζ : the Gamma factor and the pole at $s = 1$. Under RH every zero contributes $F(\gamma)$ with γ real, so the right side is $\int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for two positive measures: μ on the real ordinates of the zeros and ν on the points $\log n/(2\pi)$, $n \geq 2$.

Positivity of this kind drives the explicit-formula method for lower bounds of discriminants and conductors: the analytic method of Stark and Odlyzko, recast by Serre through the explicit formula and refined by Poitou [75, 61, 63, 73, 66, 65] (see [62, pp. 120–121]). A test function with $F \geq 0$ on $\mathbb{R}$ and $\widehat{F} \geq 0$ on the relevant range makes the zero side and the prime side non-negative, and (1.1) then bounds the archimedean data, in particular the conductor. Odlyzko asked which test functions give the best bounds [62, Open Problem 2.2, p. 123], how the prime ideals and the zeros share the explicit formula for minimal discriminants [62, Open Problem 6.1 and Table 5], and whether the bounds obtained under GRH remain valid without it [62, Open Problem 6.3, p. 132]. For $n = 1$ his tables of root-discriminant bounds, that is, for data of the type of ζ , give 0.997, both under GRH and unconditionally [62, Table 4, p. 134]; these values are based on his 1976 tables [61] (which print 0.996), optimised over a one-parameter family of kernels. The true conductor is 1.

This paper studies the linear problem behind these bounds for the data of ζ itself, as a problem about *all* positive solutions of (1.1). Its main idea is that the two questions it raises, how sharp the optimal conductor bound is and how many positive solutions there are, are linked by complementary slackness: an extremal test function for the conductor bound pins down the support of every positive solution (Lemma 2.13). Whether, conversely, uniqueness forces such an extremal function is open (Problem (4)).

1.2. Admissible pairs, the slack, and Conjectures U and S. An *admissible pair* (Definition 2.4) is a pair (μ, ν) of positive measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, where $\xi_2 = \log 2/(2\pi)$, such that $\mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for every F in a natural test class $\mathcal{T}$ (Definition 2.1). Nothing is assumed about where μ and ν live, beyond the gap $(-\xi_2, \xi_2)$ in the prime measure, and nothing about integrality. We write $\mathcal{K}$ for the set of admissible pairs and $p_{\zeta} = (\mu_{\zeta}, \nu_{\zeta})$ for the pair of ζ : μ_{ζ} counts the real zero ordinates with multiplicity and $\nu_{\zeta} = \sum_n \Lambda(n)n^{-1/2} \delta_{\log n/2\pi}$. The pair p_{ζ} is admissible if and only if RH holds. We consider three statements:

$$(E) : \mathcal{K} \neq \emptyset, \quad (U) : \mathcal{K} \subseteq \{p_{\zeta}\}, \quad (S) : \kappa^* \leq 0,$$

where the *slack* κ^* is the infimum of $\mathcal{A}(F)$ over the cone $\mathcal{C}$ of $F \in \mathcal{T}$ with $F \geq 0$ on $\mathbb{R}$, $\widehat{F} \geq 0$ on $[\xi_2, \infty)$ and $\int F = 1$. In the language of conductors, $q_{\min} = e^{-2\pi\kappa^*}$ is the least conductor for which the Gamma factor $\Gamma_{\mathbb{R}}$, one simple pole and a positive prime measure on $[\xi_2, \infty)$ admit a positive solution of the explicit formula, and it is the fully optimised degree-one conductor bound (Proposition 2.10).

Conjecture U. *Every admissible pair, if there is one, is ζ 's pair: $\mathcal{K} \subseteq \{p_{\zeta}\}$.*

Conjecture S. $\kappa^* \leq 0$. *Equivalently, $q_{\min} \geq 1$; equivalently, no admissible pair has a zero measure $\mu \geq \lambda dt$ with $\lambda > 0$.*

The logic of these statements is (Theorem 2.9)

$$\text{RH} \Rightarrow (E), \quad (U) \Rightarrow (S), \quad (U) \iff [(E) \Rightarrow \text{RH}] \wedge [\text{RH} \Rightarrow \mathcal{K} = \{p_{\zeta}\}].$$

So (U) is the conjunction of a prime-free criterion for RH and a uniqueness statement under RH. It does *not* imply RH: it is consistent with $\neg\text{RH} \wedge \neg(\text{E})$, and under (U) alone we only get $\text{RH} \Leftrightarrow (\text{E})$. For the field $\mathbb{Q}$, every GRH-type conductor bound given by a test function in $\mathcal{C}$ holds without RH exactly when $\mathcal{A} \geq 0$ on $\mathcal{C}$, that is, under (E); so (E) is the statement that all such bounds of Odlyzko's Open Problem 6.3 hold for $\mathbb{Q}$, and the first half of (U) is a converse to it. (The tabulated value $0.997 < 1$ holds for $\mathbb{Q}$ trivially.) Under RH, (S) is equivalent to $\kappa^* = 0$, that is, to the statement that ζ 's data are exactly critical: they cannot be tightened at all without losing every positive solution.

1.3. Main results. We write $\mathcal{M} = \{\log m/(4\pi) : m \geq 1\}$; it contains the prime-power nodes $\log n/(2\pi) = \log(n^2)/(4\pi)$.

Theorem A (Duality and criticality). (a) $\mathcal{K} \neq \emptyset$ if and only if $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C}$, if and only if $\kappa^* \geq 0$; and $q_{\min} = e^{-2\pi\kappa^*}$.
 (b) Assume (E). Then $\kappa^* = 0$ if and only if every admissible μ is purely atomic, if and only if all admissible μ are carried by one discrete set.

These are Theorem 2.7, Proposition 2.10 and Theorem 2.12; the logic displayed in Section 1.2 is Theorem 2.9. In (b) the common discrete set is the real zero set of a *weak magic function*, a limit of near-optimal test functions. Abstract sign and gap conditions alone do not force rigidity (Section 3.1); the arithmetic input is the Fourier interpolation basis of Bondarenko, Radchenko and Seip [8], which is attached to the zeros of ζ and the points $\mathcal{M}$ and does not assume RH.

Theorem B (Uniqueness near ζ 's support). *Let $E \subset \mathbb{R} \setminus Z_\zeta$ be finite. If $(\mu, \nu) \in \mathcal{K}$, μ is carried by $Z_\zeta \cup E$, and ν is carried by $\mathcal{M}$, then RH holds and $(\mu, \nu) = p_\zeta$.*

This follows from Theorem 3.7 applied to $E \cup (-E)$. That theorem uses positivity of ν only at the nodes $\log m/(4\pi)$ with $m \equiv 2 \pmod{3}$, a progression that contains none of ζ 's own prime-power nodes, and positivity of μ only at 0.

Corollary (Magic-function principle). *If some $F \in \mathcal{C}$ with $\mathcal{A}(F) = 0$ has only finitely many real zeros outside Z_ζ , and the zeros of $\widehat{F}$ in $[\xi_2, \infty)$ lie in $\mathcal{M}$, then (U) holds; if moreover (E) holds, then RH holds and $\mathcal{K} = \{p_\zeta\}$.*

This is Corollary 3.8: by complementary slackness (Lemma 2.13), every admissible pair is carried by the real zeros of F and the zeros of $\widehat{F}$ in $[\xi_2, \infty)$, and Theorem B applies. It is the chain that links the two halves of the paper: near-extremal certificates for the conductor problem, whose limit is an exact magic function with only finitely many real zeros outside Z_ζ and with the zeros of its transform in $[\xi_2, \infty)$ inside $\mathcal{M}$, would prove (U) through complementary slackness and Theorem B. Theorem D below states this for an explicit family.

Theorem C (Near-criticality). (a) $\kappa^* \leq 1.1508391 \cdot 10^{-906}$. Hence $q_{\min} \geq 1 - 7.3 \cdot 10^{-906}$; in the classical cone $\mathcal{C}_{\text{OPS}}$, $\kappa_{\text{OPS}}^* \leq 9.9462 \cdot 10^{-41}$.
 (b) $\kappa^* \geq -2.7112 \cdot 10^{-3}$.

These are Theorem 5.1, Corollary 5.2, Proposition 5.3 and Proposition 5.8; they are computer-assisted and unconditional. Together they give, unconditionally, $1 - 7.3 \cdot 10^{-906} \leq q_{\min} < 1.017182$; only under (E), for instance under RH, is $\kappa^* \geq 0$, that is, $q_{\min} \leq 1$. So Theorem C sharpens the conductor bound that the explicit-formula method can give for ζ 's data; the arithmetic conductor of ζ is 1 in any case. The certificate in (a) is a zero-killing function $\Xi^2(H + \varepsilon e^{-\pi t^2})$, where Ξ is Riemann's Ξ -function and $H > 0$ is a polynomial in t^2 . It vanishes at every zero of ζ , on or off the line, so $\mathcal{A}$ of it is a sum over prime powers alone. For the exact interpolant P_J of Section 4.2, the transform of $\Xi^2 P_J(t^2)$ has zeros of order at least two at the

first J prime powers, so that sum would start at the $(J+1)$ -st prime power. The certificate uses a rational approximation H of $P_J(t^2)$ and a small positive cushion, which leave small contributions at the first J prime powers; the proof bounds these contributions and the tail rigorously (Section 5.2). The exponent 906 is a computational budget ($J = 100$); the evidence for (S) is the ladder of Table 1, in which the certified bounds keep improving from $J = 20$ to $J = 100$. A certificate of the classical (Gaussian–Laguerre) kind shows that every admissible pair, whether or not RH holds, puts mass at most $2.4 \cdot 10^{-13}$ into each gap between the first six positive zero ordinates of ζ , shrunk by 0.1 at both ends, and at most $3.8 \cdot 10^{-8}$ into listed windows between low prime powers (Proposition 5.4). With the same gap ξ_2 , and also at their own natural gaps, the archimedean data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ have positive slack (Proposition 5.9); this indicates that near-criticality is a property of ζ ’s data rather than of the method.

Theorem D (A criterion through the zero-killing family). *If the exact Hermite family $F_J = \Xi^2 P_J(t^2)$ of Section 4.2 satisfies the asymptotic hypotheses of Theorem 4.8, then (S) and (U) hold, and $RH \Leftrightarrow (E)$.*

Here P_J is the polynomial with $P_J(0) = 1$ and degree at most $2J$ for which the transform of F_J has zeros of order at least two at the first J prime powers; the certificates of Theorem C(a) are rational approximations of these functions.

1.4. What is claimed and what is not. Theorems A and B and the Corollary are proved; Theorem C is computer-assisted and unconditional; Theorem D is proved, and its hypotheses are open. This list is the single place where the limits of the results are stated.

- (1) Nothing here proves RH, (E), (S) or (U), and Conjecture U itself does not imply RH (Section 1.2).
- (2) Only one side of criticality is certified. The inequality $\kappa^* \geq 0$, equivalently $q_{\min} \leq 1$, is (E); we know it only under RH, and unconditionally we know only $\kappa^* \geq -2.7112 \cdot 10^{-3}$, that is, $q_{\min} < 1.017182$. We never claim $\kappa^* = 0$.
- (3) The hypotheses of Theorem D concern infinitely many J , or a limit in J , and are open. Only finite- J instances are certified (Proposition 5.5); convergence of the family is neither assumed nor proved.
- (4) Uniqueness is proved only near ζ ’s support: Theorem B allows finitely many extra zeros, but no prime mass off $\mathcal{M}$.
- (5) The comparison with $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ concerns two examples, at ζ ’s gap and at their natural gaps; it is not a statement about L -functions in general (Remark 5.10).

1.5. Related work. Explicit formulas and positivity. The explicit formula goes back to Riemann, von Mangoldt, Guinand [38, 39] and Weil [81, 82]. Weil’s criterion characterises RH by positivity of a *quadratic* form on test functions of the type $f * \tilde{f}$ [81, 7]; our cone $\mathcal{C}$ is defined instead by pointwise positivity of F and $\hat{F}$, and the problem is a linear programme. The archimedean part of Weil’s form is positive on test functions supported in $[2^{-1/2}, 2^{1/2}]$ whose Fourier transform vanishes at $\pm \frac{i}{2}$; this was proved by Yoshida [83], and Connes and Consani [23] gave an operator-theoretic strengthening. Connes and Consani [24] exhibited very small eigenvalues of Weil’s form on test functions supported in a window $[-L, L]$, and Zhu [84] certifies two-sided bounds for them; from the prime side of the explicit formula alone he shows that the normalised smallest eigenvalue for $L = 2$ is at most $3.2 \cdot 10^{-283}$, a quadratic analogue of Theorem C. Connes [22] extremises a restriction of Weil’s form built from the primes $p \leq 13$ to approximate the first 50 zeros. None of these works makes a statement about positive solutions of the explicit formula. Weil’s Hermitian form and the explicit formula are also the tools of recent work on a different question, the proportion of simple zeros on the critical line

[1, 56]; the proof in [1] was found by an AI model. The problem that defines κ^* involves only the archimedean data and the gap, and so does the Gaussian–Laguerre certificate of Proposition 5.3. The certificates of Theorem C(a), in contrast, are built from arithmetic input: the factor Ξ^2 , which vanishes at the zeros of ζ , and Hermite interpolation at the prime powers.

Fourier summation formulas. Admissible pairs are pairs of positive measures in Fourier duality with fixed archimedean data. Dyson suggested that a classification of one-dimensional quasicrystals might lead to a proof of RH [34]; such pairs of sequences go back to Guinand’s theory of concordance [40]. Gonçalves [36] classified Fourier summation formulas, with ζ ’s explicit formula in his class, and his Theorem 5 classifies non-negative crystalline measures with uniformly discrete support; Gonçalves and Vedana [37] and Vedana [79] completed the classification on the line and extended it to a strip. These classifications concern formulas whose summation side is a countable sum of point masses, whereas an admissible pair may have an arbitrary positive prime measure; so they are related frameworks, which apply directly only to admissible pairs with a discrete prime measure (for instance one carried by $\mathcal{M}$, as in Theorem B). None of these works imposes positivity on both sides with fixed archimedean data.

Conductor bounds and characterisations of ζ . The method for discriminants is due to Stark and Odlyzko, with the explicit formula introduced by Serre and refinements by Poitou [75, 61, 73, 66, 65]; Odlyzko treated Artin conductors [63] and Mestre conductors of varieties [58]; see [62, 28]. In Odlyzko’s own setting, the classical cone, Proposition 5.3 raises the tabulated 0.997 to at least $1 - 6.25 \cdot 10^{-40}$; Theorem C(a) relaxes the positivity of $\hat{F}$ to $[\xi_2, \infty)$ and gives at least $1 - 7.3 \cdot 10^{-906}$. Both address [62, Open Problem 2.2] in degree one for the data of ζ . In the literature we searched we found no optimisation of this bound beyond Odlyzko’s one-parameter family, and no claim that its optimum is 1. A related extremal problem concerns the lowest zero: among Dedekind zeta functions, ζ has the highest lowest zero [62, p. 132]; Miller [59] proved the analogue under GRH for entire L -functions of real archimedean type by positivity in the explicit formula, and Bober et al. [4] showed that it fails beyond real archimedean type. Hamburger’s theorem [41, 42] (see [64, 11]) and its extensions [6, 51, 52], the classification of the degree-one Selberg class [72, 25, 49, 74], and the theorems of Beurling [3] and A. B. Dixit [29] characterise ζ among Dirichlet series, or impose positivity on Dirichlet coefficients. Conjecture U starts instead from the explicit formula, with arbitrary positive measures on both sides. The closest antecedent we found is Booker’s converse theorem for positive L -data [9, Definition 1.3, Theorem 1.7]; Theorem B is complementary to it, not a generalisation (Section 3.3).

Interpolation, linear programming and magic functions. Bondarenko, Radchenko and Seip [8] constructed a Fourier interpolation basis for the zeros of ζ and the points $\mathcal{M}$, after Radchenko and Viazovska [69]; Proposition 3.5 is the uniqueness statement implicit in [8, §4.4]. The linear programming bounds of Cohn and Elkies [15] and the magic functions of [80, 18, 19] are the model for our use of “magic function”, as is the uniqueness of the Leech lattice [17]. In that setting near-sharpness is familiar, and our results have direct analogues there: Theorem C corresponds to the certificate of Cohn and Kumar that the linear programming bound in dimension 24 is sharp up to a factor $1 + 1.65 \cdot 10^{-30}$ [17]; the explicit admissible pairs of Propositions 5.8 and 5.9 correspond to the dual certificates with which Cohn and Triantafyllou proved non-sharpness in other dimensions [21]; and Theorem D with Conjecture 6.1 corresponds to the sequences that Cohn and Miller conjectured to converge to magic functions [20]. The cone $\mathcal{C}$ is a sign-uncertainty cone [10, 16, 12], and Bourgain, Clozel and Kahane already related such problems to zeta functions and discriminant bounds [10, §4]. Linear programmes over trace and crossing formulas, with near-sharp bounds attained by special arithmetic objects, appear in the bootstrap [43, 54]. Fourier optimisation with the explicit formula is usually done under GRH [13, 14]; our problem is unconditional on the archimedean side and asks about all positive solutions. A test function whose transform contains ζ as a factor kills every non-trivial zero, and Guinand’s

formula then gives an identity over prime powers alone [70]; our certificates use Ξ^2 for this purpose, with a cofactor chosen by Hermite interpolation.

What is new. As far as our literature search shows, the following are new: the linear programme over all positive solutions of the explicit formula with ζ 's data, with Conjecture U, its logic and the link between criticality and atomicity (Theorem A(b)); Theorem B and the Corollary; the zero-killing certificates $\Xi^2 H$, which evaluate the dual objective through ζ 's own explicit formula (Theorem C(a)); the criterion of Theorem D; and the certified non-criticality of the data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ at their natural gaps. Proposition 3.5 is essentially contained in [8, §4.4] with [8, Corollary 1.1], and the duality theory of Section 2 is standard conic linear programming.

1.6. Organisation, notation and companion manuscripts. Section 2 sets up admissible pairs and proves duality, the logic and criticality. Section 3 proves the uniqueness results, Section 4 introduces the zero-killing family and proves Theorem D, Section 5 contains the certified bounds, and Section 6 the conjectures and open problems. Appendix A describes the certification methods, Appendix B contains the long proofs, and Appendix C indexes the ancillary files. A companion manuscript [47] studies the family of Section 4 in detail, and three notes treat extra prime-side atoms [46], the critical conductors of other Gamma factors [45], and pairs with integer zero weights [48]; nothing here depends on them.

We use $\widehat{F}(\xi) = \int F(t)e^{-2\pi i \xi t} dt$, $\xi_n = \log n/(2\pi)$ for real $n > 0$, and the multiplicative coordinate $x = e^{2\pi \xi}$, so that the prime-power node ξ_n corresponds to $x = n$ and the gap $[\xi_2, \infty)$ to $x \geq 2$. $\text{PP} = \{n_1 < n_2 < \dots\}$ is the set of prime powers and $\xi_{\text{PP}} = \{\xi_n : n \in \text{PP}\}$; Z_ζ is the set of real γ with $\zeta(\frac{1}{2} + i\gamma) = 0$; $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$; and $\psi = \Gamma'/\Gamma$ is the digamma function.

2. ADMISSIBLE PAIRS, DUALITY AND CRITICALITY

This section sets up the linear problem behind Conjecture U and proves what can be said about it with positivity, the explicit formula and soft analysis alone. Beyond the explicit formula, the only arithmetic used is the bound $O(T \log T)$ for the number of zeros up to height T (Lemma 2.3); the prime number theorem enters only in the last sentence of Lemma 2.6.

2.1. The test class and the explicit formula. We write $\widehat{F}(\xi) = \int_{\mathbb{R}} F(t)e^{-2\pi i \xi t} dt$, $\xi_n = \log n/(2\pi)$ for real $n > 0$, and $\xi_2 = \log 2/(2\pi)$ for the *gap*. The archimedean data of ζ are the density and the functional

$$(2.1) \quad \Omega_\infty(t) := \frac{1}{2\pi} \left(\text{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \log \pi \right), \quad \mathcal{A}(F) := F\left(\frac{i}{2}\right) + F\left(-\frac{i}{2}\right) + \int_{\mathbb{R}} F(t)\Omega_\infty(t) dt,$$

where $\psi = \Gamma'/\Gamma$. Since $\text{Re} \psi(\frac{1}{4} + \frac{it}{2}) = \log(|t|/2) + O(t^{-2})$ as $|t| \rightarrow \infty$, we have $|\Omega_\infty(t)| \leq C \log(2 + |t|)$. The series $\psi(z) = -\gamma_E + \sum_{n \geq 0} (\frac{1}{n+1} - \frac{1}{n+z})$ (γ_E Euler's constant) shows that $\text{Re} \psi(a + ib)$ is an increasing function of b^2 for fixed $a > 0$; hence Ω_∞ is even, increasing in $|t|$, and

$$(2.2) \quad \Omega_\infty(t) \geq \Omega_\infty(0) = \frac{\psi(\frac{1}{4}) - \log \pi}{2\pi} = -0.85500\dots$$

Definition 2.1 (Test class). For $0 < \delta < \frac{1}{2}$ let $\mathcal{T}_\delta$ be the set of even functions F , real on $\mathbb{R}$, analytic in a neighbourhood of the closed strip $\{| \text{Im } z | \leq \frac{1}{2} + \delta\}$, with

$$\|F\|_\delta := \sup_{|\text{Im } z| \leq \frac{1}{2} + \delta} (1 + |z|)^2 |F(z)| < \infty.$$

Put $\mathcal{T} := \bigcup_{\delta > 0} \mathcal{T}_\delta$. Let $\mathcal{G}$ be the set of even real-valued elements of $\text{span}_{\mathbb{C}}\{e^{-\pi(t-a)^2/s^2 + 2\pi i b t} : a, b \in \mathbb{R}, s > 0\}$ (Gaussian wave packets); then $\mathcal{G} \subset \mathcal{T}_\delta$ for every δ . An even complex-valued

function whose real and imaginary parts on $\mathbb{R}$ extend to elements of $\mathcal{T}$ is paired with measures, and with $\mathcal{A}$, by linearity.

The class $\mathcal{T}$ contains Gaussians times polynomials, the Poisson kernels $t \mapsto (s - \frac{1}{2})/((s - \frac{1}{2})^2 + t^2)$ for real $s > 1$, the functions $\Xi^2 H$ of Section 4, and the interpolation basis of Bondarenko, Radchenko and Seip (Section 3.2). Since $\mathcal{G}$ is not norm-dense in $\mathcal{T}_\delta$ (for $F(z) = 1/(1 + z^2)$ and $G \in \mathcal{G}$, $\|F - G\|_\delta \geq 1$), density arguments below use dominated convergence.

Lemma 2.2 (Elementary facts). *Let $F \in \mathcal{T}_\delta$. Then*

- (a) $|\widehat{F}(\xi)| \leq \pi \|F\|_\delta e^{-\pi(1+2\delta)|\xi|}$ for all $\xi \in \mathbb{R}$;
- (b) $F(\frac{i}{2}) + F(-\frac{i}{2}) = 2 \int_{\mathbb{R}} \widehat{F}(\xi) \cosh(\pi\xi) d\xi$;
- (c) $|\mathcal{A}(F)| \leq C_\delta \|F\|_\delta$.

In particular $\mathcal{A}(F) = 2 \int_{\mathbb{R}} \widehat{F}(\xi) \cosh(\pi\xi) d\xi + \int_{\mathbb{R}} F \Omega_\infty$ for every $F \in \mathcal{T}$.

Proof. (a) Let $\xi > 0$ and $c = \frac{1}{2} + \delta$. Since F is analytic on a neighbourhood of the closed strip and $|F(z)| \leq \|F\|_\delta (1 + |z|)^{-2}$ there, Cauchy's theorem moves the line of integration to $\text{Im } t = -c$: $\widehat{F}(\xi) = e^{-2\pi c \xi} \int_{\mathbb{R}} F(t - ic) e^{-2\pi i \xi t} dt$, and $|\int F(t - ic) e^{-2\pi i \xi t} dt| \leq \|F\|_\delta \int (1 + t^2)^{-1} dt = \pi \|F\|_\delta$. For $\xi < 0$ use $\text{Im } t = +c$. (b) By (a), $z \mapsto \int \widehat{F}(\xi) e^{2\pi i \xi z} d\xi$ converges absolutely and is analytic for $|\text{Im } z| < \frac{1}{2} + \delta$; by Fourier inversion it equals F on $\mathbb{R}$, hence on the strip. Evaluate at $z = \pm \frac{i}{2}$ and use that $\widehat{F}$ is even. (c) follows from $|F(\pm \frac{i}{2})| \leq \|F\|_\delta$ and $\int (1 + |t|)^{-2} |\Omega_\infty(t)| dt < \infty$. $\square$

Lemma 2.3 (Explicit formula on $\mathcal{T}$; [81], [8, (1.1)]). *For every $F \in \mathcal{T}$,*

$$(2.3) \quad \mathcal{A}(F) = \sum_{\rho} F(t_\rho) + \frac{1}{\pi} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \widehat{F}(\xi_n), \quad t_\rho := -i(\rho - \tfrac{1}{2}),$$

where ρ runs over the non-trivial zeros of ζ , counted with multiplicity. Both series converge absolutely. No hypothesis on the zeros is made.

Proof. Formula (1.1) of [8] is (2.3) for functions f analytic in $|\text{Im } z| < \frac{1}{2} + \varepsilon$ with $|f(z)| \ll (1 + |z|)^{-1-\eta}$ for some $\varepsilon, \eta > 0$; this class contains $\mathcal{T}$. In [8] the archimedean integral carries $\psi(\frac{1}{4} + \frac{it}{2})$ and the prime sum carries $\frac{1}{2\pi}(\widehat{F}(\xi_n) + \widehat{F}(-\xi_n))$; for even F , real on $\mathbb{R}$, these reduce to $\text{Re } \psi$ and to $\frac{1}{\pi} \widehat{F}(\xi_n)$. Absolute convergence: $|\text{Re } t_\rho| = |\text{Im } \rho|$, so $|F(t_\rho)| \leq \|F\|_\delta (1 + |\text{Im } \rho|)^{-2}$, which is summable because the number of zeros with $|\text{Im } \rho| \leq T$ is $O(T \log T)$; and $|\widehat{F}(\xi_n)| \leq \pi \|F\|_\delta n^{-1/2-\delta}$ by Lemma 2.2(a). $\square$

Write $t_\rho = \gamma - i(\beta - \frac{1}{2})$ for $\rho = \beta + i\gamma$; thus t_ρ is real exactly when ρ is on the critical line, and then $t_\rho = \gamma \in Z_\zeta$. The multiset $\{t_\rho\}$ is invariant under $t \mapsto -t$ and $t \mapsto \bar{t}$. We write

$$\mu_\zeta := \sum_{\gamma \in Z_\zeta} m_\gamma \delta_\gamma, \quad \nu_\zeta := \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta_{\xi_n}, \quad p_\zeta := (\mu_\zeta, \nu_\zeta),$$

where $Z_\zeta = \{\gamma \in \mathbb{R} : \zeta(\frac{1}{2} + i\gamma) = 0\}$ and m_γ is the multiplicity. Only real ordinates enter μ_ζ ; under RH, (2.3) reads $\mathcal{A}(F) = \int F d\mu_\zeta + \frac{1}{\pi} \int \widehat{F} d\nu_\zeta$.

2.2. Admissible pairs, a priori bounds and duality.

Definition 2.4 (Admissible pairs, cones, slack). An *admissible pair* is a pair (μ, ν) of positive Radon measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, such that for every $F \in \mathcal{T}$ the integrals $\int F d\mu$ and $\int \widehat{F} d\nu$ converge absolutely and

$$\mathcal{A}(F) = \mathcal{R}_{\mu, \nu}(F) := \int_{\mathbb{R}} F d\mu + \frac{1}{\pi} \int_{[\xi_2, \infty)} \widehat{F} d\nu.$$

$\mathcal{K}$ denotes the set of admissible pairs. We put

$$\mathcal{C} := \{F \in \mathcal{T} : F \geq 0 \text{ on } \mathbb{R}, \widehat{F} \geq 0 \text{ on } [\xi_2, \infty)\}, \quad \mathcal{C}_{\text{OPS}} := \{F \in \mathcal{T} : F \geq 0 \text{ and } \widehat{F} \geq 0 \text{ on } \mathbb{R}\},$$

$$\kappa^* := \inf \left\{ \mathcal{A}(F) : F \in \mathcal{C}, \int F = 1 \right\}, \quad \kappa_{\text{OPS}}^* := \inf \left\{ \mathcal{A}(F) : F \in \mathcal{C}_{\text{OPS}}, \int F = 1 \right\}.$$

We consider the statements

$$(E) : \mathcal{K} \neq \emptyset, \quad (U) : \mathcal{K} \subseteq \{p_\zeta\}, \quad (S) : \kappa^* \leq 0.$$

For another archimedean functional $\mathcal{A}'$ (for instance $\mathcal{A}_q$ below, or the functional of another Gamma factor in Section 5.5) the same words define $\mathcal{K}(\mathcal{A}')$ and $\kappa^*(\mathcal{A}')$.

Since $\mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}$, we have $\kappa^* \leq \kappa_{\text{OPS}}^*$. The pair p_ζ uses only the real ordinates; it is admissible if and only if RH holds (Theorem 2.9). The absolute-convergence clause in the definition is automatic once the identity holds on $\mathcal{G}$ (Lemma 2.6 and Theorem 2.7(b)).

Lemma 2.5 (Weak duality). *Let $(\mu, \nu) \in \mathcal{K}$ and $F \in \mathcal{C}$. For Borel sets $I \subseteq \mathbb{R}$ and $J \subseteq [\xi_2, \infty)$,*

$$\left(\inf_I F \right) \mu(I) + \frac{1}{\pi} \left(\inf_J \widehat{F} \right) \nu(J) \leq \mathcal{A}(F).$$

In particular $\mathcal{A}(F) \geq 0$; $\mu(I) \leq \mathcal{A}(F)$ if $F \geq 1$ on I ; $\nu(J) \leq \pi \mathcal{A}(F)$ if $\widehat{F} \geq 1$ on J ; and if $\int F = 1$, no admissible μ satisfies $\mu \geq \lambda dt$ with $\lambda > \mathcal{A}(F)$.

Proof. $\mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$, and both integrands are non-negative where the measures live. Drop everything outside I and J . For the last claim, $\mathcal{A}(F) \geq \int F d\mu \geq \lambda \int F = \lambda$. $\square$

The certified upper bounds of Section 5 are of this form.

Lemma 2.6 (A priori bounds). *There is an absolute constant C such that every $(\mu, \nu) \in \mathcal{K}$ satisfies*

$$\mu([T-1, T+1]) \leq C \log(3 + |T|) \quad (T \in \mathbb{R}), \quad \nu([X-1, X+1]) \leq C e^{\pi X} \quad (X \in \mathbb{R}).$$

Hence $\int (1+t^2)^{-1} d\mu < \infty$, $\int e^{-\pi(1+\eta)\xi} d\nu(\xi) < \infty$ for every $\eta > 0$, and $|\mathcal{R}_{\mu, \nu}(F)| \leq C_\delta \|F\|_\delta$ on $\mathcal{T}_\delta$, with C_δ independent of $(\mu, \nu) \in \mathcal{K}$. The same holds, with other constants, for positive pairs representing $\mathcal{A} + \lambda \int(\cdot)$ ($\lambda \in \mathbb{R}$) on $\mathcal{G}$. The rate $e^{\pi X}$ is attained by ν_ζ .

The proof is in Appendix B.1.2.

Theorem 2.7 (Duality). (a) *The following are equivalent: (i) $\mathcal{K} \neq \emptyset$; (ii) $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C}$; (iii) $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C} \cap \mathcal{G}$; (iv) $\kappa^* \geq 0$.*

(b) *If an even positive Radon measure μ on $\mathbb{R}$ and a positive Radon measure ν on $[\xi_2, \infty)$ satisfy $\mathcal{A}(F) = \mathcal{R}_{\mu, \nu}(F)$ for every $F \in \mathcal{G}$, then $(\mu, \nu) \in \mathcal{K}$. The set $\mathcal{K}$ is convex and compact for the product of the vague topologies.*

The equivalence (i) $\Leftrightarrow$ (ii) says that there is no duality gap: the existence of a positive solution of the explicit formula with the archimedean data of ζ is equivalent to a positivity statement that mentions no primes and no zeros.

Proof of Theorem 2.7. In (a), (i) $\Rightarrow$ (ii) is Lemma 2.5 and (ii) $\Rightarrow$ (iii) is trivial. If $F \in \mathcal{C}$ and $\int F = 0$ then $F \equiv 0$, since $F \geq 0$ is continuous; so (ii) $\Leftrightarrow$ (iv) by homogeneity. The implications (iii) $\Rightarrow$ (ii) and (ii) $\Rightarrow$ (i), and part (b), are proved in Appendix B.1.1: the first by approximating $F \in \mathcal{C}$ by Riemann sums of Gaussian convolutions, the second by Carathéodory's theorem in finite-dimensional subspaces, uniform bounds from Lemma 2.6 and Helly's selection theorem. $\square$

The proof uses only that $\mathcal{A}$ is $\|\cdot\|_\delta$ -continuous on each $\mathcal{T}_\delta$ and the two families of bounds in the proof of Lemma 2.6. It therefore applies verbatim to $\mathcal{A} + \lambda \int(\cdot)$ for every $\lambda \in \mathbb{R}$, and in particular to the functionals $\mathcal{A}_q$ of Proposition 2.10.

Proposition 2.8 (Floors). κ^* is finite and equals the largest $\lambda \in \mathbb{R}$ for which there are an even real Radon measure μ with $\mu - \lambda dt \geq 0$ and a positive Radon measure ν on $[\xi_2, \infty)$ with $\mathcal{A} = \mathcal{R}_{\mu, \nu}$ on $\mathcal{T}$; the maximum is attained. In particular $\kappa^* \geq 0$ if and only if some admissible μ dominates $\kappa^* dt$.

Proof. A pair (μ, ν) with $\mu - \lambda dt \geq 0$ represents $\mathcal{A}$ if and only if $(\mu - \lambda dt, \nu)$ is admissible for $\mathcal{A} - \lambda \int(\cdot)$. By Theorem 2.7(a) for this functional, such a pair exists if and only if $\mathcal{A}(F) \geq \lambda \int F$ for all $F \in \mathcal{C}$, that is, if and only if $\lambda \leq \kappa^*$. The value κ^* is $< \infty$ (test any Gaussian). For the lower bound let $F \in \mathcal{C}$ with $\int F = 1$. Then $|\hat{F}| \leq 1$, and $\hat{F} \geq 0$ for $|\xi| \geq \xi_2$ because $\hat{F}$ is even. By Lemma 2.2(b) and (2.2),

$$\mathcal{A}(F) \geq 2 \int_{-\xi_2}^{\xi_2} \hat{F}(\xi) \cosh(\pi\xi) d\xi + \Omega_\infty(0) \int F \geq -4\xi_2 \cosh(\pi\xi_2) + \Omega_\infty(0).$$

Here $\pi\xi_2 = \frac{1}{2} \log 2$, so $4\xi_2 \cosh(\pi\xi_2) = \frac{3 \log 2}{\sqrt{2}\pi} = 0.4680387\dots$, and $\Omega_\infty(0) = -0.8550095\dots$; hence $\kappa^* \geq -1.3230483\dots \geq -1.3231$. So κ^* is finite, the set of feasible λ is $(-\infty, \kappa^*]$, and the maximum is attained at $\lambda = \kappa^*$. If $\kappa^* \geq 0$, the pair obtained at $\lambda = \kappa^*$ is admissible; conversely an admissible pair gives (E), hence $\kappa^* \geq 0$. $\square$

A certified admissible pair for a slightly larger conductor gives a much better unconditional floor (Proposition 5.8).

2.3. The logic of (E), (U) and RH.

Theorem 2.9 (Logic). (a) *RH implies $p_\zeta \in \mathcal{K}$; hence RH implies (E).*

(b) *If $(\mu, \nu_\zeta) \in \mathcal{K}$ for some μ , then RH holds and $\mu = \mu_\zeta$.*

(c) *(U) holds if and only if both $[(E) \Rightarrow RH]$ and $[RH \Rightarrow \mathcal{K} = \{p_\zeta\}]$ hold. In particular $(E) \wedge (U) \Rightarrow RH$, and (U) implies $RH \Leftrightarrow (E) \Leftrightarrow \kappa^* \geq 0$.*

(d) *(U) $\Rightarrow$ (S). Consequently $(U) \wedge (E) \Rightarrow \kappa^* = 0$, and $(U) \wedge \neg RH \Rightarrow \mathcal{K} = \emptyset$ and $\kappa^* < 0$.*

The proof is in Appendix B.1.2. Part (b) is the heart of it: if some zero $\rho_0 = \beta_0 + i\gamma_0$ were off the line, the Gaussians $F_u = g_u(\cdot - \gamma_0) + g_u(\cdot + \gamma_0)$, $g_u(t) = e^{-\pi u t^2}$, would make $\sum_\rho F_u(t_\rho)$ grow at least like $e^{\pi u(\beta_0 - \frac{1}{2})^2}$ along a sequence $u \rightarrow \infty$, while $\int F_u d\mu$ stays bounded.

2.4. The slack, conductors and criticality. For $q > 0$ put $\mathcal{A}_q(F) := \mathcal{A}(F) + \frac{\log q}{2\pi} \int F$. This is the archimedean functional of the explicit formula for data with the Gamma factor $\Gamma_\mathbb{R}$, one simple pole at $s = 1$ and conductor q ; $\mathcal{A} = \mathcal{A}_1$.

Proposition 2.10 (Conductor form). *For $q > 0$, $\kappa^*(\mathcal{A}_q) = \kappa^* + \frac{\log q}{2\pi}$. Hence $\mathcal{K}(\mathcal{A}_q) \neq \emptyset$ if and only if $q \geq q_{\min} := e^{-2\pi\kappa^*}$. Thus $q_{\min}$ is the least conductor for which $\Gamma_\mathbb{R}$, one simple pole at $s = 1$ and a positive prime measure on $[\xi_2, \infty)$ admit a positive solution of the explicit formula, and it is the best lower bound for the conductor that can be obtained from $\mathcal{A}_q$ with test functions in $\mathcal{C}$. Moreover $(S) \Leftrightarrow q_{\min} \geq 1$, and $RH \Rightarrow q_{\min} \leq 1$.*

Proof. The conductor term is $\frac{\log q}{2\pi}$ times Lebesgue measure, and κ^* is normalised by $\int F = 1$; this gives the first identity. Theorem 2.7(a) for $\mathcal{A}_q$ gives the second. A test function $F \in \mathcal{C}$ with $\int F = 1$ certifies $\log q \geq -2\pi\mathcal{A}(F)$ for all data admitting a positive solution, and the supremum of these bounds over F is $\log q_{\min}$. Finally (S) is $\kappa^* \leq 0$, and RH implies (E), i.e. $\kappa^* \geq 0$. $\square$

The proposition is unconditional, and only $q_{\min}$ appears in it. The arithmetic conductor of ζ is 1 unconditionally; what requires RH is that ζ 's own pair, a positive solution at $q = 1$, is admissible (Theorem 2.9). Among L -functions, a degree-one bound with $\Gamma_\mathbb{R}$ and a pole applies to a single object: by the classification of the degree-one elements of the Selberg class [25, 49, 74], the only element with Gamma factor $\Gamma_\mathbb{R}(s)$, conductor 1 and a pole at $s = 1$ is ζ . The content of

(S) is that the optimum $q_{\min}$ of the linear programme is at least this arithmetic value; equality needs (E) as well, for instance RH. In the classical cone, Odlyzko's tables give 0.997 for this bound [62, Table 4, p. 134] and Proposition 5.3 gives at least $1 - 6.25 \cdot 10^{-40}$; in $\mathcal{C}$, Corollary 5.2 gives at least $1 - 7.3 \cdot 10^{-906}$.

Definition 2.11 (Magic functions). An *exact magic function* is an $F \in \mathcal{C}$ with $\int F > 0$ and $\mathcal{A}(F) = 0$. A *weak magic function* is a function $F^* : \mathbb{R} \rightarrow [0, \infty)$ for which there are $F_K \in \mathcal{C}$ with $\int F_K = 1$ and $\mathcal{A}(F_K) \rightarrow 0$, such that $F_K \rightarrow F^*$ and $\widehat{F_K} \rightarrow \widehat{F^*}$ pointwise on $\mathbb{R}$, and $\int F^* = 1$. For a function F on $\mathbb{R}$ with Fourier transform $\widehat{F}$ we write

$$Z(F) := F^{-1}(0) \cap \mathbb{R}, \quad \widehat{Z}(F) := \widehat{F}^{-1}(0) \cap [\xi_2, \infty).$$

Weak magic functions exist only if $\kappa^* \leq 0$. A weak magic function need not belong to $\mathcal{T}$; by Theorem 2.12(a) it is analytic in $|\operatorname{Im} t| < \frac{1}{2}$. Every minimising sequence satisfies the moment bounds (2.4) below, so the properties listed in Theorem 2.12(a) hold for every weak magic function.

Theorem 2.12 (Criticality and atomicity). (a) *If $\kappa^* = 0$, weak magic functions exist. Every weak magic function F^* is even, has $\widehat{F^*} \in L^1(\mathbb{R})$ with $\int |\widehat{F^*}(\xi)| \cosh(\pi\xi) d\xi < \infty$ (so F^* extends analytically to $|\operatorname{Im} t| < \frac{1}{2}$), satisfies $\widehat{F^*} \geq 0$ on $|\xi| \geq \xi_2$ and $\int F^*(t) \log(2 + |t|) dt < \infty$, and has $\int F^* d\mu = 0 = \int \widehat{F^*} d\nu$ for every $(\mu, \nu) \in \mathcal{K}$.*
 (b) *Assume (E). The following are equivalent: (i) $\kappa^* = 0$; (ii) every admissible μ is purely atomic; (iii) there is a discrete set $Z^* \subset \mathbb{R}$ carrying every admissible μ ; (iv) no admissible μ satisfies $\mu \geq \lambda dt$ with $\lambda > 0$.*
 (c) *If $\kappa^* = 0$ and F^* is a weak magic function, one may take $Z^* = Z(F^*)$ in (iii); every zero of F^* on $\mathbb{R}$ has even order; and every admissible ν is carried by the closed set $\widehat{Z}(F^*)$.*

Proof. (a) Let $F_K \in \mathcal{C}$, $\int F_K = 1$, $\mathcal{A}(F_K) \rightarrow 0$. Since $|\widehat{F_K}| \leq 1$, $\widehat{F_K} \geq 0$ for $|\xi| \geq \xi_2$, and $\Omega_\infty \geq \Omega_\infty(0)$, Lemma 2.2(b) and the computation in the proof of Proposition 2.8 give

$$(2.4) \quad 2 \int_{|\xi| \geq \xi_2} \widehat{F_K}(\xi) \cosh(\pi\xi) d\xi + \int F_K(\Omega_\infty - \Omega_\infty(0)) \leq \mathcal{A}(F_K) + 1.3231,$$

and both terms on the left are non-negative. As $\Omega_\infty(t) - \Omega_\infty(0) \geq c \log(2 + |t|) - C$, the probability densities F_K are tight with $\sup_K \int F_K \log(2 + |t|) < \infty$, and $\sup_K \int |\widehat{F_K}| \cosh(\pi\xi) d\xi < \infty$. By Prokhorov's theorem a subsequence of $F_K dt$ converges weakly to a probability measure m , and $\widehat{F_K} \rightarrow \widehat{m}$ pointwise. The uniform bound with weight $\cosh(\pi\xi)$ makes $|\widehat{F_K}|$ uniformly integrable for any weight $\cosh((1 - \varepsilon)\pi\xi)$, so $\widehat{F_K} \rightarrow \widehat{m}$ in L^1 with these weights. Hence m has the continuous density $F^*(t) = \int \widehat{m}(\xi) e^{2\pi i \xi t} d\xi$, $F_K \rightarrow F^*$ uniformly on $\mathbb{R}$, and $\int F^* = 1$; so F^* is a weak magic function. For an arbitrary weak magic function F^* with sequence F_K , (2.4) holds for F_K ; evenness, $\widehat{F^*} \geq 0$ on $|\xi| \geq \xi_2$, $\int |\widehat{F^*}| \cosh(\pi\xi) < \infty$ and $\int F^* \log(2 + |t|) < \infty$ follow by pointwise limits and Fatou's lemma. Moreover (2.4) and the pointwise convergence $\widehat{F_K} \rightarrow \widehat{F^*}$ give $\widehat{F_K} \rightarrow \widehat{F^*}$ in $L^1(\mathbb{R})$ by Vitali's theorem, so $F_K \rightarrow (\widehat{F^*})^\vee$ uniformly on $\mathbb{R}$; hence $F^* = (\widehat{F^*})^\vee$ everywhere, F^* is continuous, and Fourier inversion gives analyticity in $|\operatorname{Im} t| < \frac{1}{2}$. Finally, for $(\mu, \nu) \in \mathcal{K}$, Fatou's lemma ($F_K \geq 0$ on $\mathbb{R}$, $\widehat{F_K} \geq 0$ on $[\xi_2, \infty)$) gives $\int F^* d\mu + \frac{1}{\pi} \int \widehat{F^*} d\nu \leq \liminf \mathcal{A}(F_K) = 0$, and both terms are non-negative.

(b) (i) $\Rightarrow$ (iii): by (a) there is a weak magic F^* , and every admissible μ satisfies $\int F^* d\mu = 0$ with $F^* \geq 0$ continuous, so μ is carried by $Z(F^*)$. As F^* is real-analytic and $F^* \not\equiv 0$, $Z(F^*)$ is discrete. (iii) $\Rightarrow$ (ii) $\Rightarrow$ (iv) is clear. (iv) $\Rightarrow$ (i): (E) gives $\kappa^* \geq 0$; if $\kappa^* > 0$, Proposition 2.8 gives an admissible $\mu \geq \kappa^* dt$.

(c) The first claim was shown in (b). A non-negative real-analytic function has zeros of even order. Since $\widehat{F^*}$ is continuous and $\widehat{F^*} \geq 0$ on $[\xi_2, \infty)$, $\int \widehat{F^*} d\nu = 0$ forces ν to be carried by the closed set $\widehat{Z}(F^*)$. $\square$

The hypothesis (E) in (b) is essential: under $\neg(\text{E})$ the conditions (ii)–(iv) hold vacuously while $\kappa^* < 0$. “Purely atomic” does not mean “with integer atoms”.

2.5. Complementary slackness.

Lemma 2.13 (Complementary slackness). *If $F \in \mathcal{C}$ and $\mathcal{A}(F) = 0$, then every $(\mu, \nu) \in \mathcal{K}$ has μ carried by $Z(F)$ and ν carried by $\widehat{Z}(F)$.*

Proof. $0 = \mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$, with continuous integrands that are non-negative where the measures live. $\square$

No limit, no hypothesis (E) and no information on κ^* is needed. Combined with Theorem 3.7, this gives the magic-function principle (Corollary 3.8).

3. UNIQUENESS RESULTS

This section proves uniqueness for admissible pairs whose supports are close to those of p_ζ . Section 3.1 shows, with soft analysis only, how far positivity alone goes. From Section 3.2 on, the input is the Fourier interpolation theorem of Bondarenko, Radchenko and Seip [8], which does not assume RH. We use the following notation. $\vartheta(\tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau}$ is Jacobi’s theta function and Γ_ϑ the theta group, generated by $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and viewed, as in [8], in $\text{PSL}_2(\mathbb{R})$, so that matrices are identified up to sign and $S^2 = 1$ (as matrices, $S^2 = -I$); we write $\tau = \sigma + iv \in \mathbb{H}$ and $\omega = e^{2\pi i/3}$. μ is the Möbius function, $\mathbf{1}_\square(m) = 1$ if m is a perfect square and 0 otherwise, and $\Lambda(\sqrt{m}) := 0$ if m is not a square. Recall $\mathcal{M} = \{\log m/(4\pi) : m \geq 1\}$; note $\xi_n = \log(n^2)/(4\pi)$, so $\xi_{\text{pp}} \subset \mathcal{M}$.

3.1. Why arithmetic is needed. How far can two admissible pairs differ, if one uses only positivity and the gap $(-\xi_2, \xi_2)$ in the prime measures? For a real measure ν on $[\xi_2, \infty)$ write $\nu^s := \nu + \nu(-\cdot)$ for its even extension.

Lemma 3.1 (Homogeneous directions). *Let $\mathcal{D}$ be the real vector space of pairs $\Delta = (\Delta\mu, \Delta\nu)$, with $\Delta\mu$ an even real Radon measure on $\mathbb{R}$ and $\Delta\nu$ a real Radon measure on $[\xi_2, \infty)$, such that $|\Delta\mu|([T-1, T+1]) \ll \log(3+|T|)$, $|\Delta\nu|([X-1, X+1]) \ll e^{\pi X}$, and $\mathcal{R}_{\Delta\mu, \Delta\nu}(F) = 0$ for every $F \in \mathcal{T}$. Then:*

- (a) $\mathcal{K} - \mathcal{K} \subseteq \mathcal{D}$.
- (b) For Δ with these growth bounds, $\mathcal{R}_\Delta = 0$ on $\mathcal{T}$ if and only if $\widehat{\Delta\mu} = -\frac{1}{2\pi}(\Delta\nu)^s$ in $\mathcal{D}'(\mathbb{R})$; in that case $(\Delta\nu)^s$ is tempered.
- (c) Consequently $\widehat{\Delta\mu} = 0$ on $(-\xi_2, \xi_2)$: any two admissible pairs satisfy $\int \varphi d\mu_1 = \int \varphi d\mu_2$ for every even real φ with $\widehat{\varphi} \in C_c^\infty((-\xi_2, \xi_2))$.

The proof is in Appendix B.1.3. By (b), the elements of $\mathcal{D}$ are pairs of measures in Fourier duality with a spectral gap. Those whose measures are countable sums of point masses are Fourier summation formulas in the sense of [36, 37, 79]; in general they are not, since a smooth even density on $[\xi_2, \infty)$ and its Fourier transform give an element of $\mathcal{D}$ with two continuous sides.

Proposition 3.2 (Two-sided rigidity). *Let $\Delta = (\Delta\mu, \Delta\nu) \in \mathcal{D}$ and let $Z \subset \mathbb{R}$ be countable.*

- (a) *If $0 \notin Z$, $\Delta\mu \geq 0$ off Z , $\sum_{z \in Z} (\Delta\mu)_-(\{z\}) < \infty$, and $\Delta\nu \geq 0$ off a Lebesgue-null set, then $\Delta = 0$.*

- (b) If $0 \notin Z$, $\Delta\mu \geq 0$ off Z and $(\Delta\nu)_-$ is a finite measure, then $\Delta = 0$.
(c) If $(\Delta\mu)_-$ is carried by a finite set, then $\Delta = 0$.

The proof (Appendix B.1.3) uses only that every even $\varphi \geq 0$ with $\widehat{\varphi}$ supported in $(-\xi_2, \xi_2)$ has the same integral against $(\Delta\mu)_+$ and $(\Delta\mu)_-$, by (c). Applied to $\Delta = (\mu - \mu_\zeta, \nu - \nu_\zeta)$ under RH, with $Z = Z_\zeta$ (which does not contain 0, since $\zeta(\frac{1}{2}) \neq 0$; indeed $(1 - 2^{1-\sigma})\zeta(\sigma) = \sum_{n \geq 1} (-1)^{n-1} n^{-\sigma} > 0$ for $0 < \sigma < 1$ [33, (25.2.3)]), it shows that no admissible pair other than p_ζ arises from ζ 's pair by only adding mass on one side, or by deleting a finite amount of mass on either side. The hypotheses cannot be weakened, by the following examples.

Proposition 3.3 (Poisson examples). *Let $\beta \geq \xi_2$.*

- (a) $\Delta\mu := \beta dt - \sum_{n \in \mathbb{Z}} \delta_{n/\beta}$, $\Delta\nu := 2\pi\beta \sum_{m \geq 1} \delta_{\beta m}$ defines a non-zero $\Delta \in \mathcal{D}$ with $\Delta\nu \geq 0$ and $\Delta\mu \geq 0$ off $Z = \beta^{-1}\mathbb{Z}$, which contains 0; $(\Delta\mu)_-$ has infinite mass.
(b) $\Delta\mu := \beta dt - \sum_{n \in \mathbb{Z}} \delta_{(n+1/2)/\beta}$, $\Delta\nu := 2\pi\beta \sum_{m \geq 1} (-1)^m \delta_{\beta m}$ defines a non-zero $\Delta \in \mathcal{D}$ with $\Delta\mu \geq 0$ off $Z = \beta^{-1}(\mathbb{Z} + \frac{1}{2})$, which does not contain 0, and $\Delta\nu \geq 0$ off the null set $\{\beta m : m \text{ odd}\}$; both $(\Delta\mu)_-$ and $(\Delta\nu)_-$ have infinite mass.

Proof. By Poisson summation, $\widehat{\sum_n \delta_{n/\beta}} = \beta \sum_m \delta_{\beta m}$ and $\widehat{\sum_n \delta_{(n+1/2)/\beta}} = \beta \sum_m (-1)^m \delta_{\beta m}$, while $\widehat{\beta dt} = \beta \delta_0$. Hence $\widehat{\Delta\mu} = -\frac{1}{2\pi}(\Delta\nu)^s$ in both cases, the growth bounds are clear, $\Delta\nu$ lives on $[\beta, \infty) \subseteq [\xi_2, \infty)$, and Lemma 3.1(b) gives $\Delta \in \mathcal{D}$. The sign statements are read off. $\square$

So the sign and gap conditions of Proposition 3.2, taken alone, do not force homogeneous rigidity. The examples are not shown to be differences of admissible pairs for ζ 's data, so they do not rule out every soft proof of (U); they show that such a proof needs more than these conditions. For $\beta = \xi_2$ the exceptional set in (b) lies inside the prime-power nodes, so the obstruction is on the zero side. The rest of this section supplies arithmetic through the basis of [8].

3.2. Uniqueness on the support of ζ 's pair. For $s \in \mathbb{C}$ let $F(\tau, s) = \sum_{N \geq 0} \alpha_N(s) e^{\pi i N \tau}$ be the unique 2-periodic analytic function of moderate growth on $\mathbb{H}$ satisfying

$$(3.1) \quad F(\tau, s) + (\tau/i)^{-1/2} F(-1/\tau, s) = \mathfrak{p}_s(\tau) := (\tau/i)^{-s} + (\tau/i)^{s-1/2}$$

(principal branches); this is $F_{1/2}^-(\tau, s)$ of [8, Theorem 3.1]. Its coefficients grow polynomially in N , are entire in s , and satisfy $\alpha_N(\frac{1}{2} - s) = \alpha_N(s)$ and $\alpha_N(\bar{s}) = \overline{\alpha_N(s)}$ (both by uniqueness in (3.1)); hence $\alpha_N(s) \in \mathbb{R}$ on the line $\text{Re } s = \frac{1}{4}$. Let b_N ($N \geq 0$) be the even Fourier eigenfunctions with eigenvalue +1 of Radchenko and Viazovska [69] (the functions b_N^- of [8, §3.4.1]), so that $b_N(\sqrt{n}) = \delta_{Nn}$ for $n \geq 1$, and put

$$B_N(x) := \sum_{k \geq 1} b_N(kx) \quad (x > 0).$$

Definition 3.4 (The BRS basis). By [8, Theorem 1.1], which assumes neither RH nor simplicity of the zeros, there are even entire functions U_m ($m \geq 1$) and $V_{\rho,j}$ (ρ a non-trivial zero with $\text{Im } \rho > 0$, $0 \leq j < m(\rho)$, where $m(\rho)$ is the multiplicity) such that

$$\begin{aligned} U_m^{(j)}(t_\rho) &= 0 \quad (j < m(\rho)), & \widehat{U_m}\left(\frac{\log m'}{4\pi}\right) &= \delta_{mm'}, \\ V_{\rho,j}^{(j')}(t_{\rho'}) &= \delta_{(\rho,j),(\rho',j')}, & \widehat{V_{\rho,j}}\left(\frac{\log m}{4\pi}\right) &= 0 \end{aligned}$$

for all $m, m' \geq 1$. They decay rapidly in vertical strips: see the paragraph on the rapid decay of the basis functions in the proof of [8, Theorem 1.1] ([8, §4.3]). The bounds used there for ζ^* and for the coefficients $\alpha_{m,1/2}^\pm$ are uniform for $\text{Re } s$ in compact sets, so the decay is uniform on

every closed strip $|\operatorname{Im} z| \leq \frac{1}{2} + \delta$, and the real and imaginary parts of these functions lie in $\mathcal{T}$. Explicitly [8, §§4.2–4.3],

$$(3.2) \quad U_m(z) = \frac{m^{-1/4}}{2\pi} h_m(\tfrac{1}{2} + iz), \quad h_m(s) = \frac{\Gamma_{\mathbb{R}}(s)\zeta(s)}{2} \sum_{d^2|m} \boldsymbol{\mu}(d) \alpha_{m/d^2}(s/2),$$

and $\widehat{U}_m(\xi) = m^{-1/4} e^{\pi\xi} u_m(e^{2\pi\xi})$ with $u_m(x) = \sum_{d^2|m} \boldsymbol{\mu}(d) B_{m/d^2}(x)$. We call U_m and $V_{\rho,j}$ the *BRS functions*.

By Lemma 2.3, applied to real and imaginary parts,

$$(3.3) \quad \mathcal{A}(U_m) = \frac{1}{\pi} \Lambda(\sqrt{m}) m^{-1/4}, \quad \mathcal{A}(V_{\rho,0}) = 2m(\rho), \quad \mathcal{A}(V_{\rho,j}) = 0 \quad (j \geq 1).$$

Indeed U_m vanishes at every t_ρ and $\widehat{U}_m(\xi_n) = \widehat{U}_m(\log(n^2)/4\pi) = \delta_{m,n^2}$. For $V_{\rho,j}$ the prime sum vanishes; the zeros with negative imaginary part are the $1 - \rho'$ with $\operatorname{Im} \rho' > 0$, with $t_{1-\rho'} = -t_{\rho'}$, so by evenness $\sum_{\rho'} V_{\rho,j}(t_{\rho'}) = 2 \sum_{\operatorname{Im} \rho' > 0} m(\rho') V_{\rho,j}(t_{\rho'}) = 2m(\rho) \delta_{j0}$. These are the values of [8, §4.4], with two slips corrected: there the first value is printed as $\pi^{-1} \Lambda(n) n^{-1/2}$ for a square $m = n$, whereas (3.3) gives $\pi^{-1} \Lambda(k) k^{-1/2}$ at $m = k^2$, and the second as 2 instead of $2m(\rho)$.

Proposition 3.5 (Uniqueness on ζ 's support). *Let $(\mu, \nu) \in \mathcal{K}$ with μ carried by Z_ζ and ν carried by $\mathcal{M}$. Then RH holds and $(\mu, \nu) = p_\zeta$. Positivity of (μ, ν) is not used beyond making the identity meaningful.*

This is essentially the observation made in [8, §4.4], that the Riemann–Weil functional has the values (3.3) in the interpolation basis, combined with the uniqueness statement [8, Corollary 1.1]; we record it as a corollary of [8, Theorem 1.1].

Proof. Write $\mu = \sum_{\gamma \in Z_\zeta} w_\gamma \delta_\gamma$ and $\nu = \sum_{m \geq 1} \nu(\{\frac{\log m}{4\pi}\}) \delta_{\log m/4\pi}$. Every pairing below is a finite sum, by the interpolation properties. Suppose ρ_0 is a zero with $\operatorname{Im} \rho_0 > 0$ and $\operatorname{Re} \rho_0 \neq \frac{1}{2}$. Every real ordinate $\gamma \in Z_\zeta$ is $\pm t_\rho$ for an on-line zero $\rho \neq \rho_0$ with $\operatorname{Im} \rho > 0$, so $V_{\rho_0,0}(\gamma) = 0$; and $\widehat{V}_{\rho_0,0}$ vanishes on $\mathcal{M}$. Hence $\mathcal{R}_{\mu,\nu}(V_{\rho_0,0}) = 0$, whereas $\mathcal{A}(V_{\rho_0,0}) = 2m(\rho_0) \neq 0$ by (3.3). So RH holds. For $\gamma > 0$ in Z_ζ and $\rho = \frac{1}{2} + i\gamma$, testing on $V_{\rho,0}$ gives $w_\gamma + w_{-\gamma} = 2m(\rho)$, i.e. $w_\gamma = m_\gamma$ since μ is even; so $\mu = \mu_\zeta$. Testing on U_m gives $\frac{1}{\pi} \nu(\{\frac{\log m}{4\pi}\}) = \frac{1}{\pi} \Lambda(\sqrt{m}) m^{-1/4}$, i.e. $\nu = \nu_\zeta$. $\square$

The proof uses the identity $\mathcal{A} = \mathcal{R}_{\mu,\nu}$ only for $F \in \{U_m, V_{\rho,j}\}$, and no positivity. In this form it applies to real measures μ (even) and ν carried by Z_ζ and $\mathcal{M}$, and we use it so in the proof of Theorem 3.7.

3.3. Finitely many extra zeros. Fix a finite symmetric set $E \subset \mathbb{R} \setminus Z_\zeta$, and put $E_+ := E \cap [0, \infty)$ and $s_e := \frac{1}{4} + \frac{ie}{2}$.

Lemma 3.6 (Structure of pairs with extra atoms). *Let μ be an even real measure carried by $Z_\zeta \cup E$ and ν a real measure carried by $\mathcal{M}$ such that $\mathcal{A}(F) = \mathcal{R}_{\mu,\nu}(F)$ for $F = U_m$ ($m \geq 1$) and $F = V_{\rho,j}$. Write $a_e := \mu(\{e\})$, $w_\gamma := \mu(\{\gamma\})$, $\ell_m := m^{1/4} \nu(\{\frac{\log m}{4\pi}\})$, $C_N := \sum_{k^2|N} \ell_{N/k^2}$,*

$$\tilde{q}_e := \frac{a_e \Xi(e)}{\frac{1}{4} + e^2} \quad (e > 0), \quad \tilde{q}_0 := 2a_0 \Xi(0), \quad c_N(E) := \sum_{e \in E_+} \tilde{q}_e \alpha_N(s_e) \in \mathbb{R}.$$

Then for every $N \geq 1$

$$(3.4) \quad C_N = c_N(E) + \frac{1}{2}(\log N) \mathbf{1}_\square(N).$$

Moreover: $c_1 = c_2 = c_3 = 0$ when ν is carried by $[\xi_2, \infty)$; $w_\gamma = m_\gamma - \frac{1}{2} \sum_{e \in E} a_e V_{\rho,0}(e)$ for $\gamma > 0$ and $\rho = \frac{1}{2} + i\gamma$; $\frac{1}{2} \sum_e a_e V_{\rho,0}(e) = m(\rho)$ for every zero ρ off the line with $\operatorname{Im} \rho > 0$;

$\sum_e a_e V_{\rho,j}(e) = 0$ for $1 \leq j < m(\rho)$; and $(\mu, \nu) = p_\zeta$ if and only if $a = 0$, if and only if $c(E) \equiv 0$. If $(\mu, \nu) \in \mathcal{K}$, then $C_N \geq 0$ for all N ; in particular $c_N(E) \geq 0$ for every non-square N .

The proof is in Appendix B.1.4.

At a non-square N there is no room: $c_N(E) \geq 0$ outright. At squares, ζ 's own prime mass $\frac{1}{2} \log N$ gives a positive margin, which is why a soft argument cannot work there.

Theorem 3.7 (Finitely many extra zeros). *Let $E \subset \mathbb{R} \setminus Z_\zeta$ be finite and symmetric. Let μ be an even real measure carried by $Z_\zeta \cup E$ and ν a real measure carried by $\mathcal{M}$ such that $\mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for every BRS function F (in particular, if the identity holds on $\mathcal{T}$ with absolutely convergent integrals). Assume $\nu(\{\frac{\log m}{4\pi}\}) \geq 0$ for every $m \equiv 2 \pmod{3}$, and $\mu(\{0\}) \geq 0$ if $0 \in E$. Then RH holds, $\mu = \mu_\zeta$ and $\nu = \nu_\zeta$. In particular every admissible pair carried by $(Z_\zeta \cup E) \times \mathcal{M}$ is p_ζ .*

Proof. By Lemma 3.6, (3.4) holds. If $N \equiv 2 \pmod{3}$ and $k^2 \mid N$, then $3 \nmid k$, so $N/k^2 \equiv 2 \pmod{3}$; hence $c_N(E) = C_N = \sum_{k^2 \mid N} \ell_{N/k^2} \geq 0$ for all $N \equiv 2 \pmod{3}$ (no such N is a square). Since $a_0 \geq 0$ and $\Xi(0) = -\frac{1}{8}\pi^{-1/4}\Gamma(\frac{1}{4})\zeta(\frac{1}{2}) > 0$, we have $\tilde{\varrho}_0 \geq 0$. Lemma B.3 in Appendix B.1.5 now gives $\tilde{\varrho}_e = 0$ for every $e \in E_+$, so $a \equiv 0$, μ is carried by Z_ζ , and the proof of Proposition 3.5 (which uses no positivity) gives RH, $\mu = \mu_\zeta$ and $\nu = \nu_\zeta$. $\square$

The argument in more detail. Positivity enters only through the residue class 2 mod 3, and the work is to see the extra zeros in the coefficients $c_N(E)$ of that class. Their generating function

$$\Theta_E(\tau) := \sum_{N \geq 1} c_N(E) e^{\pi i N \tau} = \sum_{e \in E_+} \tilde{\varrho}_e (F(\tau, s_e) - \alpha_0(s_e))$$

is a finite combination of the modular integrals of (3.1), and by orthogonality of additive characters the class 2 mod 3 is picked out by three cusps:

$$\Theta_E^{(2)}(v) := \sum_{N \equiv 2(3)} c_N(E) e^{-\pi N v} = \frac{1}{3} \left[\Theta_E(iv) + \omega \Theta_E\left(\frac{2}{3} + iv\right) + \bar{\omega} \Theta_E\left(\frac{4}{3} + iv\right) \right].$$

The points 0 , $\frac{2}{3}$ and $\frac{4}{3}$ are cusps of Γ_ϑ equivalent to ∞ . At 0 , (3.1) itself gives the expansion of $F(iv, s)$ as $v \rightarrow 0^+$. At $\frac{2}{3}$, the word ST^2ST^2S maps $\frac{2}{3}$ to ∞ , and unrolling (3.1) along it writes $F(\frac{2}{3} + iv, s)$ as $-\alpha_0(s)\vartheta(\frac{2}{3} + iv)$ plus explicit powers of v , a constant and an error $O(v^{3/4})$; at $\frac{4}{3}$ one uses $F(-\bar{\tau}, s) = \overline{F(\tau, s)}$ (Lemmas B.1 and B.2). Two cancellations follow. Each cusp contributes a theta term of size $v^{-1/2}$, and these combine to $-\alpha_0(s) \sum_{n^2 \equiv 2(3)} e^{-\pi n^2 v}$, which vanishes identically because no square is $\equiv 2 \pmod{3}$; and the constants $\alpha_0(s_e)$ cancel because $1 + \omega + \bar{\omega} = 0$. What remains is the exact expansion

$$\begin{aligned} \Theta_E^{(2)}(v) &= \sum_{e \in E_+, e \neq 0} (B_e v^{-s_e} + \overline{B_e} v^{-\bar{s}_e}) + B_0 v^{-1/4} + C + O(v^{3/4}) \quad (v \rightarrow 0^+), \\ B_e &= \frac{\tilde{\varrho}_e}{3} (1 - 3 \cdot 9^{-s_e}), \quad B_0 = \frac{2\tilde{\varrho}_0}{3} (1 - \sqrt{3}). \end{aligned}$$

Since $|3 \cdot 9^{-s_e}| = \sqrt{3} \neq 1$, $B_e = 0$ only if $\tilde{\varrho}_e = 0$. Landau's theorem concludes. The Dirichlet series $D(w) := \sum_{N \equiv 2(3)} c_N(E) N^{-w}$ has non-negative coefficients, and $\pi^{-w} \Gamma(w) D(w) = \int_0^\infty \Theta_E^{(2)}(v) v^{w-1} dv$. By the expansion, D continues meromorphically to $\operatorname{Re} w > -\frac{3}{4}$, with poles at most at the non-real points $s_e, \bar{s}_e$ and at $\frac{1}{4}$, where the residue has the sign of $B_0 \leq 0$. If some $B_e \neq 0$, the continuation has a pole on the line $\operatorname{Re} w = \frac{1}{4}$, so the abscissa of convergence of D is at least $\frac{1}{4}$, and in particular finite (it is not $-\infty$). A Dirichlet series with non-negative coefficients is singular at the real point of its abscissa of convergence [60, Theorem 1.7]; as the

continuation is analytic on $(\frac{1}{4}, \infty)$, that point is $\frac{1}{4}$, so $B_0 \neq 0$ and $D(\sigma) \rightarrow -\infty$ as $\sigma \downarrow \frac{1}{4}$, which is impossible since $D(\sigma) \geq 0$. So every B_e vanishes.

Related results. The identities of Lemma 3.6 are the explicit-formula form of a remark in [8, §4.1]: by Bochner’s converse theorem [5], a Dirichlet series with the functional equation of the kernel and finitely many poles is determined up to the kernels themselves, which strengthens Knopp’s abundance principle [52]. The restriction to a residue class is a twist by additive characters, the standard tool of the degree-one classification of the Selberg class [25, 49, 74]; there it is not combined with positivity or with Landau’s theorem. The closest uniqueness theorem we found is Booker’s converse theorem for positive L -data [9, Definition 1.3, Theorem 1.7]. There the zero multiplicities are real, integral outside a finite set, and non-negative outside a finite set, and positive data of degree less than $5/3$ are 1 or shifted Dirichlet L -functions; the prime side is a priori supported on $\{\log n\}$ with growth conditions on the coefficients. Theorem 3.7 is complementary to it, not a generalisation: it allows non-integral weights at infinitely many ordinates and prime mass at the non-square points of $\mathcal{M}$, imposes no growth condition, and uses positivity on the prime side, on one residue class, instead of on the zero side; but it requires the zero measure to be carried by Z_ζ and finitely many further points, and the prime measure by $\mathcal{M}$. In the language of Fourier summation formulas [36], the signed pairs that the BRS basis produces on $Z_\zeta \cup E$ are exactly what the theorem excludes under positivity.

3.4. The magic-function principle.

Corollary 3.8 (Magic-function principle). *Suppose some $F \in \mathcal{C}$ satisfies $\mathcal{A}(F) = 0$, has only finitely many real zeros outside Z_ζ , and has $\widehat{Z}(F) \subseteq \mathcal{M}$. Then $\mathcal{K} \subseteq \{p_\zeta\}$, that is, (U) holds. If (E) holds as well, then RH holds and $\mathcal{K} = \{p_\zeta\}$.*

Proof. By Lemma 2.13, every admissible pair is carried by $(Z_\zeta \cup E) \times \mathcal{M}$ with $E := Z(F) \setminus Z_\zeta$, which is finite and symmetric because F is even. Theorem 3.7 applies. $\square$

As in the proof that the Leech lattice is the unique optimal lattice in its dimension [17], the zeros of an extremal function determine the extremal configuration. The prime-side condition $\widehat{Z}(F) \subseteq \mathcal{M}$ is needed (Remark 3.9). The converse, that (U) implies the existence of such an F , is not known: given (E) it amounts to a strict complementarity statement for an infinite-dimensional linear programme (Problem (4)). The functions of Section 4 have the form $\Xi^2 H$, for which $Z(\Xi^2 H) = Z_\zeta \cup \{\text{real zeros of } H\}$.

3.5. Integer weights and extra prime atoms. With integer zero weights, as for an L -function, (U) becomes a uniqueness problem for Beurling generalised number systems with Riemann’s functional equation. For prime measures on $\{\xi_n : n \geq 2\}$ this is Hamburger’s theorem [41], and for prime measures that generate a uniformly discrete multiplicative semigroup, a setting related to Lagarias’s study of Beurling integers with the Delone property [55], it is treated in [48] (in preparation) with the theorem of Lev and Olevskii on measures with uniformly discrete support and spectrum [57]; Problem (1) asks for the general case.

Remark 3.9 (Extra prime atoms). Theorem 3.7 allows extra zeros but no prime mass off $\mathcal{M}$, and this is where the method stops. An extra prime-side atom at η enters the analogue of the identities (3.4) through the numbers $B_N(x_\eta)$, $x_\eta = e^{2\pi\eta}$, which are the Fourier coefficients of a second modular integral built from $\vartheta(x_\eta^2 \tau)$. Its sums over a class of quadratic non-residues modulo an odd prime have a leading term of order $v^{-1/2}$ at the cusps, given by Gauss sums, when x_η^2 is rational; when x_η^2 is irrational that leading term vanishes. So an atom with $x_\eta^2 \notin \mathbb{Q}$ is invisible to the leading-order argument behind Landau’s theorem; see [46].

4. THE ZERO-KILLING FAMILY AND THEOREM D

Every function of the form $F = \Xi^2 H$, with H analytic on a strip containing $|\operatorname{Im} t| \leq \frac{1}{2}$, vanishes at every non-trivial zero of ζ , on or off the critical line. We call such functions *zero-killing*. For such F the explicit formula loses its zero side, and $\mathcal{A}(F)$ becomes a sum over the prime powers (Corollary 4.4). Section 4.1 gives the Fourier transform of $\Xi^2 H$ in closed form; Section 4.2 defines the functions $\Xi^2 P_J(t^2)$ whose Fourier transforms have zeros of order at least two at the first J prime powers; and Section 4.3 proves Theorem D.

Here $\xi(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$ with $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$, $\Xi(t) = \xi(\frac{1}{2} + it)$ is Riemann's Ξ -function, and $\Xi_{\infty}(t) := \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)$ at $s = \frac{1}{2} + it$ is its archimedean part. We use the coordinates $x = e^{2\pi\xi}$ and $y = 2\pi x$, so that the prime power n sits at $\xi = \xi_n$, at $x = n$ and at $y = 2\pi n$, and $\theta = x \frac{d}{dx} = y \frac{d}{dy} = \frac{1}{2\pi} \frac{d}{d\xi}$. For $a < \pi/2$, $\mathcal{H}_a$ is the class of even functions H , real on $\mathbb{R}$ and analytic on some closed strip $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ ($\delta > 0$), with $|H(t)| \leq Ce^{a|\operatorname{Re} t|}$ there; $\mathcal{H} := \bigcup_{a < \pi/2} \mathcal{H}_a$.

4.1. The Bessel form. We use two classical facts about Ξ . First, Riemann's representation $\Xi(t) = \int_{\mathbb{R}} \Phi(u) e^{iut} du$ with $\Phi(u) = 2 \sum_{n \geq 1} (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2}) \exp(-\pi n^2 e^{2u})$, where Φ is even [67, p. 305, (2)–(3) and the remark after (4)]. Second, Stirling's formula and the convexity bound for ζ give, for every $b \in (0, 1]$, a constant C_b with

$$(4.1) \quad |\Xi(t)| \leq C_b (1 + |\operatorname{Re} t|)^3 e^{-\pi|\operatorname{Re} t|/4} \quad (|\operatorname{Im} t| \leq b)$$

[78]. (The same bound holds for $b \leq \frac{5}{2}$; only $b < 1$ is used.) By (4.1), $\Xi^2 H \in \mathcal{T}_{\delta}$ whenever $H \in \mathcal{H}_a$ is analytic near $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ (the decay $e^{-(\pi/2-a)|\operatorname{Re} t|}$ absorbs any power of $1 + |z|$); in particular $\Xi^2 P(t^2) \in \mathcal{T}_{\delta}$ for every polynomial P and every $\delta \in (0, \frac{1}{2})$.

Lemma 4.1 (Positivity of the kernel). $\Psi(\xi) = \widehat{\Xi^2}(\xi) > 0$ for every $\xi \in \mathbb{R}$.

Proof. For $u \geq 0$ every term of the series for Φ is positive, because $2\pi n^2 e^{2u} \geq 2\pi > 3$; as Φ is even, $\Phi > 0$ on $\mathbb{R}$ (see [27, Theorem A(i)], which credits Pólya [68]). Hence $\widehat{\Xi}(\xi) = 2\pi\Phi(2\pi\xi) > 0$, and $\Psi = \widehat{\Xi} * \widehat{\Xi} > 0$. $\square$

Write K_0, K_1 for the modified Bessel functions and

$$(4.2) \quad W_0(z) := z^2[(z^2 + 9)K_0(z) - 6zK_1(z)] = \theta^2(\theta + 1)^2 K_0(z), \quad \theta = z \frac{d}{dz}.$$

The second expression follows from $\theta K_0 = -zK_1$ and $\theta(zK_1) = -z^2 K_0$: on the basis $e_0 = K_0$, $e_1 = zK_1$ one has $(\theta + 1)^2 e_0 = (1 + z^2)e_0 - 2e_1$, then $\theta(\theta + 1)^2 e_0 = 4z^2 e_0 - (1 + z^2)e_1$, and finally $\theta^2(\theta + 1)^2 e_0 = z^2(z^2 + 9)e_0 - 6z^2 e_1$.

Proposition 4.2 (Bessel form of Ψ). For every real ξ , with $x = e^{2\pi\xi}$,

$$\Psi(\xi) = \int_{\mathbb{R}} \Xi(t)^2 e^{-2\pi i \xi t} dt = 2\pi\sqrt{x} \sum_{N \geq 1} d(N) W_0(2\pi N x),$$

the series converging absolutely and uniformly on compact sets, together with all its ξ -derivatives. In particular Ψ is even and real, and $\Psi(\xi_n) = (C_{\infty} + o(1))n^4 e^{-2\pi n}$ as $n \rightarrow \infty$, with $C_{\infty} = \pi(2\pi)^4 = (2\pi)^{9/2}(\pi/2)^{1/2} = 4896.3\dots$. The Gamma-only analogue $\Psi_1(\xi) := \widehat{\Xi_{\infty}^2}(\xi) = 2\pi\sqrt{x} W_0(2\pi x)$ also holds, and extends to complex x with $|\arg x| < \pi/2$.

Here Ξ_{∞} is not real on $\mathbb{R}$, so Ψ_1 is real but not even, and $\Xi_{\infty}^2 P(t^2)$ is not a test function; Ψ_1 enters only through Lemma 4.6. The proof is in Appendix B.2.1. One has $\int_0^{\infty} \sum_N d(N) K_0(2\pi N \sqrt{\lambda}) \lambda^{s/2-1} d\lambda = \frac{1}{2} \Gamma_{\mathbb{R}}(s)^2 \zeta(s)^2$; in the variable u with $\lambda = e^{2u}$, the operator $(\partial_u^2 - \frac{1}{4})^2$ removes the double poles at $s = 0$ and $s = 1$, and Fourier inversion identifies the result with Ξ^2 .

Remark 4.3 (Attribution). Proposition 4.2 is a differentiated form of the extended Koshliakov formula. Koshliakov's formula for $\sum_N d(N)K_0(2\pi Nx)$ [53] had been found earlier by Ramanujan [2, Ch. 3]; its extended form, with an integral of Ξ^2 , is due to Dixit [30], and in [31, Theorem 1.4] it expresses $\sqrt{x}\sum_N d(N)K_0(2\pi Nx)$, up to an explicit polar term, as a cosine transform of $\Xi(t/2)^2/(1+t^2)^2$. The operator $(\theta^2 - \frac{1}{4})^2$ removes both the polar term and the factor $(t^2 + \frac{1}{4})^{-2}$. For more general Koshliakov-kernel identities of this type see [32, Theorem 1.2]. We include the proof (Appendix B.2.1) because the normalisation matters for the certificates of Section 5.

Corollary 4.4 (The explicit formula for zero-killing functions). *Let $F = \Xi^2 H$ with H analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and assume $F \in \mathcal{T}$ (for instance $H \in \mathcal{H}$, or H a polynomial in t^2). Then*

$$\mathcal{A}(F) = \frac{1}{\pi} \sum_{n \in \text{PP}} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n).$$

In particular, if $\widehat{F}$ vanishes at $\xi_{n_1}, \dots, \xi_{n_J}$, then $\mathcal{A}(F) = \frac{1}{\pi} \sum_{n \in \text{PP}, n > n_J} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n)$. No hypothesis on the zeros of ζ is used.

Proof. Apply Lemma 2.3. For every non-trivial zero ρ , on or off the line, $t_\rho = -i(\rho - \frac{1}{2})$ satisfies $|\operatorname{Im} t_\rho| < \frac{1}{2}$ and $\Xi(t_\rho) = \frac{1}{2}\rho(\rho-1)\Gamma_{\mathbb{R}}(\rho)\zeta(\rho) = 0$, so $F(t_\rho) = \Xi(t_\rho)^2 H(t_\rho) = 0$ and the zero side vanishes. $\square$

4.2. The exact family. Let $n_1 < n_2 < \dots$ be the prime powers, $\text{PP} = \{n_j\}$, and put $m_k := \widehat{t^{2k}\Xi^2}$; since $(P(t^2)F)^\wedge = P(-\theta^2)\widehat{F}$, $m_k = (-\theta^2)^k \Psi$. All derivatives below are taken in ξ .

Proposition 4.5 (The exact family is a linear Hermite interpolation). *Let $J \geq 1$. If the matrix $M_J = [m_k(\xi_{n_j}); m'_k(\xi_{n_j})]_{j \leq J, 1 \leq k \leq 2J}$ is non-singular, that is $J \in \mathcal{J}$, there is exactly one polynomial P_J with $P_J(0) = 1$ and $\deg P_J \leq 2J$ such that $\widehat{F}_J := \Xi^2 \widehat{P_J(t^2)}$ has zeros of order at least two at $\xi_{n_1}, \dots, \xi_{n_J}$. Its coefficient vector is $p^{(J)} = -M_J^{-1}b_J$, with $b_J = [m_0(\xi_{n_j}); m'_0(\xi_{n_j})]$. Equivalently, $\widehat{F}_J$ is the error of Hermite interpolation of Ψ by $\text{span}\{m_1, \dots, m_{2J}\}$ at the nodes. Each $F_J = \Xi^2 H_J$, $H_J(t) = P_J(t^2)$, lies in $\mathcal{T}$, and $\mathcal{A}(F_J) = \frac{1}{\pi} \sum_{n \in \text{PP}, n > n_J} \Lambda(n) n^{-1/2} \widehat{F}_J(\xi_n)$.*

Proof. We have $\widehat{F}_P = \sum_k p_k m_k$ for $F_P = \Xi^2 P(t^2)$, so the $2J$ conditions are the linear system $M_J(p_1, \dots, p_{2J})^\top = -b_J$. The element $-\sum_{k \geq 1} p_k^{(J)} m_k$ of $\text{span}\{m_1, \dots, m_{2J}\}$ has the data of $m_0 = \Psi$, which is the second description. The rest is the remark before Lemma 4.1 and Corollary 4.4. $\square$

Every real polynomial with $P(0) = 1$ factors over $\mathbb{R}$ as $\prod_j (1 + r_j u + s_j u^2)$, by pairing complex conjugate roots and pairing the real roots arbitrarily (with one factor having $s_j = 0$ if the degree is odd). The certified representatives of Section 5 use this parametrisation of P_J .

Lemma 4.6 (An integral representation). *For a polynomial P put $Q_P(\theta) := P(-(\theta + \frac{1}{2})^2)\theta^2(\theta + 1)^2$ and $Q_P(y) := e^y Q_P(\theta) e^{-y}$, with $\theta = y \frac{d}{dy}$. Then Q_P is a polynomial in y of degree $2 \deg P + 4$, with leading coefficient $(-1)^{\deg P}$ times that of P , and $Q_P(0) = 0$. For $z > 0$,*

$$g_P(z) := [P(-(\theta + \frac{1}{2})^2)W_0](z) = \int_z^\infty Q_P(y) e^{-y} (y^2 - z^2)^{-1/2} dy,$$

and $\widehat{\Xi^2 P(t^2)}(\xi) = 2\pi\sqrt{x} \sum_N d(N) g_P(2\pi Nx)$.

The proof is in Appendix B.2.2. The certificates of Section 5.2 evaluate $\widehat{\Xi^2 H}$ and its derivatives through Proposition 4.2 and this lemma.

4.3. Robust compactness and the proof of Theorem D. The next lemma is the compactness step behind every conditional theorem below. It needs no convergence of the family and no exact positivity at finite J .

Lemma 4.7 (Robust compactness). *Let H_J , J in an infinite index set, be even, real on $\mathbb{R}$ and analytic on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, and put $F_J := \Xi^2 H_J$. Assume:*

- (i) $|H_J(t)| \leq C e^{a|\operatorname{Re} t|}$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$, with C and $a < \pi/2$ independent of J ;
- (ii) $H_J(0) = 1$;
- (iii) $\liminf_J H_J(t) \geq 0$ for every real t ;
- (iv) $\liminf_J \widehat{F}_J(\xi) \geq 0$ for every $\xi \geq \xi_2$;
- (v) $\widehat{F}_J(\xi_n) \rightarrow 0$ for every prime power n .

Then a subsequence H_J converges locally uniformly on $\{|\operatorname{Im} t| < \frac{1}{2} + \delta\}$ to some H_∞ , and $F_\infty := \Xi^2 H_\infty \in \mathcal{C} \cap \mathcal{T}$, with $\mathcal{A}(F_\infty) = 0$ and $\widehat{F}_\infty(0) = \int F_\infty > 0$. Hence $\kappa^ \leq 0$; moreover $\mathcal{A}(F) \geq 0$ for every $F \in \mathcal{C}$ of the form $\Xi^2 H$, so the infimum of $\mathcal{A}(F)/\int F$ over such F is 0 and is attained at F_∞ .*

The proof (Appendix B.2.3) combines Montel's theorem with dominated convergence: by (i) and (4.1), $|F_J(t)| \leq C(1 + |\operatorname{Re} t|)^6 e^{-(\pi/2 - a)|\operatorname{Re} t|}$ on the strip, uniformly in J .

The certified functions of Section 5 satisfy (ii)–(v) only approximately; (iv) and (v) are asymptotic conditions, so cushioned or rationalised representatives may be used, provided the uniform bound (i) holds for them. The hypotheses of Lemma 4.7, for some sequence, are equivalent to the existence of an exact magic function of the form $\Xi^2 H$ with $H \in \mathcal{H}$ and $H(0) \neq 0$ (take the constant sequence $H/H(0)$), which is a priori stronger than (S).

For the exact family $(P_J)_{J \in \mathcal{J}}$ of Proposition 4.5 we consider the following hypotheses.

- (N) $\mathcal{J}$ is infinite.
- (G_a) There are $\delta \in (0, \frac{1}{2})$, $a < \pi/2$ and C with $|H_J(t)| \leq C e^{a|\operatorname{Re} t|}$ on $|\operatorname{Im} t| \leq \frac{1}{2} + \delta$ for all $J \in \mathcal{J}$.
- (Z_∞) No real point is a limit of zeros of H_{J_k} with $J_k \in \mathcal{J}$, $J_k \rightarrow \infty$.
- (Mg) For every $\xi \in [\xi_2, \infty) \setminus \xi_{\text{PP}}$, $\liminf_{J \in \mathcal{J}} \widehat{F}_J(\xi) > 0$.

Given (G_a), (Z_∞) is equivalent to: for every $T > 0$, $\liminf_{J \in \mathcal{J}} \min_{|t| \leq T} H_J(t) > 0$ (Appendix B.2.3). It holds, for instance, if there are $\eta > 0$ and J_0 such that every zero of H_J , $J \in \mathcal{J}$, $J \geq J_0$, has $|\operatorname{Im} t| \geq \eta$. Since a bound on a strip implies the same bound on every narrower strip, the size of δ in (G_a) plays no role. Finite- J instances of (N), and of the coefficient bound (4.3) below, which would give (G_a) and (Z_∞), are certified for $J \in \{10, 60, 61, 110, 111\}$ (Proposition 5.5); (Mg) is a lim inf in J and has no finite- J instance.

Theorem 4.8 (Theorem D). *Assume (N), (G_a), (Z_∞) and (Mg) for the exact family. Then some subsequential limit $F_\infty = \Xi^2 H_\infty$, with $H_\infty \in \mathcal{H}$ and $H_\infty > 0$ on $\mathbb{R}$, satisfies $F_\infty \in \mathcal{C} \cap \mathcal{T}$, $\mathcal{A}(F_\infty) = 0$, $F_\infty^{-1}(0) \cap \mathbb{R} = Z_\zeta$ and $\widehat{F}_\infty^{-1}(0) \cap [\xi_2, \infty) = \xi_{\text{PP}}$. Hence (S) and (U) hold, and $\text{RH} \Leftrightarrow \text{(E)}$; under either, $\kappa^* = 0$ and $\mathcal{K} = \{p_\zeta\}$.*

Proof. Hypotheses (i), (ii) of Lemma 4.7 are (G_a) and $P_J(0) = 1$. For (v): if $n = n_j$, then $\widehat{F}_J(\xi_n) = 0$ for every $J \in \mathcal{J}$ with $J \geq j$. For (iv): at $\xi \in \xi_{\text{PP}}$ this holds by (v), elsewhere by (Mg). For (iii): by the equivalent form of (Z_∞), for each T there is $c_T > 0$ with $H_J \geq c_T$ on $[-T, T]$ for all large $J \in \mathcal{J}$. So Lemma 4.7 applies, and its limit satisfies $H_\infty \geq c_T > 0$ on $[-T, T]$ for every T . Hence $F_\infty^{-1}(0) \cap \mathbb{R} = Z_\zeta$. For $\xi \in [\xi_2, \infty) \setminus \xi_{\text{PP}}$, $\widehat{F}_\infty(\xi) = \lim \widehat{F}_J(\xi) \geq \liminf_{J \in \mathcal{J}} \widehat{F}_J(\xi) > 0$, so the zeros of $\widehat{F}_\infty$ on $[\xi_2, \infty)$ are exactly $\xi_{\text{PP}} \subset \mathcal{M}$. Lemma 4.7 gives (S), and Corollary 3.8, applied to F_∞ , gives (U), and RH under (E); conversely RH implies (E) (Theorem 2.9). Under either, $\kappa^* \geq 0$ (Theorem 2.7), so $\kappa^* = 0$, and $\mathcal{K} = \{p_\zeta\}$. $\square$

No convergence of the family is assumed, and the proof never divides by Ξ^2 : H_∞ inherits the bound (G_a) , so $F_\infty \in \mathcal{T}$ directly. The strict inequality $a < \pi/2$ is essential for the domination. Further details (norm convergence, the equivalent forms of (Z_∞)) are in Appendix B.2.3. In sphere packing, Cohn and Miller conjectured, with numerical evidence, that explicit sequences of functions converge to the magic functions in dimensions 8 and 24 [20]; Theorem 4.8 asks for the same kind of convergence, for an explicit family, in the explicit formula.

Hypotheses (G_a) and (Z_∞) concern the polynomials P_J alone. They follow from a bound on the coefficients.

Proposition 4.9 (Coefficient bounds control the zero side). *Suppose that for all $J \in \mathcal{J}$ with $J \geq J_0$,*

$$(4.3) \quad 0 \leq p_k^{(J)} \leq C \frac{a^{2k}}{(2k)!} \quad \text{for all } k, \quad \text{with } 0 \leq a < \pi/2 \text{ and } C \text{ independent of } J.$$

Then for these J : (1) $H_J(t) \geq 1$ for all real t , so (Z_∞) holds; (2) $|H_J(t)| \leq C \cosh(a|t|)$ for all $t \in \mathbb{C}$, so (G_a) holds on every strip; (3) $\{H_J\}$ is normal on $\mathbb{C}$, and every limit H_∞ is entire of exponential type at most a , with $0 \leq [t^{2k}]H_\infty \leq Ca^{2k}/(2k)!$ and $H_\infty \geq 1$ on $\mathbb{R}$; (4) $|P_J(u)| \leq P_J(|u|)$, and every root u of P_J satisfies $|u| \geq \min_k p_{k-1}^{(J)}/p_k^{(J)}$ (minimum over k with $p_k^{(J)} > 0$).

The proof is in Appendix B.2.4.

Remark 4.10 (Variants). (a) In Theorem 4.8, (Z_∞) may be replaced by a lower bound $H_J \geq h$ on $\mathbb{R}$ for all large $J \in \mathcal{J}$, with $h : \mathbb{R} \rightarrow (0, \infty)$; the pointwise form $\liminf_{\mathcal{J}} H_J(t) > 0$ for every real t suffices, and the limit then satisfies $H_\infty \geq h > 0$. By Proposition 4.9(1), the coefficient bound (4.3) for all large J gives this with $h \equiv 1$; so (N), (Mg) and (4.3) for all large J imply (S) and (U). (b) If $\kappa^* = 0$ and some weak magic function F^* has only finitely many real zeros outside Z_ζ and satisfies $\widehat{Z}(F^*) \subseteq \mathcal{M}$, then RH holds and $\mathcal{K} = \{p_\zeta\}$: $\kappa^* = 0$ gives (E) (Theorem 2.7), by Theorem 2.12(c) every admissible pair is carried by $Z(F^*) \times \widehat{Z}(F^*)$, and Theorem 3.7 applies to each of them.

5. CERTIFIED BOUNDS

5.1. Method. Every statement of this section is computer-assisted, and this paragraph states the conventions once for the whole paper. Each proof names the function that is certified, the lemmas it uses and the arithmetic. Appendix C gives, for each statement, the script and its input, the command that re-runs it, the run time and the decisive output line; the SHA-256 hashes of the inputs are listed there or in the file SHA256SUMS of the ancillary files. *Exact* computations use integers and rationals (FLINT [76]); *ball* computations use Arb midpoint-radius arithmetic with outward rounding [44], and a sign is accepted only when the ball excludes 0. Every input is an exact rational parameter file whose hash the verifier recomputes. Printed upper bounds are rounded up and lower bounds down. Statements labelled “Numerical observation” are computed but not certified, and are never used in a proof. The common lemmas (positivity certificates, validated Bessel functions, the cushion) are in Appendix A. Theorem 5.1, Propositions 5.3 and 5.4 and the upper bounds in Proposition 5.9 use only weak duality (Lemma 2.5) and the exact membership of an explicit function in $\mathcal{C}$ or $\mathcal{C}_{\text{Ops}}$: Proposition 5.4 bounds masses, and the others bound slacks from above. Proposition 5.8 and the lower bounds in Proposition 5.9 use explicit admissible pairs (Lemma 5.7). Proposition 5.5 concerns the exact family. No statement about the zeros of ζ is used.

5.2. The slack.

Theorem 5.1 (A certified upper bound for the slack). $\kappa^* \leq 1.1508391 \cdot 10^{-906}$. More precisely, for each J in Table 1 there are explicit rationals r_j, s_j ($j \leq J$) and $\varepsilon > 0$ such that, with $H(t) := \prod_{j \leq J} (1 + r_j t^2 + s_j t^4)$, the function

$$F_{\text{rep}}(t) := \Xi(t)^2 (H(t) + \varepsilon e^{-\pi t^2})$$

lies in $\mathcal{C}$ and satisfies $\mathcal{A}(F_{\text{rep}})/\int F_{\text{rep}} \leq \kappa_J$, with κ_J as in the table; in particular $\kappa_{100} = 1.1508391 \cdot 10^{-906}$.

TABLE 1. Certified upper bounds κ_J for κ^* (Theorem 5.1); ε is the weight of the cushion and x_{far} the start of the far region in the proof.

J	κ_J (rounded up)	ε	x_{far}
20	$2.0151627 \cdot 10^{-62}$	10^{-67}	55
23	$1.1963682 \cdot 10^{-80}$	10^{-86}	66
24	$5.2941806 \cdot 10^{-91}$	10^{-93}	70
30	$7.7522383 \cdot 10^{-132}$	10^{-138}	93
35	$9.9924044 \cdot 10^{-177}$	10^{-183}	116
40	$9.3828982 \cdot 10^{-220}$	10^{-226}	137
50	$1.8493714 \cdot 10^{-311}$	10^{-318}	182
60	$1.0640143 \cdot 10^{-417}$	10^{-424}	231
100	$1.1508391 \cdot 10^{-906}$	10^{-911}	452

Proof (computer-assisted; ball arithmetic; Appendix C, Table 4, first row). Write $H(t) = P(t^2)$. Four facts are proved. (1) $H > 0$ on $\mathbb{R}$, exactly: $s_j > 0$ and $r_j^2 < 4s_j$ for every j . Hence $F_{\text{rep}} \geq 0$ on $\mathbb{R}$, and $F_{\text{rep}} \in \mathcal{T}$ (Section 4.1). (2) $\widehat{F}_{\text{rep}} \geq 0$ on $[\xi_2, \infty)$. The transform of $\Xi^2 H$ and its ξ -derivatives are evaluated through Proposition 4.2 and Lemma 4.6, with K_0, K_1 from Lemma A.3: the operator $P(-(\theta + \frac{1}{2})^2)\theta^2(\theta + 1)^2$ acts exactly on $\text{span}\{z^k K_0, z^k K_1\}$, and the divisor sum is truncated with a rigorous tail bound. Near each of the J prime powers where $\widehat{\Xi^2 H}$ has a zero of order at least two, up to the rationalisation of the parameters, including the gap edge ξ_2 , a Taylor model centred exactly at ξ_n gives a quadratic lower bound; elsewhere on $[\xi_2, \xi_{\text{far}}]$, Taylor models on pieces with dyadic endpoints give lower bounds, with a rigorous bound for the remainder. The cushion term $\varepsilon \widehat{\Xi^2 e^{-\pi t^2}}$ is positive everywhere and is at least $c(\xi) := \varepsilon I_\delta e^{-\pi(\xi+\delta)^2}$ for $\xi \geq 0$, with $\delta = \frac{1}{20}$ and I_δ as in Lemma A.4; it makes the transform positive at and near the nodes, where the rationalisation may leave $\widehat{\Xi^2 H}$ slightly negative. *The far region.* By Lemma 4.6, $\widehat{\Xi^2 H}(\xi) = 2\pi\sqrt{x} \sum_N d(N)(A(2\pi Nx)K_0(2\pi Nx) + B(2\pi Nx)K_1(2\pi Nx))$ with explicit polynomials A, B of degree at most D . Write $|A|, |B|$ for the polynomials whose coefficients are the absolute values of those of A, B , and $z = 2\pi x$. From $d(N) \leq 2\sqrt{N}$, $0 < K_0 \leq K_1 \leq K_{3/2}$ with $K_{3/2}(w) = (\pi/2w)^{1/2}e^{-w}(1 + 1/w)$, and a geometric series over N (ratio at most $q := 2^D e^{-z} < \frac{1}{2}$), one gets $|\widehat{\Xi^2 H}(\xi)| \leq \Upsilon(x) := 4\pi(|A| + |B|)(z)e^{-z}(1 + 1/z)/(1 - q)$. The verifier takes the first integer x_{far} beyond the last node with $2\pi x_{\text{far}} > D + 4$ and $\Upsilon(x_{\text{far}}) < c(\xi_{\text{far}})$, and checks $\xi_{\text{far}} + \delta < 3$, where $\xi_{\text{far}} = \log x_{\text{far}}/2\pi$. Since $\frac{d}{dz} \log \Upsilon \leq D/z - 1$ and $\frac{d}{dz} \log c = -(\xi + \delta)/z$, we get $\frac{d}{dz}(\log \Upsilon - \log c) \leq (D + \xi + \delta)/z - 1$. This is negative for every $x \geq x_{\text{far}}$, because $z - \xi$ increases with z and $z_{\text{far}} > D + 4 > D + \xi_{\text{far}} + \delta$. Hence $\Upsilon < c$, and the transform is positive, on all of $[x_{\text{far}}, \infty)$. (3) $\mathcal{A}(F_{\text{rep}})$ is bounded above using Corollary 4.4 for $\Xi^2 H$ and the value of

$\mathcal{A}(\Xi^2 e^{-\pi t^2})$ in Lemma A.4. The prime-power sum is evaluated term by term up to $x_{\text{far}} + 1$; beyond, its terms are at most $\log n \cdot \Upsilon(n)$, and their ratios are checked to be below $\frac{1}{2}$, so the tail is bounded by a geometric series. (4) $\int F_{\text{rep}} \geq \widehat{\Xi^2 H}(0)$, which is bounded below. Since $F_{\text{rep}}/\int F_{\text{rep}} \in \mathcal{C}$ has integral 1, $\kappa^* \leq \mathcal{A}(F_{\text{rep}})/\int F_{\text{rep}}$, and the printed bound is the upper end of the resulting ball, rounded up. $\square$

Every F_{rep} vanishes at every non-trivial zero of ζ , on or off the line, and H is a rational approximation of the exact polynomial P_J of Proposition 4.5, factored as in Section 4.2. By Corollary 4.4, $\mathcal{A}(F_{\text{rep}})$ is essentially a tail sum starting at the first prime power n_{J+1} not killed by H ; its first term is of the size of $\Psi(\xi_{n_{J+1}}) \approx C_{\infty} n_{J+1}^4 e^{-2\pi n_{J+1}}$ (Proposition 4.2) times a cofactor. Table 1 gives $-\log \kappa_J/n_{J+1} = 3.30, 3.82, 4.17, 4.55, 4.95$ at $J = 20, 30, 40, 60, 100$ ($n_{J+1} = 43, 79, 121, 211, 421$), against $2\pi \approx 6.28$ for the factor $e^{-2\pi n_{J+1}}$ alone. So the number of certified digits is governed by n_{J+1} , and it grows with J for as long as positivity can be certified.

Corollary 5.2 (Conductor form; infeasible deformations). *(a) No admissible pair has a zero measure $\mu \geq \lambda dt$ with $\lambda > 1.1508391 \cdot 10^{-906}$. (b) If $\log q < -2\pi \cdot 1.1508391 \cdot 10^{-906}$, in particular if $q \leq 1 - 7.3 \cdot 10^{-906}$, the data $\mathcal{A}_q$ admit no admissible pair. Equivalently, replacing $\log \pi$ by $\log \pi + \eta$ in Ω_{∞} leaves no admissible pair once $\eta > 7.2309354 \cdot 10^{-906}$. Hence $q_{\min} = e^{-2\pi \kappa^*} \geq 1 - 7.3 \cdot 10^{-906}$. Under RH, $q_{\min} \leq 1$, so then $1 - 7.3 \cdot 10^{-906} \leq q_{\min} \leq 1$.*

Proof. (a) is Lemma 2.5 applied to $F_{\text{rep}}/\int F_{\text{rep}}$ with $J = 100$. (b) For that function, $\mathcal{A}_q = \mathcal{A} + (\log q/2\pi) \int$ is negative when $\log q < -2\pi \kappa_{100}$, which excludes admissible pairs by Lemma 2.5; replacing $\log \pi$ by $\log \pi + \eta$ is the case $\log q = -\eta$, and $2\pi \kappa_{100} \leq 7.2309354 \cdot 10^{-906}$. Then $q_{\min} \geq e^{-7.2309354 \cdot 10^{-906}} \geq 1 - 7.3 \cdot 10^{-906}$ (Proposition 2.10). Under RH, $p_{\zeta} \in \mathcal{K}$, so $\kappa^* \geq 0$ (Theorems 2.9 and 2.7). $\square$

5.3. The classical cone and low-height rigidity. Let $f_k(u) := L_k^{(-1/2)}(2\pi u^2)e^{-\pi u^2}$; these are even Hermite functions, with $\widehat{f}_k = (-1)^k f_k$. They form the Laguerre basis that Cohn and Gonçalves used for sign uncertainty [16].

Proposition 5.3 (The classical cone). $\kappa^* \leq \kappa_{\text{OPS}}^* \leq 9.9462 \cdot 10^{-41}$. Explicitly, with $s = 13$, $K = 128$, explicit rationals $c_0, \dots, c_K$ and $\varepsilon = 10^{-56}$, the function

$$F_{\text{rep}}(t) := \sum_{k \leq K} c_k f_k(t/s) + \varepsilon(e^{-\pi t^2/(16s^2)} + e^{-16\pi t^2/s^2})$$

satisfies $F_{\text{rep}} > 0$ and $\widehat{F}_{\text{rep}} > 0$ on all of $\mathbb{R}$, and $\mathcal{A}(F_{\text{rep}})/\int F_{\text{rep}} \leq 9.9461827002 \cdot 10^{-41}$. Consequently the classical explicit-formula conductor bound in degree 1, optimised over $\mathcal{C}_{\text{OPS}}$, is at least $1 - 6.25 \cdot 10^{-40}$.

Proof (computer-assisted; exact Sturm sequences and ball arithmetic; Appendix C). Write $X = 2\pi t^2/s^2$, $Y = 2\pi s^2 \xi^2$ and $F := \sum_{k \leq K} c_k f_k(t/s)$. Then $F = e^{-X/2} P(X)$ and $\widehat{F} = s e^{-Y/2} Q(Y)$ with $P, Q \in \mathbb{Q}[X]$ of degree 128. Exact Sturm sequences show that P and Q have no root in $(0, \infty)$, and $P(0), Q(0)$ and the leading coefficients are positive; hence $F > 0$ and $\widehat{F} > 0$ on $\mathbb{R}$. The two Gaussians of the repair term are positive with positive Fourier transforms, so $F_{\text{rep}} \geq F$ and $\widehat{F}_{\text{rep}} \geq \widehat{F}$, and $F_{\text{rep}} \in \mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}$. (Here the repair term is not needed for positivity, since P and Q have no positive roots; its contribution to $\mathcal{A}(F_{\text{rep}})$ is included in the computed value.) Then $\mathcal{A}(F_{\text{rep}})$ is one rigorous Arb integral of F_{rep} against the digamma weight on a finite interval, plus an explicit tail bound (Lemma A.5), and $\int F_{\text{rep}} = \widehat{F}_{\text{rep}}(0)$ is enclosed exactly up to a ball. Finally $2\pi \cdot 9.9461827002 \cdot 10^{-41} \leq 6.2494 \cdot 10^{-40}$ and $e^{-a} \geq 1 - a$. $\square$

This bound needs no gap at ξ_2 , so it is directly comparable with Odlyzko's tables [62, Table 4]. A certificate of the same kind bounds the mass that *any* admissible pair can put away from the low zeros of ζ and away from the prime powers.

Proposition 5.4 (Low-height rigidity of every admissible pair). *There is an explicit $F_{80} \in \mathcal{C}_{\text{OPS}}$ of the form of Proposition 5.3, with $K = 80$, $s = 12$ and $\varepsilon = 10^{-30}$, such that $\mathcal{A}(F_{80}) \leq 7.72525424595 \cdot 10^{-25}$ and $\int F_{80} \geq 0.9999999999999999444$. Consequently every admissible pair (μ, ν) satisfies the mass bounds of Table 2.*

TABLE 2. Mass bounds valid for every admissible pair (Proposition 5.4). The intervals I_k are the gaps between the first six positive zero ordinates of ζ , shrunk by 0.1 at both ends and rounded inward to ten decimals; the prime-side windows are $[\xi_a, \xi_b]$.

Zero side			Prime side	
k	I_k	bound for $\mu(I_k)$	$[a, b]$	bound for ν
1	[14.2347251418, 20.9220396387]	$2.75 \cdot 10^{-17}$	[2.05, 2.95]	$6.81 \cdot 10^{-19}$
2	[21.1220396388, 24.9108575801]	$1.91 \cdot 10^{-16}$	[5.05, 6.95]	$6.94 \cdot 10^{-14}$
3	[25.1108575802, 30.3248761258]	$4.81 \cdot 10^{-15}$	[7.05, 7.95]	$2.72 \cdot 10^{-12}$
4	[30.5248761259, 32.8350615877]	$2.28 \cdot 10^{-14}$	[10.05, 10.95]	$5.95 \cdot 10^{-10}$
5	[33.0350615878, 37.4861781588]	$2.39 \cdot 10^{-13}$	[12.05, 12.95]	$3.80 \cdot 10^{-8}$
			[14, 15]	$1.82 \cdot 10^{-8}$
			atom at $\xi_{2.5}$	$1.53 \cdot 10^{-21}$
			atom at ξ_6	$6.37 \cdot 10^{-18}$

No hypothesis on the zeros is used: whether or not RH holds, every admissible pair puts at most these masses into windows that avoid the low zeros of ζ and the low prime powers. Positivity in the explicit formula has long been used to locate low zeros [62, p. 132], [59, 4]; here the positivity of both measures takes the place of GRH.

Proof (computer-assisted; exact arithmetic and ball arithmetic; Appendix C). As in the proof of Proposition 5.3, exact Sturm sequences give $F_{80} \in \mathcal{C}_{\text{OPS}}$, and ball arithmetic gives the bounds for $\mathcal{A}(F_{80})$ and $\int F_{80}$. For $F \in \mathcal{C}$, μ even and $I \subset (0, \infty)$, Lemma 2.5 gives $2 \min_I F \cdot \mu(I) \leq \mathcal{A}(F)$, and $\min_I \hat{F} \cdot \nu(I) \leq \pi \mathcal{A}(F)$. Lower bounds $F_{80} \geq m$ on 40 intervals in t and $\hat{F}_{80} \geq m$ on 19 windows in ξ are proved in exact arithmetic: $e^{X/2}$ is bounded by a rational Taylor polynomial with Lagrange remainder, and positivity of the resulting rational polynomials is shown by Descartes' rule after a Möbius map, with bisection (Lemma A.2). The values $\hat{F}_{80}(\xi_n)$ at the two atoms are enclosed in ball arithmetic. With the certified upper bound for $\mathcal{A}(F_{80})$, the two inequalities give the bounds of Table 2, rounded up. $\square$

5.4. The exact family at finite J .

Proposition 5.5 (Certified finite- J instances). *For $J \in \{10, 60, 61, 110, 111\}$ the Hermite matrix M_J is non-singular, every coefficient of the exact P_J is positive, and $\max_{1 \leq k \leq 2J} ((2k)! p_k^{(J)})^{1/2k} \leq 0.71636$, that is, $0 < p_k^{(J)} \leq 0.71636^{2k} / (2k)!$. Hence $H_J \geq 1$ on $\mathbb{R}$ and $|H_J(t)| \leq \cosh(0.71636|t|)$ on $\mathbb{C}$.*

Proof (computer-assisted; ball arithmetic; Appendix C). Through Proposition 4.2, the moments $(-1)^k (2\pi)^{-2k} \Psi(2k+\varepsilon)(\xi_n)$, with $\varepsilon \in \{0, 1\}$, are enclosed rigorously; rows and columns are scaled by exact powers of two, and the ball system is solved with Arb's preconditioned solver. A

successful solve proves that every matrix in the ball, in particular M_J , is non-singular, and the output balls contain the exact $p_k^{(J)}$; positivity holds because every ball is positive. The certified bounds for $\max_k ((2k)!p_k)^{1/2k}$ are 0.697078 ($J = 10$), 0.716207, 0.716214, 0.716351 and 0.716352; the bounds on H_J follow from Proposition 4.9 with $C = 1$. Each enclosure is compared with a second enclosure computed by a different implementation, and every pair of balls overlaps. $\square$

Uniform versions of these facts are the hypotheses (N), (G_a) and (Z_∞) of Theorem 4.8, through Proposition 4.9.

Numerical observation 5.6 (The exact family for $J \leq 110$). The following were computed in high-precision arithmetic (at most 4600 bits); the floating-point table is `computed/family_J1_110.tsv` in the ancillary files. For every $1 \leq J \leq 110$ the Hermite system is non-singular and $p_{2J}^{(J)} > 0$. All coefficients $p_k^{(J)}$ are positive for $J \in \{1, 2\}$ and for $10 \leq J \leq 110$; for $3 \leq J \leq 9$ the coefficient $p_{2J-1}^{(J)}$ is non-positive, and for $4 \leq J \leq 8$ so is $p_{2J-3}^{(J)}$. The quantity $a_J := \max_{1 \leq k \leq 2J} ((2k)!|p_k^{(J)}|)^{1/2k}$ (for $J \leq 110$ the maximum is attained at a positive coefficient, so the table's values are the same if only positive coefficients are used) stays below 0.71 for $J \leq 9$ and increases with J for $10 \leq J \leq 110$, up to 0.7164 at $J = 110$ (attained near $k \approx 20$ –24); so (4.3) holds numerically with $a = 0.7164$ and $C = 1$ for $10 \leq J \leq 110$. Dense scans at $J = 20, 30, 40, 50, 60$ ($x \in [2, 1.15n_J + 5]$, step 0.05, and up to $x = 425$ at $J = 60$) found no sign change of $\hat{F}_J$ off the nodes.

5.5. A lower bound for the slack, and other archimedean data. Lower bounds for slacks come from explicit admissible pairs, which are feasible points of the dual linear programme; Cohn and Triantafyllou used dual feasible points in the same way to prove that linear programming bounds for sphere packing are not sharp [21]. We build them for the data of ζ with a conductor q , that is, for $\mathcal{A}_q = \mathcal{A} + \frac{\log q}{2\pi} \int$ (Section 2.4), and, as a control, for the archimedean data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$. For $\mathfrak{g} \in \{\Gamma_{\mathbb{R}}, \Gamma_{\mathbb{R}}^2, \Gamma_{\mathbb{C}}\}$ let $(k_j)_{j \leq d}$ be (0) , $(0, 0)$ and $(0, 1)$ respectively, put

$$\Omega_{\mathfrak{g}}(t) := \frac{1}{2\pi} \sum_{j \leq d} \left(\operatorname{Re} \psi\left(\frac{\frac{1}{2} + k_j + it}{2}\right) - \log \pi \right),$$

$$\mathcal{A}_{\mathfrak{g},q}(F) := F\left(\frac{i}{2}\right) + F\left(-\frac{i}{2}\right) + \int_{\mathbb{R}} F(t) \left(\Omega_{\mathfrak{g}}(t) + \frac{\log q}{2\pi} \right) dt,$$

so that $\Omega_{\Gamma_{\mathbb{R}}} = \Omega_{\infty}$ and $\mathcal{A}_{\Gamma_{\mathbb{R}},q} = \mathcal{A}_q$. Admissible pairs, $\mathcal{C}$ and κ^* are defined for $\mathcal{A}_{\mathfrak{g},q}$ exactly as for ζ , with the same test class and the same gap ξ_2 (Definition 2.4); Lemma 2.5 holds with the same proof, and $\kappa^*(\mathcal{A}_{\mathfrak{g},q}) = \kappa^*(\mathcal{A}_{\mathfrak{g},1}) + \frac{\log q}{2\pi}$. More generally, for a gap $g > 0$ we write $\mathcal{C}_g$ and κ_g^* for the cone and the slack defined with $[g, \infty)$ in place of $[\xi_2, \infty)$, both for the prime measures and for the condition $\hat{F} \geq 0$; Lemma 2.5 holds with the same proof, and $\mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}_g$ for every g .

Lemma 5.7 (Herglotz form of an admissible pair). *Let $\tilde{\nu} \geq 0$ be a measure on $[2, \infty)$ such that $\sigma := \tilde{\nu} - \mathbf{1}_{[2, \infty)}(x) dx$ satisfies $\int x^{-1/2} d|\sigma|(x) < \infty$, and put $\mathcal{L}(s) := \int x^{-s} d\tilde{\nu}(x)$ for $\operatorname{Re} s > 1$. Then $\mathcal{L}$ extends continuously to $\operatorname{Re} s \geq \frac{1}{2}$, $s \neq 1$, as $2^{1-s}/(s-1) + \int x^{-s} d\sigma$. Let $\mu(t) := \Omega_{\mathfrak{g}}(t) + \frac{\log q}{2\pi} - \frac{1}{\pi} \operatorname{Re} \mathcal{L}\left(\frac{1}{2} + it\right)$, and let ν be the image of $x^{-1/2}\tilde{\nu}$ under $x \mapsto \log x/2\pi$. Then $\mathcal{A}_{\mathfrak{g},q}(F) = \int F \mu dt + \frac{1}{\pi} \int \hat{F} d\nu$ for every $F \in \mathcal{T}$. Hence $(\mu dt, \nu)$ is an admissible pair for $\mathcal{A}_{\mathfrak{g},q}$ if and only if $\mu \geq 0$.*

Proof. Let $F \in \mathcal{T}_{\delta}$ and $\frac{1}{2} < y < \frac{1}{2} + \delta$. Since $\hat{F}(\log x/2\pi) = \int F(t) x^{-it} dt$, shifting the contour to $\operatorname{Im} t = -y$ and using Fubini (note $\int x^{-1/2-y} d\tilde{\nu} < \infty$) gives $\frac{1}{\pi} \int \hat{F} d\nu = \frac{1}{\pi} \int_{\operatorname{Im} w = -y} F(w) \mathcal{L}\left(\frac{1}{2} + iw\right) dw$. The integrand is meromorphic in $-y \leq \operatorname{Im} w \leq 0$ with one simple pole, at $w = -i/2$, where $\mathcal{L}\left(\frac{1}{2} + iw\right) \sim 1/(i(w + \frac{i}{2}))$; F decays like $(1 + |w|)^{-2}$ and $\mathcal{L}$ is bounded away from the

pole. Moving the contour to $\text{Im } w = -\eta$, $0 < \eta < \frac{1}{2}$, picks up $2\pi i \cdot F(-\frac{i}{2})/i$. As $\eta \downarrow 0$ the integrals over $\text{Im } w = -\eta$ converge to the integral over $\mathbb{R}$, by dominated convergence: $\mathcal{L}(\frac{1}{2} + iw)$ is bounded on $-\frac{1}{4} \leq \text{Im } w \leq 0$ away from the pole and continuous up to $\text{Im } w = 0$. So $\frac{1}{\pi} \int \widehat{F} d\nu = \frac{1}{\pi} \int_{\mathbb{R}} F(t) \mathcal{L}(\frac{1}{2} + it) dt + 2F(-\frac{i}{2})$. As F is even and real and $\mathcal{L}(\frac{1}{2} - it) = \overline{\mathcal{L}(\frac{1}{2} + it)}$, the integral equals $\frac{1}{\pi} \int F \text{Re } \mathcal{L}(\frac{1}{2} + it) dt$, and $2F(-\frac{i}{2}) = F(\frac{i}{2}) + F(-\frac{i}{2})$. Adding $\int F \mu$ gives $\mathcal{A}_{g,q}(F)$. The measure ν is positive and carried by $[\xi_2, \infty)$, and μ is even; so admissibility is equivalent to $\mu \geq 0$. $\square$

Lemma 5.7 is a representation of Herglotz–Nevanlinna type: the zero density is read off from the boundary values of $\text{Re } \mathcal{L}$ on the critical line, much as Vedana’s classification of summation formulas on a strip describes the continuous part of the Fourier-side measure through a generalised Nevanlinna representation of a generating function [79].

We use measures $\tilde{\nu}$ equal to $\mathbf{1}_{[2,\infty)} dx$ minus densities on finitely many cells, plus finitely many atoms and polynomial bumps, minus tails $a_m x^{-m} dx$ ($m \geq 1$) beyond some X ; then $\mathcal{L}$ and μ are explicit. The verifier checks $\tilde{\nu} \geq 0$ and $\text{supp } \tilde{\nu} \subseteq [2, \infty)$, and proves $\mu > 0$ on a range $[0, T]$ by mean-value steps $\mu(t) \geq \mu(t_0) - h \sup |\mu'|$, with μ' enclosed in ball arithmetic on each whole step. On $[T, \infty)$ it uses monotone majorants: Ω_g is increasing in $t \geq 0$, while the contributions of the cells, bumps and tails to μ are bounded by explicit constants times the decreasing functions $(\frac{1}{4} + t^2)^{-1/2}$, $\min(1, (k/\beta t)^k)$ (for bumps of half-width β built from k uniform variables) and $((m - \frac{1}{2})^2 + t^2)^{-1/2}$. The tail coefficients are checked to satisfy $a_m \geq 0$, so that $1 - \sum_m a_m x^{-m}$ is increasing and $\tilde{\nu} \geq 0$ needs to be checked only at $x = X$.

Proposition 5.8 (An unconditional lower bound for the slack). *For ζ ’s data with conductor $q = e^{0.02}$ there is an admissible pair of the form of Lemma 5.7 with $\mu \geq 4.71949 \cdot 10^{-4}$ on $\mathbb{R}$. Hence $q_{\min} \leq e^{0.02}$ and*

$$\kappa^* \geq 4.71949 \cdot 10^{-4} - \frac{0.02}{2\pi} \geq -2.7112 \cdot 10^{-3}.$$

Proof (computer-assisted; ball arithmetic; Appendix C). The pair is certified as described above, with mean-value steps on $[0, 2000]$ and monotone majorants beyond; Lemma 5.7 gives admissibility for $\mathcal{A}_q$, so $q_{\min} \leq e^{0.02}$. For $F \in \mathcal{C}$ with $\int F = 1$, Lemma 2.5 gives $\mathcal{A}_q(F) \geq \int F \mu dt \geq 4.71949 \cdot 10^{-4}$; so $\kappa^*(\mathcal{A}_q) \geq 4.71949 \cdot 10^{-4}$, and $\kappa^* = \kappa^*(\mathcal{A}_q) - 0.02/(2\pi)$ by Proposition 2.10. Finally $0.02/(2\pi) \leq 3.1831 \cdot 10^{-3}$. $\square$

Proposition 5.9 (The data of $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$, also at their natural gaps). (a) *For the archimedean data $\mathcal{A}_{\Gamma_{\mathbb{R},5}^2}$ of $\mathbb{Q}(\sqrt{5})$ and every gap $g \in (0, \xi_{4.04915}]$, in particular for ζ ’s gap ξ_2 and for the natural gap ξ_4 of the field: $1.64 \cdot 10^{-5} \leq \kappa_g^* \leq 4.83 \cdot 10^{-4}$. In particular some admissible pair has zero measure $\geq 1.64 \cdot 10^{-5} dt$, and the set of admissible pairs is not a singleton.* (b) *For the data $\mathcal{A}_{\Gamma_{\mathbb{C},3}} of $\mathbb{Q}(\sqrt{-3})$ and every gap $g \in (0, \xi_{3.011664}]$, in particular for ξ_2 and for the natural gap ξ_3 : $1.50 \cdot 10^{-4} \leq \kappa_g^* \leq 9.99 \cdot 10^{-4}$, with the same consequences.$*

Proof (computer-assisted; exact Sturm sequences and ball arithmetic; Appendix C). (a) For the upper bound, an explicit F of the form of Proposition 5.3, a combination of $f_0(t/7), \dots, f_{80}(t/7)$ plus the two Gaussians, has $\mathcal{A}_{\Gamma_{\mathbb{R},1}^2}(F)/\int F \leq k_0 := -0.25566705789831547610$; positivity of F and $\widehat{F}$ on $\mathbb{R}$ is shown by exact Sturm sequences, so $F \in \mathcal{C}_{\text{OPS}} \subseteq \mathcal{C}_g$ for every g , and the functional is one Arb integral. Hence $\kappa_g^*(\mathcal{A}_{\Gamma_{\mathbb{R},5}^2}) \leq k_0 + \log 5/2\pi \leq 4.83 \cdot 10^{-4}$. For the lower bound, a pair of the form of Lemma 5.7 for $\mathcal{A}_{\Gamma_{\mathbb{R},q_0}^2}$, $\log q_0 = 1.6093347792651136$, has $\tilde{\nu} \geq 0$, and $\tilde{\nu}$ vanishes on $[2, x_0)$ for some $x_0 \geq 4.049150$, a bound read off exactly from its parameters; its zero density satisfies $\mu \geq 6.59 \cdot 10^{-9}$ on $[0, 800]$ and $\mu \geq 1.79$ on $[800, \infty)$. So the pair is admissible for every gap $g \leq \xi_{x_0}$. Read at $q = 5$, the same prime measure gives the zero density

$\mu + (\log 5 - \log q_0)/2\pi \geq 1.64 \cdot 10^{-5}$, and Lemma 2.5 for the gap g gives $\kappa_g^* \geq 1.64 \cdot 10^{-5}$. Adding to $\tilde{v}$ a small atom $w\delta_{x_1}$, $x_1 \geq x_0$, changes μ by $-(w/\pi)x_1^{-1/2} \cos(t \log x_1)$; for small $w > 0$ the result is a second admissible pair. The two certificates are combined in exact arithmetic, which gives $1.6420748 \cdot 10^{-5} \leq \kappa_g^* \leq 4.8294147 \cdot 10^{-4}$. (b) Identical, with a dual certificate for $\Gamma_{\mathbb{C}}$ ($K = 64$, functions $f_k(t/9)$), also in $\mathcal{C}_{\text{OPS}}$, and a pair of the form of Lemma 5.7 at $\log q_0 = 1.0976698108720329$ whose prime measure vanishes below $x_0 \geq 3.011664$ and whose zero density satisfies the certified floor $\mu \geq 6.67 \cdot 10^{-8}$ on $\mathbb{R}$. The lower bound is $(\log 3 - \log q_0)/2\pi + \inf \mu$; the floor is needed, since $(\log 3 - \log q_0)/2\pi$ alone is slightly below $1.5 \cdot 10^{-4}$. The combination gives $1.5006672 \cdot 10^{-4} \leq \kappa_g^* \leq 9.9854968 \cdot 10^{-4}$. $\square$

Remark 5.10 (The gap and the margins). The prime-ideal norms of $\mathbb{Q}(\sqrt{5})$ start at 4, and those of $\mathbb{Q}(\sqrt{-3})$ at 3, so ξ_4 and ξ_3 are the gaps that match the fields' own explicit formulas. Proposition 5.9 holds at these natural gaps as well as at ξ_2 , so for these two fields the comparison with ζ is like-for-like. Under GRH for the Dedekind zeta function, the field's own pair is admissible at its natural gap. Since $\kappa_g^* > 0$ there, these data are not critical, and their admissible pairs are not unique: the field's own pair, whose zero measure is atomic, and the certified pair, whose zero measure is at least $c dt$ with $c > 0$, are two distinct pairs. This indicates that near-criticality is a property of ζ 's data rather than of the method. These are two examples; we make no claim about L -functions in general. The margins are small. At every gap $g \leq \xi_{4.04915}$ the certificates in (a) place the critical conductor of the data $\Gamma_{\mathbb{R}}^2$ (the infimum of the q for which $\mathcal{A}_{\Gamma_{\mathbb{R}}^2, q}$ has an admissible pair at the gap g) in $[4.9848, 4.9995]$: the dual function excludes every $q < 4.9848$ by weak duality, and the pair is admissible at $q_0 < 4.9995$. So the conductor 5 of $\mathbb{Q}(\sqrt{5})$ exceeds it by a factor between $1 + 1.0 \cdot 10^{-4}$ and $1 + 3.1 \cdot 10^{-3}$. For $\Gamma_{\mathbb{C}}$ and every gap $g \leq \xi_{3.011664}$ the same argument gives $[2.9812, 2.9972]$, and the conductor 3 of $\mathbb{Q}(\sqrt{-3})$ exceeds it by a factor between $1 + 9.4 \cdot 10^{-4}$ and $1 + 6.3 \cdot 10^{-3}$. The slack as a function of the gap is treated in [45].

6. CONJECTURES AND OPEN PROBLEMS

For each conjecture, Table 3 gives what it would imply and through which result, the evidence, and finite computations that would refute it; these are sufficient tests, not a complete list of the ways in which the conjecture could fail.

TABLE 3. Conjectures of the paper: consequence, evidence, and sufficient finite refutation tests.

Statement	Gives (via)	Evidence	Refuted, for example, by
Conj. U	RH $\Leftrightarrow$ (E) (Thm. 2.9)	Section 3	a second admissible pair
Conj. S	$\kappa^* = 0$ under (E) (Thm. 2.7(a))	Table 1	an admissible pair at $q = 1$ with $\mu \geq \lambda dt$, $\lambda > 0$; the finite certificates here cannot confirm it
Conj. 6.1	(S), (U) (Thm. 4.8)	Prop. 5.5; Obs. 5.6	a singular M_J ; $p_k^{(J)} \leq 0$ with $J \geq 10$; $a_J \geq \pi/2$; no finite prefix of the family alone refutes (c)

6.1. A prime-power magic-function conjecture. Let P_J , $H_J = P_J(t^2)$ and $F_J = \Xi^2 H_J$ be the exact family of Proposition 4.5. The following conjecture is of the same kind as the conjectures of Cohn and Miller on explicit sequences converging to magic functions [20].

Conjecture 6.1 (Prime-power magic function). (a) M_J is non-singular for every $J \geq 1$.
 (b) For every $J \geq 10$ all coefficients $p_k^{(J)}$, $0 \leq k \leq 2J$, are positive, and $a^* := \sup_{J \geq 10} \max_{1 \leq k \leq 2J} ((2k)! p_k^{(J)})^{1/(2k)} < \pi/2$.
 (c) $P_J \rightarrow P_\infty$ coefficientwise, and $F_\infty := \Xi^2 P_\infty(t^2)$ satisfies $\widehat{F}_\infty(\xi) > 0$ for every $\xi \in [\xi_2, \infty) \setminus \xi_{\text{PP}}$.

Implication. The conjecture implies (S) and (U) unconditionally, hence $\text{RH} \Leftrightarrow (\text{E})$. Indeed (b) is (4.3) with $a = a^*$ and $C = 1$, so Proposition 4.9 gives $H_J \geq 1$ on $\mathbb{R}$ and $|H_J(t)| \leq \cosh(a^*|t|)$; with this bound, coefficientwise convergence gives $H_J \rightarrow H_\infty$ locally uniformly with a uniform exponential majorant of type $a^* < \pi/2$, so $\widehat{F}_J \rightarrow \widehat{F}_\infty$ pointwise by dominated convergence, and (c) gives (Mg). Remark 4.10(a) applies.

Evidence. Parts (a) and (b) are certified at $J \in \{10, 60, 61, 110, 111\}$ (Proposition 5.5) and computed for $1 \leq J \leq 110$ and $10 \leq J \leq 110$ respectively (Numerical observation 5.6), which also shows why (b) starts at $J = 10$.

Refutation. Each of the following, at some finite J , refutes the conjecture: a singular M_J ; a non-positive coefficient with $J \geq 10$; $a_J := \max_{1 \leq k \leq 2J} ((2k)! |p_k^{(J)}|)^{1/2k} \geq \pi/2$. These are sufficient tests, not a complete list. Part (b) asks for $\sup_{J \geq 10} a_J < \pi/2$, which fails if $a_J \rightarrow \pi/2$, even when $a_J < \pi/2$ for every J . Part (c) concerns the limit: no finite prefix of the computed family alone refutes it, although a finite computation combined with rigorous bounds for the rest of the family could.

6.2. Sharpness and uniqueness. The certificates of Section 5 bound κ^* above by positive numbers, so no finite collection of them proves (S); a proof needs more, for instance an exact magic function (Corollary 3.8, Theorem 4.8), which a finite computer-assisted argument for a suitable uniform statement might supply. A finite computation can refute (S): an explicit admissible pair at $q = 1$ whose zero measure is certified to satisfy $\mu \geq \lambda dt$ with $\lambda > 0$ would give $\kappa^* \geq \lambda > 0$ by Lemma 2.5. Proposition 5.9 certifies pairs of this kind for other archimedean data, at ξ_2 and at their natural gaps. Under (E), (S) says $\kappa^* = 0$, that is, every admissible μ is purely atomic on one discrete set (Theorem 2.12), and the optimised degree-one explicit-formula conductor bound is exactly 1. Conjecture U would be refuted by a single admissible pair other than p_ζ ; the results of Section 3 exclude such pairs on ζ 's support and with finitely many extra zeros.

6.3. Open problems.

- (1) *Beurling case.* Conjecture U for zero measures with integer weights and prime measures that generate a multiplicative semigroup which is not uniformly discrete; the uniformly discrete case is treated in [48] (in preparation), using [57]; compare [55].
- (2) *Extra primes.* Finitely many extra prime-side atoms off $\mathcal{M}$ (Remark 3.9), in particular atoms at positions η with $e^{4\pi\eta}$ irrational.
- (3) *Infinitely many extra zeros* on the zero side, for instance countable, locally finite sets E with $\sum_{e \in E} \mu(\{e\}) |\zeta(\frac{1}{2} + ie)| (1 + |e|)^{-1/4+\delta} < \infty$ for some $\delta > 0$.
- (4) *Strict complementarity.* Does (U) imply the existence of an exact magic function without spurious zeros?
- (5) *The existence statement* (E), that is $\kappa^* \geq 0$, without assuming RH: Odlyzko's question whether explicit-formula bounds proved under GRH hold unconditionally [62], specialised to $\mathbb{Q}$.

AI-USE STATEMENT

This paper was produced with substantial use of AI. The mathematical arguments, the numerical computations and the text were generated by instances of Claude Opus 5.5 (Anthropic) working as coordinated agents: one set of agents developed the arguments (building on [8] and on the explicit-formula method for discriminant and conductor bounds [75, 61, 73, 66, 65]), separate agents acted as independent hostile referees of each component, and the referees' requested repairs were incorporated. The numerical certificates were computed by AI-written scripts that are included as supplementary material (Appendix C) and can be rerun. The author initiated and directed this project, designed its verification process, and reviewed the results and the verification record, but did not independently check every step of the written proofs. The author takes full responsibility for the content of this paper, including any errors, and welcomes independent verification and corrections. Section 1.4 states precisely what is claimed.

APPENDIX A. CERTIFICATION METHODS

This appendix describes the arithmetic, the validated special functions and the cushion used in the certificates of Section 5; the conventions are stated in Section 5.1.

A.1. Exact and ball arithmetic.

Remark A.1 (Ball arithmetic and rigorous solves). *Exact* computations use integers and rationals (FLINT `fmpz`, `fmpq` and their polynomial and matrix types [76]). *Interval* computations use Arb ball arithmetic [44]: a real number is represented by a ball $[m \pm r]$ that is guaranteed to contain it, and every operation returns a ball containing all possible results. Both are accessed through python-flint. Constants such as π and Euler's constant γ_E are balls. Linear systems are solved by Arb's ball solver, which either returns balls containing the exact solution of every system in the input balls, and thereby certifies that the exact matrix is invertible, or fails. The non-rigorous library mpmath [77] is used only in self-tests, never in a certificate.

Positivity of polynomials is certified by coefficient signs.

Lemma A.2 (Positive-coefficient certificates). (a) (Descartes at c .) *If $p \in \mathbb{R}[t]$ and every coefficient of $p(c+t) \in \mathbb{R}[t]$ is non-negative with positive constant term, then $p > 0$ on $[c, \infty)$. If p vanishes to exact order m at c and the coefficients of $t^m, t^{m+1}, \dots$ in $p(c+t)$ are positive, then $p > 0$ on (c, ∞) .*

(b) *A polynomial with non-negative coefficients in $v \in [0, \infty)^m$ is non-negative there, and positive at every point at which some monomial with a positive coefficient is positive.*

With ball coefficients, "positive" means that the ball lies in $(0, \infty)$.

Proof. Both statements follow from the fact that a polynomial with non-negative coefficients takes non-negative values on the non-negative orthant, with the indicated strictness. $\square$

In Proposition 5.4 the lemma is applied, after a Möbius map of an interval onto $[0, \infty)$, to polynomials with exact rational coefficients; Taylor shifts of ball polynomials, where needed, are computed by Horner's scheme at a ball containing c , which encloses the exact shifted coefficients.

A.2. Validated Bessel functions. All certificates built on Ξ^2 (Theorem 5.1 and Proposition 5.5) evaluate K_0 and K_1 on positive real balls. We do not use the built-in Bessel and confluent hypergeometric functions of Arb for this purpose: at moderate z their enclosures can lose about e^{2z} in relative accuracy. Instead the shipped module `lib/besselk.py` uses two classical representations with explicit error bounds.

Lemma A.3 (Validated evaluation of K_0 and K_1). *Let $z > 0$ and $\nu \in \{0, 1\}$. (a) K_0 and K_1 are given by the convergent series [33, 10.31.2, 10.31.1]*

$$K_0(z) = -(\log \frac{z}{2} + \gamma_E) I_0(z) + \sum_{k \geq 1} \frac{H_k q^k}{(k!)^2},$$

$$K_1(z) = \frac{1}{z} + \log \frac{z}{2} I_1(z) - \frac{z}{4} \sum_{k \geq 0} \frac{(\psi(k+1) + \psi(k+2)) q^k}{k! (k+1)!},$$

with $q = z^2/4$, $I_0(z) = \sum_k q^k/(k!)^2$, $I_1(z) = \frac{z}{2} \sum_k q^k/(k!(k+1)!)$ and H_k the harmonic numbers. (b) For $\ell \geq \max(|\nu| - \frac{1}{2}, 1)$, $K_\nu(z) = \sqrt{\pi/(2z)} e^{-z} (\sum_{k < \ell} a_k(\nu) z^{-k} + R_\ell)$ with $a_k(\nu) = \prod_{j=1}^k (4\nu^2 - (2j-1)^2)/(k! 8^k)$ and $|R_\ell| \leq |a_\ell(\nu)| z^{-\ell}$ [33, 10.40.2 and §10.40(ii)]. (c) $K'_0 = -K_1$ and $K'_1 = -K_0 - K_1/z$; hence $|K'_0| = K_1$ and $|K'_1| \leq K_0 + K_1/z$ on $(0, \infty)$.

Proof. (a) and (b) are the cited formulas, specialised to $\nu = 0, 1$; (c) follows from the recurrences for K'_ν and the positivity of K_0, K_1 on $(0, \infty)$. $\square$

Implementation. For a ball $[m \pm r] \subset (0, \infty)$ the module evaluates K_0, K_1 at the exact midpoint m and adds r times an upper bound for $|K'_\nu|$ on the ball, obtained from (c) and $(\log K_1)' \geq -1 - 1/w$. At the midpoint it uses (b), summed until the remainder bound falls below 2^{-P-20} relative, when this is possible, and otherwise (a) at working precision $P + 2.885z + 64$ bits, since the series for K_ν cancels the growing I_ν -terms to a relative size e^{-2z} . The series is truncated at an index $k \geq 1$ with $4q \leq (k+1)^2$. From there on every ratio of consecutive terms is at most $\frac{1}{2}$: the factor $q/(k+1)^2$ is at most $\frac{1}{4}$ and decreases, and the coefficient ratios (H_{k+1}/H_k , and the ratios of the values $\psi(k+1) + \psi(k+2)$, which are positive for $k \geq 1$) are at most 2. So each tail is at most twice its first neglected term, hence at most the last computed term; twice the last computed term is added as a ball radius. Evaluating the series directly on a ball would multiply the radius by about e^{2z} , which the midpoint evaluation avoids.

A.3. The cushion. An explicit zero-killing function with rational parameters vanishes at every zero of ζ but has its zeros of order at least two at the prime powers only up to the rationalisation, so $\widehat{F}$ can be very slightly negative near a node. A small multiple of $\Xi^2 e^{-\pi t^2}$ repairs this.

Lemma A.4 (The cushion). *Let $\delta > 0$. (a) $\widehat{\Xi^2 e^{-\pi t^2}}(\xi) = (\Psi * e^{-\pi(\cdot)^2})(\xi) \geq e^{-\pi(|\xi| + \delta)^2} I_\delta > 0$ for every $\xi \in \mathbb{R}$, where $I_\delta := \int_{-\delta}^\delta \Psi(\eta) d\eta$. (b) $\mathcal{A}(\widehat{\Xi^2 e^{-\pi t^2}}) = \frac{1}{\pi} \sum_{n \geq 2} \Lambda(n) n^{-1/2} \widehat{\Xi^2 e^{-\pi t^2}}(\xi_n) > 0$. Moreover $\mathcal{A}(\widehat{\Xi^2 e^{-\pi t^2}}) \in [0.930646274515431 \pm 2.3 \cdot 10^{-16}]$, and $I_{1/20} \geq 0.09782880564$; the latter is a lower bound obtained from the term $N = 1$ of Proposition 4.2 alone, not an approximation of $I_{1/20}$. (c) Let $H \in \mathcal{H}$, $H \geq 0$ on $\mathbb{R}$, $F_H := \Xi^2 H$ and $\varepsilon \geq 0$. If $\widehat{F_H}(\xi) \geq -\varepsilon e^{-\pi(\xi + \delta)^2} \int_{-\delta}^\delta \Psi$ for $\xi \geq \xi_2$, then $F_{\text{rep}} := F_H + \varepsilon \Xi^2 e^{-\pi t^2} \in \mathcal{C}$ and*

$$\kappa^* \leq \frac{\mathcal{A}(F_H) + \varepsilon \mathcal{A}(\Xi^2 e^{-\pi t^2})}{\int F_H + \varepsilon \int \Xi^2 e^{-\pi t^2}} \leq \frac{\mathcal{A}(F_H) + \varepsilon \mathcal{A}(\Xi^2 e^{-\pi t^2})}{\int F_H}$$

when the numerator is non-negative.

Proof. (a) $\widehat{\Xi^2 e^{-\pi t^2}} = \widehat{\Xi^2} * \widehat{e^{-\pi t^2}}$, and $e^{-\pi t^2}$ is its own Fourier transform. Since $\Psi > 0$ (Lemma 4.1), restricting the convolution integral to $|\eta| \leq \delta$, where $|\xi - \eta| \leq |\xi| + \delta$, gives the bound. (b) $\Xi^2 e^{-\pi t^2} \in \mathcal{T}$ vanishes at every non-trivial zero of ζ , so Corollary 4.4 gives the identity, and positivity follows from (a). The two constants are computer-assisted: $\mathcal{A}(\widehat{\Xi^2 e^{-\pi t^2}})$ is computed from the archimedean side by Arb's rigorous numerical integration on $[0, 8]$ plus an explicit tail bound (`kappa/cushion_A.py`); the integral of Ψ uses the term $N = 1$ of Proposition 4.2 with monotone bounds for K_0, K_1 on subintervals, the terms $N \geq 2$ being positive (`lib/cushion.py`);

`log kappa/logs/cushion.log`. The slack certificates use the upper endpoint of the Arb ball for $\mathcal{A}(\Xi^2 e^{-\pi t^2})$, which is at most 0.930646274516. (c) $F_{\text{rep}} \geq 0$ on $\mathbb{R}$, $F_{\text{rep}} \in \mathcal{T}$, and for $\xi \geq \xi_2$ (a) gives $\widehat{F_{\text{rep}}}(\xi) \geq -\varepsilon e^{-\pi(\xi+\delta)^2} \int \Psi + \varepsilon e^{-\pi(\xi+\delta)^2} \int \Psi = 0$; so $F_{\text{rep}} \in \mathcal{C}$, and the bound is the definition of κ^* applied to F_{rep} (with $\int F_{\text{rep}} = \widehat{F_{\text{rep}}}(0)$). $\square$

The cushion is not optional: with rationalised parameters $\widehat{F}_H$ need not be non-negative at the nodes; for example $\widehat{F}_H(\xi_2) \approx -1.9 \cdot 10^{-119}$ in the certificate for $J = 20$. In the shipped certificates of Section 5.2 the values $|\widehat{F}_H(\xi_{n_j})|$ are at most $1.9 \cdot 10^{-119}$, $1.1 \cdot 10^{-501}$ and $2.4 \cdot 10^{-983}$ for $J = 20, 60, 100$, while the cushion term near the nodes is larger by more than forty orders of magnitude (Appendix C).

A.4. The digamma weight. The Gaussian–Laguerre certificates (Propositions 5.3, 5.4 and 5.9) integrate against the digamma weight $\Omega_{\mathfrak{g}}$ on a finite interval and bound the rest by an explicit tail, which uses the following estimate.

Lemma A.5 (The digamma function on vertical lines). *For $w > 0$ and $\tau > 0$, $|\operatorname{Re} \psi(w + i\tau) - \log \sqrt{w^2 + \tau^2}| \leq 1/\tau$. In particular, if $0 < w \leq 5\sqrt{3}$ and $t \geq 10$, then $0 < \operatorname{Re} \psi(w + \frac{it}{2}) \leq \log t + \frac{2}{t}$.*

Proof. Let $f(x) := (x+w)/((x+w)^2 + \tau^2) = \operatorname{Re} (x+w+i\tau)^{-1}$. The series for ψ in Section 2 gives $\operatorname{Re} \psi(w + i\tau) = \lim_{N \rightarrow \infty} (\log N - \sum_{n=0}^{N-1} f(n))$, and $\int_0^N f = \log \sqrt{(N+w)^2 + \tau^2} - \log \sqrt{w^2 + \tau^2}$. For each n , $|f(n) - \int_n^{n+1} f|$ is at most the total variation of f on $[n, n+1]$. On $[0, \infty)$, $f > 0$ increases to its maximum $1/(2\tau)$, attained where $x+w = \tau$ if that point lies in $[0, \infty)$, and then decreases to 0; so its total variation there is at most $1/\tau$. Hence $|\sum_{n < N} f(n) - \int_0^N f| \leq 1/\tau$ for every N , and letting $N \rightarrow \infty$ gives the first claim. For the second take $\tau = t/2 \geq 5$: then $\log \sqrt{w^2 + \tau^2} \geq \log 5 > \frac{1}{5} \geq \frac{1}{\tau}$, and $\log \sqrt{w^2 + \tau^2} \leq \log t$ because $w^2 \leq 75 \leq \frac{3}{4}t^2$. $\square$

The tail estimates of `general/verify_general.py` use the consequence $|\operatorname{Re} \psi(w + \frac{it}{2})| \leq \log t + 2 \leq t$ for $t \geq 10$. The verifier enforces $T \geq 10$ and, for every Gamma shift k_j , $0 \leq k_j$ and $(\frac{1}{2} + k_j)^2 \leq 300$, checked exactly; that is, $w_j = \frac{1}{2}(\frac{1}{2} + k_j) \in [\frac{1}{4}, 5\sqrt{3}]$, where the lemma applies, and it rejects other inputs. The data used here have $k_j \in \{0, 1\}$, that is, $w = \frac{1}{4}$ and $w = \frac{3}{4}$. The restriction is needed: for fixed t , $\operatorname{Re} \psi(w + \frac{it}{2})$ grows like $\log w$.

A.5. Provenance. Every input of a certificate is an exact object (integers, rationals, or decimal strings read as exact rationals) stored in a parameter file; for the slack certificates the file also stores the SHA-256 of a canonical string of the parameters, which the verifier recomputes. The scripts check the rounding direction of every printed decimal exactly. Each shipped log was produced by a fresh run of the shipped script and records the command, the date, the software versions, the exit code and the wall time.

APPENDIX B. LONG PROOFS

This appendix contains the longer proofs, in the order of the sections; each statement in the main text says where its proof is. The heart of Theorem 3.7 is in Appendix B.1.5: Lemma B.2 computes the expansions at the three cusps, and Lemma B.3 is the Landau step.

B.1. Proofs for Sections 2 and 3.

B.1.1. Duality. We complete the proof of Theorem 2.7. Throughout, $g_\sigma(t) := \sigma^{-1}e^{-\pi t^2/\sigma^2}$, so that $\widehat{g_\sigma}(\xi) = e^{-\pi\sigma^2\xi^2}$.

Approximation. Let $F \in \mathcal{T}_\delta$, $0 < \sigma \leq 1$, $F_\sigma := F * g_\sigma$, and for $N \geq 1$ let $R_{\sigma,N}(t) := N^{-1} \sum_{|k| \leq N^2} F(k/N) g_\sigma(t - k/N)$, an element of $\mathcal{G}$. Since $1 + |t| \leq (1 + |u|)(1 + |t - u|)$, we have $|F_\sigma(t)| \leq C(1 + |t|)^{-2}$ uniformly in $\sigma \leq 1$, and $|R_{\sigma,N}(t)| \leq C_\sigma(1 + |t|)^{-2}$ uniformly in N . As $N \rightarrow \infty$, $R_{\sigma,N} \rightarrow F_\sigma$ pointwise (Riemann sums), and $F_\sigma \rightarrow F$ pointwise as $\sigma \rightarrow 0$. On the Fourier side $\widehat{F_\sigma} = \widehat{F} \widehat{g_\sigma}$ and $\widehat{R_{\sigma,N}} = \widehat{g_\sigma} S_N$ with $S_N(\xi) := N^{-1} \sum_{|k| \leq N^2} F(k/N) e^{-2\pi i \xi k/N}$, a real function with $|S_N| \leq N^{-1} \sum_k |F(k/N)| \leq C$. By Poisson summation $N^{-1} \sum_{k \in \mathbb{Z}} F(k/N) e^{-2\pi i \xi k/N} = \sum_{j \in \mathbb{Z}} \widehat{F}(\xi + jN)$, so Lemma 2.2(a) and $|F(t)| \leq \|F\|_\delta(1 + |t|)^{-2}$ give

$$(B.1) \quad |S_N(\xi) - \widehat{F}(\xi)| \leq \varepsilon_N := C_F(e^{-\pi N^2/2} + N^{-1}) \quad (|\xi| \leq N/2).$$

In particular $S_N \rightarrow \widehat{F}$ pointwise. By Lemma 2.2(b), $\mathcal{A}(G) = 2 \int \widehat{G} \cosh(\pi \xi) d\xi + \int G \Omega_\infty$ for $G \in \mathcal{T}$; with the dominating functions $C(1 + |t|)^{-2} |\Omega_\infty(t)|$, $C \widehat{g_\sigma}(\xi) \cosh(\pi \xi)$ and $|\widehat{F}(\xi)| \cosh(\pi \xi)$, dominated convergence gives $\lim_{\sigma \rightarrow 0} \lim_{N \rightarrow \infty} \mathcal{A}(R_{\sigma,N}) = \mathcal{A}(F)$. The same dominating functions show that $\lim_\sigma \lim_N \mathcal{R}_{\mu,\nu}(R_{\sigma,N}) = \mathcal{R}_{\mu,\nu}(F)$ whenever $\int (1 + t^2)^{-1} d|\mu| < \infty$ and $\int e^{-\pi(1+2\delta)|\xi|} d|\nu| < \infty$.

Proof of (a), (iii) $\Rightarrow$ (ii). Let $F \in \mathcal{C} \cap \mathcal{T}_\delta$, and fix $0 < \sigma' < \sigma \leq 1$. Then $R_{\sigma,N} \geq 0$. On $[\xi_2, N/2]$, (B.1) and $\widehat{F} \geq 0$ give $\widehat{R_{\sigma,N}} \geq -\varepsilon_N \widehat{g_\sigma} \geq -\varepsilon_N \widehat{g_{\sigma'}}$; for $|\xi| > N/2$, $|\widehat{R_{\sigma,N}}| \leq C \widehat{g_\sigma} \leq C e^{-\pi(\sigma^2 - \sigma'^2)N^2/4} \widehat{g_{\sigma'}}$. Hence, with $\eta_N := \max(\varepsilon_N, C e^{-\pi(\sigma^2 - \sigma'^2)N^2/4}) \rightarrow 0$, the function $R_{\sigma,N} + \eta_N g_{\sigma'}$ lies in $\mathcal{C} \cap \mathcal{G}$. By (iii) $\mathcal{A}(R_{\sigma,N}) + \eta_N \mathcal{A}(g_{\sigma'}) \geq 0$; letting $N \rightarrow \infty$ and then $\sigma \rightarrow 0$ gives $\mathcal{A}(F) \geq 0$.

Proof of (b). The proof of Lemma 2.6 used the identity only on $\mathcal{G}$, and only for the non-negative functions A_T, B_X ; so the a priori bounds hold for (μ, ν) . For $F \in \mathcal{T}$, the approximation above gives $\mathcal{R}_{\mu,\nu}(F) = \lim \lim \mathcal{R}_{\mu,\nu}(R_{\sigma,N}) = \lim \lim \mathcal{A}(R_{\sigma,N}) = \mathcal{A}(F)$, and the integrals converge absolutely. Convexity of $\mathcal{K}$ is clear. By Lemma 2.6 the masses of all $(\mu, \nu) \in \mathcal{K}$ on compact sets are uniformly bounded, so $\mathcal{K}$ is relatively compact for the vague topologies. It is closed: if $(\mu_k, \nu_k) \rightarrow (\mu, \nu)$ vaguely with $(\mu_k, \nu_k) \in \mathcal{K}$, then for $F \in \mathcal{G}$ the contributions of $|t| > R$ and of $\xi > R$ to $\mathcal{R}_{\mu_k, \nu_k}(F)$ tend to 0 as $R \rightarrow \infty$, uniformly in k (Gaussian decay of F and $\widehat{F}$ against the uniform bounds), so the identity passes to the limit on $\mathcal{G}$, and the limit pair is admissible by the first part of (b).

Proof of (a), (ii) $\Rightarrow$ (i). Every element of $\mathcal{G}$ is a real combination of the functions $e^{-\pi(t-a)^2/s^2} \cos(2\pi bt + \varphi) + e^{-\pi(t+a)^2/s^2} \cos(2\pi bt - \varphi)$. Let $\{F_j\}_{j \geq 1}$ enumerate the rational combinations of these functions with rational parameters, together with all A_T, B_X ($T, X \in \mathbb{Z}$) of the proof of Lemma 2.6. Every $F \in \mathcal{G}$ is a limit of a sequence F_{j_k} with $\|F_{j_k} - F\|_\delta \rightarrow 0$ and with $|F_{j_k}(t)| \leq C e^{-ct^2}$, $|\widehat{F_{j_k}}(\xi)| \leq C e^{-c\xi^2}$ for some $C, c > 0$ independent of k . Put

$$P_W := \sum_{T \in \mathbb{Z}} (1 + |T|)^{-4} A_T, \quad Q_W := \sum_{X \in \mathbb{Z}} e^{-2\pi |X|} B_X;$$

since $\|A_T\|_{\delta'} \ll (1 + |T|)^2$ and $\|B_X\|_{\delta'} \ll e^{\pi(1+2\delta')|X|}$, both series converge in the norm $\|\cdot\|_{\delta'}$ for every $\delta' < \frac{1}{2}$. Their terms are entire, so the sums are analytic on the open strip $|\operatorname{Im} z| < 1$, which contains the closed strip $|\operatorname{Im} z| \leq \frac{1}{2} + \delta$ for every $\delta < \frac{1}{2}$; hence $P_W, Q_W \in \mathcal{T}_\delta$ for every $\delta < \frac{1}{2}$, and $P_W, Q_W \in \mathcal{C}$ with $P_W(t) \geq c(1 + |t|)^{-4}$ and $\widehat{Q_W}(\xi) \geq c e^{-2\pi |\xi|}$. Let $V_K := \operatorname{span}\{F_1, \dots, F_K, P_W, Q_W\}$, $C_K := \mathcal{C} \cap V_K$, and $\mathcal{N}(F) := \int F(t) e^{-t^2} dt + \int_{\xi_2}^\infty \widehat{F}(\xi) e^{-\xi^2} d\xi$. If $F \in C_K \setminus \{0\}$ then $\mathcal{N}(F) > 0$, so by (ii) the functional $\mathcal{A}_\varepsilon := \mathcal{A} + \varepsilon \mathcal{N}$ ($0 < \varepsilon \leq 1$) is strictly positive on $C_K \setminus \{0\}$. In V_K^* , let Γ_K be the convex cone generated by the functionals $F \mapsto F(t)$ ($t \in \mathbb{R}$) and $F \mapsto \widehat{F}(\xi)/\pi$ ($\xi \geq \xi_2$). Its dual cone in V_K is C_K , so the dual cone of C_K is $\overline{\Gamma_K}$

(bipolar theorem). By compactness of $\{F \in C_K : \|F\| = 1\}$, every functional close to $\mathcal{A}_\varepsilon|_{V_K}$ is positive on $C_K \setminus \{0\}$; hence $\mathcal{A}_\varepsilon|_{V_K}$ is an interior point of $\overline{\Gamma_K}$, and so of Γ_K [71, Theorem 6.3]. By Carathéodory's theorem [71, Theorem 17.1] there are finitely supported measures $\mu_K \geq 0$ on $\mathbb{R}$, which we may take even since V_K consists of even functions, and $\nu_K \geq 0$ on $[\xi_2, \infty)$ with $\mathcal{A}_\varepsilon = \mathcal{R}_{\mu_K, \nu_K}$ on V_K . Testing on P_W and Q_W gives $\int (1+|t|)^{-4} d\mu_K \leq B$ and $\int e^{-2\pi\xi} d\nu_K \leq B$, with B independent of K and $\varepsilon \leq 1$. By Helly's selection theorem a subsequence converges vaguely to some $(\mu^\varepsilon, \nu^\varepsilon)$. For fixed j the identity $\mathcal{R}_{\mu_K, \nu_K}(F_j) = \mathcal{A}_\varepsilon(F_j)$ holds for $K \geq j$, and the Gaussian decay of F_j and $\widehat{F_j}$ against the bounds just proved controls the tails uniformly in K ; so $\mathcal{R}_{\mu^\varepsilon, \nu^\varepsilon}(F_j) = \mathcal{A}_\varepsilon(F_j)$ for every j , and the two bounds persist. Letting $\varepsilon \rightarrow 0$ along a sequence in the same way gives (μ, ν) with $\mathcal{R}_{\mu, \nu}(F_j) = \mathcal{A}(F_j)$ for every j . By the density property of $\{F_j\}$ and dominated convergence the identity holds on $\mathcal{G}$, and $(\mu, \nu) \in \mathcal{K}$ by (b).

B.1.2. *A priori bounds and the logic.*

Proof of Lemma 2.6. Let $g(t) = e^{-\pi t^2}$, so $\widehat{g} = g$, and put

$$A_T := g(\cdot - T) + 2g + g(\cdot + T), \quad B_X := 2g(t)(1 + \cos 2\pi X t).$$

Both lie in $\mathcal{G}$ and are non-negative, with $\widehat{A_T}(\xi) = 2g(\xi)(1 + \cos 2\pi T\xi) \geq 0$ and $\widehat{B_X} = 2g + g(\cdot - X) + g(\cdot + X) \geq 0$. Moreover $A_T \geq e^{-\pi}$ on $[T-1, T+1]$ and $\widehat{B_X} \geq e^{-\pi}$ on $[X-1, X+1]$. Since all integrands are non-negative, Lemma 2.5 applies (for the second bound with $F = B_X$, $J = [X-1, X+1] \cap [\xi_2, \infty)$) and gives $\mu([T-1, T+1]) \leq e^\pi \mathcal{A}(A_T)$ and $\nu([X-1, X+1]) \leq \pi e^\pi \mathcal{A}(B_X)$. Now $|A_T(\pm \frac{i}{2})| \leq 4e^{\pi/4}$ and $\int A_T \Omega_\infty \leq 2 \int g(t-T) C \log(2+|t|) dt + C \leq C' \log(3+|T|)$; also $B_X(\pm \frac{i}{2}) = 2e^{\pi/4}(1 + \cosh \pi X)$ and $|\int B_X \Omega_\infty| \leq C$. This proves the two local bounds. Summing them over unit intervals gives the integrability statements, and then Lemma 2.2(a) and $|F(t)| \leq \|F\|_\delta (1+|t|)^{-2}$ give the bound for $\mathcal{R}_{\mu, \nu}$. Only the identity on the two families $A_T, B_X \subset \mathcal{G}$ was used; for $\mathcal{A} + \lambda f(\cdot)$ the extra terms $\lambda \int A_T = 4\lambda$ and $\lambda \int B_X \leq 4|\lambda|$ are bounded. Finally $\nu_\zeta([X-1, X+1]) = \sum_{e^{2\pi(X-1)} \leq n \leq e^{2\pi(X+1)}} \Lambda(n) n^{-1/2} \asymp e^{\pi X}$ by the prime number theorem. $\square$

Proof of Theorem 2.9. (a) Under RH every t_ρ is real, and Lemma 2.3 says $\mathcal{A} = \mathcal{R}_{\mu_\zeta, \nu_\zeta}$ on $\mathcal{T}$, with absolutely convergent integrals; μ_ζ is even and positive and ν_ζ is positive on $[\xi_2, \infty)$.

(b) Let $(\mu, \nu_\zeta) \in \mathcal{K}$. By Lemma 2.3, $\int F d\mu = \sum_\rho F(t_\rho)$ for all $F \in \mathcal{T}$. Suppose that $\rho_0 = \beta_0 + i\gamma_0$ is a zero with $\beta_0 \neq \frac{1}{2}$, and put $\eta_\rho := \beta - \frac{1}{2}$ for $\rho = \beta + i\gamma$, $y := \gamma_0$, $g_u(t) := e^{-\pi u t^2}$ and $F_u := g_u(\cdot - y) + g_u(\cdot + y) \in \mathcal{G}$ ($u \geq 1$). Since $\{t_\rho\}$ is symmetric under $t \mapsto -t$, $\sum_\rho F_u(t_\rho) = 2S(u)$ with $S(u) := \sum_\rho g_u(t_\rho - y)$, and

$$g_u(t_\rho - y) = e^{\pi u \epsilon_\rho} e^{i u \omega_\rho}, \quad \epsilon_\rho := \eta_\rho^2 - (\gamma - y)^2, \quad \omega_\rho := 2\pi(\gamma - y)\eta_\rho.$$

If $\epsilon_\rho > 0$ then $|\gamma - y| < |\eta_\rho| < \frac{1}{2}$, so only finitely many ρ have $\epsilon_\rho > 0$; the others contribute at most $\sum_\rho e^{\pi \epsilon_\rho} \leq e^{\pi/4} \sum_\rho e^{-\pi(\gamma-y)^2} < \infty$, uniformly in $u \geq 1$. Let $\epsilon_{\max} := \max_\rho \epsilon_\rho \geq \epsilon_{\rho_0} = \eta_{\rho_0}^2 > 0$. The zeros with $\epsilon_\rho = \epsilon_{\max}$ contribute $e^{\pi u \epsilon_{\max}} P(u)$ with $P(u) = \sum_\omega m_\omega e^{i \omega u}$, a finite sum over distinct real ω with integer coefficients $m_\omega \geq 0$, not all zero; the remaining zeros with $\epsilon_\rho > 0$ contribute $O(e^{\pi u \epsilon'})$ with $\epsilon' < \epsilon_{\max}$. The Bohr mean of $|P|^2$ is $\sum_\omega m_\omega^2 \geq 1$, so $|P(u)| \geq \frac{1}{2}$ for arbitrarily large u , and $|S(u)|$ is unbounded. But $0 \leq \int F_u d\mu \leq \int F_1 d\mu < \infty$ for $u \geq 1$, because $g_u \leq g_1$ on $\mathbb{R}$. This contradiction proves RH. Then $\int F d\mu = \int F d\mu_\zeta$ for all $F \in \mathcal{G}$. Two even positive measures μ_1, μ_2 with $\int (1+t^2)^{-1} d\mu_i < \infty$ that agree on $\mathcal{G}$ are equal: for an even $\varphi \in C_c(\mathbb{R})$ and $g_\sigma(t) = \sigma^{-1} e^{-\pi t^2/\sigma^2}$, Fubini gives $\int \varphi * g_\sigma d\mu_i = \frac{1}{2} \int \varphi(a) \int (g_\sigma(t-a) + g_\sigma(t+a)) d\mu_i(t) da$, which is the same for $i = 1, 2$; let $\sigma \rightarrow 0$ by dominated convergence. Hence $\mu = \mu_\zeta$.

(c) Assume (U). If (E) holds, some $(\mu, \nu) \in \mathcal{K}$ exists, and $(\mu, \nu) = p_\zeta$ by (U); then (b) gives RH. If RH holds, then $p_\zeta \in \mathcal{K}$ by (a), and $\mathcal{K} = \{p_\zeta\}$ by (U). Conversely, assume the two implications.

Under RH, $\mathcal{K} = \{p_\zeta\}$. Under $\neg$ RH, the first implication gives $\neg(\text{E})$, so $\mathcal{K} = \emptyset$. Either way (U) holds. The last assertion follows from (a), the first implication, and Theorem 2.7(a).

(d) If $\kappa^* > 0$, Proposition 2.8 gives an admissible pair with $\mu \geq \kappa^* dt$; such a μ is not purely atomic, so the pair is not p_ζ , and (U) fails. The consequences follow from (c) and Theorem 2.7(a). $\square$

B.1.3. Homogeneous directions and two-sided rigidity.

Proof of Lemma 3.1. (a) follows from Lemma 2.6. (b) An even F with $\widehat{F} \in C_c^\infty(\mathbb{R})$ is entire and decays rapidly in every horizontal strip, so $F \in \mathcal{T}$; for it $\int F d\Delta\mu = \langle \widehat{\Delta\mu}, \widehat{F} \rangle$ and $\frac{1}{\pi} \int \widehat{F} d\Delta\nu = \frac{1}{2\pi} \langle (\Delta\nu)^s, \widehat{F} \rangle$. Both distributions are even, so $\mathcal{R}_\Delta = 0$ on these F is the stated identity tested on even, hence on all, test functions; and $\widehat{\Delta\mu}$ is tempered. Conversely, assume the identity and let $F \in \mathcal{T}_\delta$. Fix an even $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ near 0, and for $0 < \varepsilon \leq 1 \leq k$ let $F_{\varepsilon,k} := (Fe^{-\pi\varepsilon^2 t^2}) * k\check{\chi}(k \cdot)$, whose transform $(\widehat{F} * \gamma_\varepsilon)\chi(\cdot/k)$, with $\gamma_\varepsilon(\xi) = \varepsilon^{-1}e^{-\pi\xi^2/\varepsilon^2}$, lies in C_c^∞ . By Lemma 2.2(a), $|F_{\varepsilon,k}(t)| \leq C(1 + |t|)^{-2}$ and $|\widehat{F_{\varepsilon,k}}(\xi)| \leq Ce^{-\pi(1+2\delta)|\xi|}$ uniformly, and $F_{\varepsilon,k} \rightarrow F$, $\widehat{F_{\varepsilon,k}} \rightarrow \widehat{F}$ pointwise as $k \rightarrow \infty$ and then $\varepsilon \rightarrow 0$. Dominated convergence and the growth bounds give $\mathcal{R}_\Delta(F) = \lim_\varepsilon \lim_k \mathcal{R}_\Delta(F_{\varepsilon,k}) = 0$. (c) $(\Delta\nu)^s$ vanishes on $(-\xi_2, \xi_2)$. $\square$

Proof of Proposition 3.2. We use the *low-pass balance*: if $\varphi \geq 0$ is even with $\widehat{\varphi} \in C_c^\infty((-\xi_2, \xi_2))$, then $\int \varphi d(\Delta\mu)_+ = \int \varphi d(\Delta\mu)_-$, because $\mathcal{R}_\Delta(\varphi) = \int \varphi d\Delta\mu = 0$. Fix an even $\chi_0 \in C_c^\infty((-\frac{\xi_2}{2}, \frac{\xi_2}{2}))$ with $\chi_0 \geq 0$, $\chi_0 \not\equiv 0$, and let $\varphi_0 := (\check{\chi}_0)^2$. Then $\varphi_0 \geq 0$ is even, $\widehat{\varphi_0} = \chi_0 * \chi_0$ is supported in $(-\xi_2, \xi_2)$, and $\varphi_0 \geq \eta > 0$ on some $[-r, r]$. The function $\Phi := \sum_{k \in \mathbb{Z}} \varphi_0(\cdot - 2rk)$ is even, bounded and $\geq \eta$ on $\mathbb{R}$, and each symmetric partial sum is admissible in the low-pass balance. By monotone convergence,

$$(B.2) \quad \eta(\Delta\mu)_+(\mathbb{R}) \leq \int \Phi d(\Delta\mu)_+ = \int \Phi d(\Delta\mu)_- \leq \|\Phi\|_\infty(\Delta\mu)_-(\mathbb{R}).$$

(a) As $\Delta\mu \geq 0$ off Z , $(\Delta\mu)_-$ is carried by Z and has finite mass; by (B.2), $\Delta\mu$ is a finite measure. So $\widehat{\Delta\mu}$ is a bounded continuous function and, by Lemma 3.1(b), $(\Delta\nu)^s$ is absolutely continuous; on $[\xi_2, \infty)$ it equals $\Delta\nu$. The negative part of $\Delta\nu$ is then absolutely continuous and carried by a null set, hence zero. So $\widehat{\Delta\mu} \leq 0$ everywhere. The finite measure $m := -\Delta\mu$ has $\widehat{m} \geq 0$, and Wiener's formula $m(\{x\}) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \widehat{m}(\xi) e^{2\pi i \xi x} d\xi$ (dominated convergence applied to $\frac{1}{2T} \int_{-T}^T e^{2\pi i \xi(x-t)} d\xi$; cf. [50, I.7.13] for the circle) gives $|m(\{x\})| \leq m(\{0\})$. Since $0 \notin Z$, $m(\{0\}) = -(\Delta\mu)_+(\{0\}) \leq 0$; hence $m(\{0\}) = 0$ and m has no atoms. As $(\Delta\mu)_-$ is purely atomic and singular to $(\Delta\mu)_+$, every atom of $(\Delta\mu)_-$ is an atom of m ; so $(\Delta\mu)_- = 0$. Now (B.2) gives $\Delta\mu = 0$, and then $(\Delta\nu)^s = 0$, i.e. $\Delta\nu = 0$.

(b) Let $w := \frac{1}{2\pi}((\Delta\nu)_-)^s$, a finite even positive measure, and $\check{w}(t) := \int e^{2\pi i \xi t} dw(\xi)$, a bounded continuous positive-definite function. By Lemma 3.1(b), $(\Delta\nu)^s$ is tempered, hence so is the positive measure $((\Delta\nu)_+)^s$. The tempered real measure $P := \check{w} dt - \Delta\mu$ has $\widehat{P} = w - \widehat{\Delta\mu} = \frac{1}{2\pi}((\Delta\nu)_+)^s \geq 0$, so P is positive definite (Bochner–Schwartz [35, Ch. II, §3]). For a positive-definite Radon measure, $|P(\{x\})| \leq P(\{0\})$: let $h \geq 0$ be smooth with support in $[-\frac{1}{2}, \frac{1}{2}]$ and $\int h^2 = 1$, $g_\varepsilon := \varepsilon^{-1/2}h(\cdot/\varepsilon)$, and $k_\varepsilon(t) := \int g_\varepsilon(s)g_\varepsilon(s-t) ds$, so that $0 \leq k_\varepsilon \leq 1 = k_\varepsilon(0)$ and $\text{supp } k_\varepsilon \subseteq [-\varepsilon, \varepsilon]$. The Hermitian form $(f, g) \mapsto \langle P, f * \check{g} \rangle$, $\check{g}(t) = \overline{g(-t)}$, is positive semidefinite, and the Cauchy–Schwarz inequality applied to $g_\varepsilon(\cdot - x)$ and g_ε gives $|\langle P, k_\varepsilon(\cdot - x) \rangle| \leq \langle P, k_\varepsilon \rangle$; let $\varepsilon \rightarrow 0$ by dominated convergence. Since $\check{w} dt$ has no atoms and $0 \notin Z$, $P(\{0\}) = -\Delta\mu(\{0\}) \leq 0$, so $P(\{0\}) = 0$ and P , hence $\Delta\mu$, has no atoms. But $(\Delta\mu)_-$ is carried by the countable set Z , hence purely atomic, and singular to $(\Delta\mu)_+$; so $(\Delta\mu)_- = 0$, and (B.2) gives $\Delta\mu = 0$ and then $\Delta\nu = 0$.

(c) Let $(\Delta\mu)_- = \sum_{i=1}^N c_i \delta_{z_i}$ with $c_i > 0$ and distinct z_i . By (B.2), $(\Delta\mu)_+$ is finite, so $\widehat{\Delta\mu}$ is continuous and vanishes on $(-\xi_2, \xi_2)$. For $\chi \in C_c^\infty((-\frac{\xi_2}{2}, \frac{\xi_2}{2}))$,

$$\int |\check{\chi}|^2 d(\Delta\mu)_+ - \int |\check{\chi}|^2 d(\Delta\mu)_- = \iint \chi(a) \overline{\chi(b)} \widehat{\Delta\mu}(b-a) da db = 0.$$

For any $x \notin \{z_1, \dots, z_N\}$, the map $\chi \mapsto (\check{\chi}(z_1), \dots, \check{\chi}(z_N), \check{\chi}(x))$ is onto $\mathbb{C}^{N+1}$, since a non-trivial linear relation would make a finite sum of distinct exponentials vanish on an interval. Choosing $\check{\chi}(z_i) = 0$ for all i and $\check{\chi}(x) = 1$ gives $(\Delta\mu)_+(\{x\}) = 0$; choosing only $\check{\chi}(z_i) = 0$ shows that $(\Delta\mu)_+$ is carried by the countable set $\check{\chi}^{-1}(0) \cap \mathbb{R}$. Hence $(\Delta\mu)_+$ is carried by $\{z_1, \dots, z_N\}$, where it vanishes by mutual singularity. Thus $\Delta\mu = -\sum c_i \delta_{z_i}$ and $\widehat{\Delta\mu}(\xi) = -\sum c_i e^{-2\pi i z_i \xi}$ vanishes on an interval, so all $c_i = 0$. Then $\Delta\mu = 0$ and $\Delta\nu = 0$. $\square$

B.1.4. Structure of pairs with extra atoms.

Proof of Lemma 3.6. Since U_m vanishes on Z_ζ , $\widehat{U_m}(\frac{\log m'}{4\pi}) = \delta_{mm'}$ and (3.3) holds, testing on U_m gives

$$\sum_{e \in E} a_e U_m(e) + \frac{1}{\pi} \nu(\{\frac{\log m}{4\pi}\}) = \frac{1}{\pi} \Lambda(\sqrt{m}) m^{-1/4}.$$

Multiply by $\pi m^{1/4}$. By (3.2) and $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \zeta(\frac{1}{2} + it) = -2\Xi(t)/(\frac{1}{4} + t^2)$, $\pi m^{1/4} a_e U_m(e) = -\frac{a_e \Xi(e)}{2(\frac{1}{4} + e^2)} \sum_{d^2|m} \mu(d) \alpha_{m/d^2}(s_e)$. The atoms at $\pm e$ give equal terms, because $a_{-e} = a_e$, Ξ is even and $\alpha_N(s_{-e}) = \alpha_N(\overline{s_e}) = \alpha_N(s_e)$; so $\pi m^{1/4} \sum_e a_e U_m(e) = -\sum_{d^2|m} \mu(d) c_{m/d^2}(E)$. Hence

$$\ell_m = \Lambda(\sqrt{m}) \mathbf{1}_{\square}(m) + \sum_{d^2|m} \mu(d) c_{m/d^2}(E).$$

Möbius inversion over squares ($f_m = \sum_{d^2|m} \mu(d) g_{m/d^2}$ for all m if and only if $g_N = \sum_{k^2|N} f_{N/k^2}$ for all N) and $\sum_{k^2|N} \Lambda(\sqrt{N/k^2}) \mathbf{1}_{\square}(N/k^2) = \sum_{k|M} \Lambda(M/k) = \log M$ for $N = M^2$ give (3.4). If ν lives on $[\xi_2, \infty)$ then $\ell_1 = \ell_2 = \ell_3 = 0$, and (3.4) at $N \leq 3$ gives $c_N = 0$. Testing on $V_{\rho,j}$, whose transform vanishes on $\mathcal{M}$, and using (3.3) gives the three identities for w_γ and $\sum_e a_e V_{\rho,j}(e)$, exactly as in the proof of Proposition 3.5. If $a = 0$, that proof gives $(\mu, \nu) = p_\zeta$; conversely $\mu_\zeta(E) = 0$. Trivially $a = 0$ implies $c \equiv 0$; if $c \equiv 0$ then $\tilde{\rho}_e = 0$ for all e by Lemma B.1 in Appendix B.1.5, and $\tilde{\rho}_e = 0$ forces $a_e = 0$ since $\Xi(e) \neq 0$ for $e \notin Z_\zeta$. Finally, if $(\mu, \nu) \in \mathcal{K}$ then $\ell_m \geq 0$, so $C_N \geq 0$. $\square$

B.1.5. Cusp expansions. We use $\vartheta(-1/\tau) = (\tau/i)^{1/2} \vartheta(\tau)$, $\vartheta(\tau+2) = \vartheta(\tau)$, and the fact that ϑ has no zeros in $\mathbb{H}$ (see [8, §3.2]); principal branches are used throughout, and $(\tau/i)^\alpha > 0$ for $\tau \in i(0, \infty)$. The cusps $\frac{2}{3}$ and $\frac{4}{3}$ are reduced to ∞ by explicit words in S and $T^{\pm 2}$, and the modular integrals are transported along these words by the following twisted action. Let $\chi : \Gamma_\vartheta \rightarrow \{\pm 1\}$ be the character with $\chi(S) = -1$ and $\chi(T^2) = 1$; it is well defined because, in $\mathrm{PSL}_2(\mathbb{R})$, S and T^2 generate Γ_ϑ with the single relation $S^2 = 1$ [8, §3.1] (as matrices, $S^2 = -I$; the action below depends only on the Möbius maps). For $f : \mathbb{H} \rightarrow \mathbb{C}$ and $W \in \Gamma_\vartheta$ put

$$(f|W)(\tau) := \chi(W) \frac{\vartheta(\tau)}{\vartheta(W\tau)} f(W\tau),$$

a right action: $(f|g)|h = f|(gh)$. As $\vartheta(\tau)/\vartheta(-1/\tau) = (\tau/i)^{-1/2}$, the functional equations (3.1) and periodicity read $f - f|S = \mathfrak{p}$ and $f|T^2 = f$, for $(f, \mathfrak{p}) = (F(\cdot, s), \mathfrak{p}_s)$. If $W = g_m \cdots g_1$ with $g_k \in \{S, T^{\pm 2}\}$ and $w_k := g_k \cdots g_1$ ($w_0 = 1$), then telescoping $f|w_{k-1} - f|w_k = (f - f|g_k)|w_{k-1}$ gives

$$(B.3) \quad f = f|W + \sum_{k: g_k=S} \mathfrak{p}|w_{k-1}.$$

For the cusp $\frac{2}{3}$ we use $W = ST^2ST^2S = \begin{pmatrix} -2 & 1 \\ 3 & -2 \end{pmatrix}$, which maps $\frac{2}{3}$ to ∞ , with S in positions $k = 1, 3, 5$ and $w_2 = T^2S$, $w_4 = T^2ST^2S = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$. For $\tau = \frac{2}{3} + iv$ one finds $T^2S\tau = 2 - 1/\tau \rightarrow \frac{1}{2}$, $w_4\tau = 9iv/(1 + 6iv)$ and $W\tau = -\frac{2}{3} + \frac{i}{9v}$; moreover $\chi(W) = -1$, $\chi(T^2S) = -1$, $\chi(w_4) = 1$,

$$(B.4) \quad \frac{\vartheta(\tau)}{\vartheta(T^2S\tau)} = (\tau/i)^{-1/2}, \quad \frac{\vartheta(\tau)}{\vartheta(w_4\tau)} = (\tau/i)^{-1/2}((2 - 1/\tau)/i)^{-1/2},$$

and both are analytic near $\tau = \frac{2}{3}$, the second with value $(\frac{3}{2})^{1/2}e^{i\pi/4} \cdot \sqrt{2}e^{i\pi/4} = i\sqrt{3}$ there.

Lemma B.1 (Expansion at the cusp 0). *For $s \in \mathbb{C}$ and $0 < v \leq 1$, $F(iv, s) = -\alpha_0(s)\vartheta(iv) + v^{-s} + v^{s-1/2} + O_s(v^{-1/2}e^{-\pi/v})$. Consequently, if $E_+ \subset [0, \infty)$ is finite, $\tilde{\varrho}_e \in \mathbb{R}$, and $\sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e) = 0$ for every $N \geq 1$, then $\tilde{\varrho}_e = 0$ for all e .*

Proof. By (B.3) with $W = S$, $F(iv) = -\frac{\vartheta(iv)}{\vartheta(i/v)}F(i/v) + \mathbf{p}_s(iv)$, and $\mathbf{p}_s(iv) = v^{-s} + v^{s-1/2}$. Since the $\alpha_N(s)$ grow polynomially, $F(i/v)/\vartheta(i/v) = \alpha_0(s) + O(e^{-\pi/v})$, and $|\vartheta(iv)| \leq Cv^{-1/2}$. For the consequence, $0 = \sum_e \tilde{\varrho}_e (F(iv, s_e) - \alpha_0(s_e))$; with $A := \sum_e \tilde{\varrho}_e \alpha_0(s_e)$ and $\vartheta(iv) = v^{-1/2}(1 + O(e^{-\pi/v}))$ this reads

$$\sum_{e \in E_+, e > 0} \tilde{\varrho}_e (v^{-s_e} + v^{-\bar{s}_e}) + 2\tilde{\varrho}_0 v^{-1/4} - Av^{-1/2} - A = O(v^{-1/2}e^{-\pi/v}),$$

because $s_e - \frac{1}{2} = -\bar{s}_e$ (with $\tilde{\varrho}_0 := 0$ if $0 \notin E_+$). Multiplying by $v^{1/2}$ and letting $v \rightarrow 0$ gives $A = 0$. Then $v^{1/4}$ times the left side is $\sum_{e > 0} \tilde{\varrho}_e \cdot 2 \cos(\frac{e}{2} \log v) + 2\tilde{\varrho}_0 + o(1)$; an almost periodic function of $u = \log(1/v)$ with distinct frequencies that tends to 0 has zero Bohr mean of its square, so all coefficients vanish. $\square$

Lemma B.2 (The class 2 mod 3). *Let $\operatorname{Re} s = \frac{1}{4}$. As $v \rightarrow 0^+$,*

$$\begin{aligned} & \frac{1}{3} \left[F(iv, s) + \omega F(\tfrac{2}{3} + iv, s) + \bar{\omega} F(\tfrac{4}{3} + iv, s) \right] \\ &= \frac{1}{3} (1 - 3 \cdot 9^{-s}) v^{-s} + \frac{1}{3} (1 - 3 \cdot 9^{s-1/2}) v^{s-1/2} + c(s) + O_s(v^{3/4}), \end{aligned}$$

with a constant $c(s)$. Consequently, if $E_+ \subset [0, \infty)$ is finite, $\tilde{\varrho}_e \in \mathbb{R}$ and $c_N := \sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e)$, then

$$\Theta_E^{(2)}(v) := \sum_{N \equiv 2(3)} c_N e^{-\pi N v} = \sum_{e \in E_+, e > 0} (B_e v^{-s_e} + \bar{B}_e v^{-\bar{s}_e}) + B_0 v^{-1/4} + C + O(v^{3/4}),$$

with $B_e = \frac{\tilde{\varrho}_e}{3} (1 - 3 \cdot 9^{-s_e})$ for $e > 0$, $B_0 = \frac{2\tilde{\varrho}_0}{3} (1 - \sqrt{3})$ (and $B_0 = 0$ if $0 \notin E_+$), and a real constant C . In particular $\Theta_E^{(2)}(v) = O(v^{-1/4})$.

Proof. At $\tau = \frac{2}{3} + iv$, (B.3) gives $F(\tau) = (F|W)(\tau) + \mathbf{p}_s(\tau) + (\mathbf{p}_s|T^2S)(\tau) + (\mathbf{p}_s|w_4)(\tau)$. First, $(F|W)(\tau) = -\vartheta(\tau)F(W\tau)/\vartheta(W\tau) = -\alpha_0(s)\vartheta(\tau) + O(v^{-1/2}e^{-\pi/(9v)})$, as $\operatorname{Im} W\tau = 1/(9v)$. Second, $\mathbf{p}_s$ is analytic near $\frac{2}{3}$, and $(\mathbf{p}_s|T^2S)(\tau) = -(\tau/i)^{-1/2}\mathbf{p}_s(2 - 1/\tau)$ is analytic near $\frac{2}{3}$ because $2 - 1/\tau \rightarrow \frac{1}{2}$; together they give $c_{2/3}(s) + O(v)$. Third, by (B.4) and $w_4\tau/i = 9v/(1 + 6iv)$, which has positive real part,

$$\begin{aligned} (\mathbf{p}_s|w_4)(\tau) &= (i\sqrt{3} + O(v)) \left[(9v)^{-s} (1 + 6iv)^s + (9v)^{s-1/2} (1 + 6iv)^{1/2-s} \right] \\ &= i\sqrt{3} 9^{-s} v^{-s} + i\sqrt{3} 9^{s-1/2} v^{s-1/2} + O(v^{3/4}). \end{aligned}$$

Since the $\alpha_N(s)$ are real, $F(-\bar{\tau}, s) = \overline{F(\tau, s)}$, so $F(\frac{4}{3} + iv) = \overline{F(\frac{2}{3} + iv)}$; with $9^{-s} = 9^{s-1/2}$ this gives coefficients $-i\sqrt{3} 9^{-s}$ and $-i\sqrt{3} 9^{s-1/2}$ at $\frac{4}{3}$. Combining with Lemma B.1 at 0: the theta terms give $-\alpha_0(s) \cdot \frac{1}{3} \sum_{j=0}^2 e^{-4\pi i j/3} \vartheta(\frac{2j}{3} + iv) = -\alpha_0(s) \sum_{n \equiv 2(3)} e^{-\pi n^2 v} = 0$, exactly; and since

$\omega - \bar{\omega} = i\sqrt{3}$, the coefficient of v^{-s} is $\frac{1}{3}(1 + i\sqrt{3}9^{-s}(\omega - \bar{\omega})) = \frac{1}{3}(1 - 3 \cdot 9^{-s})$, and similarly for $v^{s-1/2}$. For the consequence, the constants $\alpha_0(s_e)$ subtracted in $\Theta_E = \sum_e \tilde{\varrho}_e(F(\cdot, s_e) - \alpha_0(s_e))$ cancel because $1 + \omega + \bar{\omega} = 0$; for $e > 0$, $v^{s_e-1/2} = v^{-\bar{s}_e}$ and $\frac{1}{3}(1 - 3 \cdot 9^{s_e-1/2}) = \frac{1}{3}(1 - 3 \cdot 9^{-s_e})$; and for $e = 0$ both exponents equal $-\frac{1}{4}$ and $9^{-1/4} = 3^{-1/2}$. $\square$

Lemma B.3 (Landau step). *Let $E_+ \subset [0, \infty)$ be finite, $\tilde{\varrho}_e \in \mathbb{R}$ with $\tilde{\varrho}_0 \geq 0$ if $0 \in E_+$, and $c_N := \sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e)$. If $c_N \geq 0$ for every $N \equiv 2 \pmod{3}$, then $\tilde{\varrho}_e = 0$ for every $e \in E_+$.*

Proof. The Dirichlet series $D(w) := \sum_{N \equiv 2 \pmod{3}} c_N N^{-w}$ has non-negative coefficients of polynomial growth, so it converges in some half-plane; let $\sigma_c \in [-\infty, \infty)$ be its abscissa of convergence, which equals its abscissa of absolute convergence. (Polynomial growth alone does not exclude $\sigma_c = -\infty$, for instance if only finitely many c_N are non-zero.) As $\Theta_E^{(2)}(v) = O(e^{-2\pi v})$ for $v \geq 1$, Lemma B.2 gives, for $\operatorname{Re} w > \max(\sigma_c, \frac{1}{4})$,

$$\pi^{-w} \Gamma(w) D(w) = \int_0^\infty \Theta_E^{(2)}(v) v^{w-1} dv = \sum_{e>0} \left(\frac{B_e}{w - s_e} + \frac{\overline{B_e}}{w - \bar{s}_e} \right) + \frac{B_0}{w - \frac{1}{4}} + \frac{C}{w} + H(w),$$

with H analytic in $\operatorname{Re} w > -\frac{3}{4}$. Call $\tilde{D}(w)$ the right side multiplied by $\pi^w \Gamma(w)^{-1}$: it is meromorphic in $\operatorname{Re} w > -\frac{3}{4}$, with poles at most at the non-real points $s_e, \bar{s}_e$ and at $\frac{1}{4}$ (the pole of C/w is cancelled by the zero of $1/\Gamma$), and, by analytic continuation, $D = \tilde{D}$ on $\operatorname{Re} w > \max(\sigma_c, -\frac{3}{4})$ away from these points. Since the points s_e are distinct and $\pi^{s_e} \Gamma(s_e)^{-1} \neq 0$, $\tilde{D}$ has a pole at s_e and $\bar{s}_e$ exactly when $B_e \neq 0$ ($e > 0$), and at $\frac{1}{4}$ exactly when $B_0 \neq 0$. Suppose that some $B_e \neq 0$. Then $\tilde{D}$ has a pole on the line $\operatorname{Re} w = \frac{1}{4}$; as D is analytic in $\operatorname{Re} w > \sigma_c$ and equals $\tilde{D}$ there, $\sigma_c \geq \frac{1}{4}$, and in particular σ_c is finite. By Landau's theorem [60, Theorem 1.7], D is singular at the real point σ_c . As $\tilde{D}$ is analytic on the real half-line $(\frac{1}{4}, \infty)$, $\sigma_c = \frac{1}{4}$ and D is singular at $\frac{1}{4}$; so $B_0 \neq 0$, hence $\tilde{\varrho}_0 > 0$ and $B_0 < 0$. But then, for real $\sigma \downarrow \frac{1}{4}$, $0 \leq D(\sigma) = \tilde{D}(\sigma) = \pi^{1/4} \Gamma(\frac{1}{4})^{-1} B_0 (\sigma - \frac{1}{4})^{-1} + O(1) \rightarrow -\infty$, a contradiction. Hence $B_e = 0$ for all e ; as $|3 \cdot 9^{-s_e}| = \sqrt{3} \neq 1$ and $1 - \sqrt{3} \neq 0$, every $\tilde{\varrho}_e = 0$. $\square$

B.2. Proofs for Section 4.

B.2.1. The Bessel form.

Proof of Proposition 4.2. Mellin step. For $\operatorname{Re} s > 1$ and $a > 0$, the substitution $\lambda = (t/a)^2$ and $\int_0^\infty K_0(t) t^{s-1} dt = 2^{s-2} \Gamma(s/2)^2$ [33, 10.43.19] give $\int_0^\infty K_0(a\sqrt{\lambda}) \lambda^{s/2-1} d\lambda = \frac{1}{2} (a/2)^{-s} \Gamma(s/2)^2$. With $a = 2\pi N$ and $\sum_N d(N) N^{-s} = \zeta(s)^2$ (Tonelli), we get $\int_0^\infty W(\lambda) \lambda^{s/2-1} d\lambda = \frac{1}{2} \Gamma_{\mathbb{R}}(s)^2 \zeta(s)^2$, where $W(\lambda) := \sum_N d(N) K_0(2\pi N \sqrt{\lambda})$. Put $\lambda = e^{2u}$, $s = \sigma + \frac{1}{2}$ and

$$g(u) := e^{u/2} W(e^{2u}) = e^{u/2} \sum_N d(N) K_0(2\pi N e^u).$$

Then

$$(B.5) \quad \int_{\mathbb{R}} g(u) e^{u\sigma} du = \frac{1}{4} \Gamma_{\mathbb{R}}(\sigma + \frac{1}{2})^2 \zeta(\sigma + \frac{1}{2})^2 \quad (\operatorname{Re} \sigma > \frac{1}{2}).$$

Regularisation. The right side of (B.5) is meromorphic in σ , with double poles only at $\sigma = \pm \frac{1}{2}$: the double poles of $\Gamma(s/2)^2$ at $s = -2k$ ($k \geq 1$) are cancelled by the double zeros of $\zeta(s)^2$ at the trivial zeros. It decays faster than any power on vertical lines (Stirling), so Mellin inversion and a shift of the contour to $\operatorname{Re} \sigma = -M$ give, as $u \rightarrow -\infty$, $g(u) = (\alpha u + \beta) e^{-u/2} + (\alpha' u + \beta') e^{u/2} + O_M(e^{-M|u|})$ for every M , with the same expansion for derivatives; as $u \rightarrow +\infty$, g and its derivatives decay like $\exp(-2\pi e^u)$. Since $(\partial_u^2 - \frac{1}{4}) e^{\pm u/2} = 0$ and $(\partial_u^2 - \frac{1}{4})(u e^{\pm u/2}) = \pm e^{\pm u/2}$, the function $\Phi_2 := (\partial_u^2 - \frac{1}{4})^2 g$ decays faster than any exponential at both ends.

Identification. For $\operatorname{Re} \sigma > \frac{1}{2}$, integrating by parts four times (the boundary terms vanish) gives $\int \Phi_2(u) e^{u\sigma} du = (\sigma^2 - \frac{1}{4})^2 \cdot \frac{1}{4} \Gamma_{\mathbb{R}}(\sigma + \frac{1}{2})^2 \zeta(\sigma + \frac{1}{2})^2$, which is $\Xi(-i\sigma)^2$, because $\frac{1}{2}s(s-1) = \frac{1}{2}(\sigma^2 - \frac{1}{4})$ for $s = \sigma + \frac{1}{2}$. Both sides are entire in σ , so equality holds for all σ ; at $\sigma = it$ we obtain $\Xi(t)^2 = \int \Phi_2(u) e^{iut} du$, hence $\Psi(\xi) = 2\pi\Phi_2(2\pi\xi)$ by Fourier inversion.

Operator form. With $u = 2\pi\xi$ we have $e^u = x$ and $\partial_u = \theta$. Since $\theta(\sqrt{x}f) = \sqrt{x}(\theta + \frac{1}{2})f$, we get $(\theta^2 - \frac{1}{4})(\sqrt{x}f) = \sqrt{x}\theta(\theta + 1)f$, so $\Phi_2 = \sqrt{x}\theta^2(\theta + 1)^2 \sum_N d(N)K_0(2\pi Nx) = \sqrt{x} \sum_N d(N)W_0(2\pi Nx)$. Here θ commutes with dilations, and the differentiated series converge uniformly on compact sets because K_0, K_1 decay exponentially. Evenness and reality of Ψ follow from those of Ξ^2 .

Asymptotics. As $z \rightarrow \infty$, $W_0(z) = z^4 K_0(z)(1 + O(1/z))$ and $K_0(z) \sim (\pi/2z)^{1/2} e^{-z}$, so the $N = 1$ term at $x = n$ is $\pi(2\pi)^4 n^4 e^{-2\pi n}(1 + O(1/n))$, and the terms $N \geq 2$ are $O(n^4 e^{-4\pi n})$.

Gamma-only analogue. Replace W by $K_0(2\pi\sqrt{\lambda})$, so that the right side of (B.5) becomes $\frac{1}{4}\Gamma_{\mathbb{R}}(\sigma + \frac{1}{2})^2$. Now $g_1(u) = e^{u/2}K_0(2\pi e^u) = (\alpha u + \beta)e^{u/2} + O(u e^{5u/2})$ as $u \rightarrow -\infty$, so $(\partial_u^2 - \frac{1}{4})^2 g_1 = O(|u|e^{5u/2})$, and its Laplace transform is analytic for $\operatorname{Re} \sigma > -\frac{5}{2}$. There it equals $(\sigma^2 - \frac{1}{4})^2 \frac{1}{4}\Gamma_{\mathbb{R}}(\sigma + \frac{1}{2})^2$, which is analytic for $\operatorname{Re} \sigma > -\frac{5}{2}$ (the double pole at $\sigma = -\frac{1}{2}$ is cancelled by $(\sigma + \frac{1}{2})^2$). At $\sigma = it$ this is $\Xi_{\infty}(t)^2$, and the rest of the argument is unchanged. Finally, $\Xi_{\infty}(t)^2 = O(|t|^4 e^{-\pi|t|/2})$, so Ψ_1 is analytic in $|\operatorname{Im} \xi| < \frac{1}{4}$, that is, for $|\arg x| < \pi/2$, and so is $\sqrt{x}W_0(2\pi x)$. $\square$

B.2.2. An integral representation.

Proof of Lemma 4.6. The polynomial $e^y \theta^m e^{-y}$ has degree m and leading coefficient $(-1)^m$, and $\deg_{\theta} \mathcal{Q}_P = 2 \deg P + 4$ with leading coefficient $(-1)^{\deg P} [u^{\deg P}]P$; this gives the degree and the leading coefficient. As $\mathcal{Q}_P(\theta)$ has the factor θ and $\theta = y \frac{d}{dy}$, $\mathcal{Q}_P(0) = 0$. By $(P(t^2)F)^{\wedge} = P(-\theta^2)\widehat{F}$ and $\theta^2(\sqrt{x}f) = \sqrt{x}(\theta + \frac{1}{2})^2 f$, Proposition 4.2 gives the transform. For the integral, start from $K_0(z) = \int_1^{\infty} e^{-zw}(w^2 - 1)^{-1/2} dw$ [33, 10.32.8]. For fixed w , θ_z acts on e^{-zw} as θ_y acts on e^{-y} at $y = zw$, so $\mathcal{Q}_P(\theta)e^{-zw} = \mathcal{Q}_P(zw)e^{-zw}$; differentiation under the integral sign is legitimate for $z > 0$, and $P(-(\theta + \frac{1}{2})^2)W_0 = \mathcal{Q}_P(\theta)K_0$ by (4.2). Substituting $y = zw$ gives the formula for g_P . $\square$

B.2.3. Robust compactness and Theorem D. Write $\Sigma_b := \{t : |\operatorname{Im} t| < b\}$.

Proof of Lemma 4.7. Montel. By (i) the family is uniformly bounded on compact subsets of $\Sigma_{1/2+\delta}$, hence normal; a subsequence converges locally uniformly to an analytic H_{∞} , which is even, real on $\mathbb{R}$ and satisfies $|H_{\infty}(t)| \leq C e^{a|\operatorname{Re} t|}$ on $\Sigma_{1/2+\delta}$ [26, VII.2.9]. *Domination.* By (4.1) with $b = \frac{1}{2} + \delta$ and (i), $|F_J(t)| \leq m(\operatorname{Re} t) := C C_b^2 (1 + |\operatorname{Re} t|)^6 e^{-(\pi/2-a)|\operatorname{Re} t|}$ on the closed strip, and the same bound holds for F_{∞} on the open strip; $(1 + |T|)^k m(T)$ is integrable for every k . *Convergence.* On $\mathbb{R}$, $F_J \rightarrow F_{\infty}$ pointwise along the subsequence, so by dominated convergence $\int |T|^k |F_J - F_{\infty}| dT \rightarrow 0$, and $\widehat{F}_J^{(k)} \rightarrow \widehat{F}_{\infty}^{(k)}$ uniformly on $\mathbb{R}$ ($k = 0, 1$). For $\delta' < \delta$, H_{∞} is analytic near $|\operatorname{Im} t| \leq \frac{1}{2} + \delta'$, and $(1 + |z|)^2 |F_{\infty}(z)| \leq (2 + |\operatorname{Re} z|)^2 m(\operatorname{Re} z)$ is bounded there, so $F_{\infty} \in \mathcal{T}_{\delta'}$. *Positivity.* By (iii), $H_{\infty} \geq 0$ on $\mathbb{R}$, so $F_{\infty} \geq 0$. By (iv) and evenness, $\widehat{F}_{\infty} \geq 0$ for $|\xi| \geq \xi_2$, so $F_{\infty} \in \mathcal{C}$. By (v), $\widehat{F}_{\infty}(\xi_n) = 0$ for every $n \in \text{PP}$, and Corollary 4.4 gives $\mathcal{A}(F_{\infty}) = 0$. Finally $\Xi(0) = \int \Phi > 0$, so $F_{\infty}(0) = \Xi(0)^2 > 0$ and $\int F_{\infty} = \widehat{F}_{\infty}(0) > 0$. Thus $\kappa^* \leq \mathcal{A}(F_{\infty})/\int F_{\infty} = 0$. If $F = \Xi^2 H \in \mathcal{C}$, then $\mathcal{A}(F) \geq 0$ by Corollary 4.4, since $\widehat{F} \geq 0$ at the prime powers. $\square$

We use the notation of Section 4.3: $(P_J)_{J \in \mathcal{J}}$ is the exact family, $H_J(t) = P_J(t^2)$, $F_J = \Xi^2 H_J$. Assume (N) and (G_a) throughout; by Montel's theorem [26, VII.2.9] every infinite subset of $\mathcal{J}$ contains a subsequence along which H_J converges locally uniformly on $\Sigma_{1/2+\delta}$, and every such limit H_{∞} satisfies $H_{\infty}(0) = 1$, so $H_{\infty} \not\equiv 0$.

Equivalent forms of (Z_∞) . The following are equivalent: (i) (Z_∞) ; (ii) no limit H_∞ as above, along any infinite subset of $\mathcal{J}$, has a real zero; (iii) for every $T > 0$, $\liminf_{J \in \mathcal{J}} \min_{|t| \leq T} H_J(t) > 0$. (i) $\Rightarrow$ (iii): suppose (iii) fails for some T . Then there are $J_k \rightarrow \infty$ in $\mathcal{J}$ and $t_k \in [-T, T]$ with $\limsup_k H_{J_k}(t_k) \leq 0$. If $H_{J_k}(t_k) \leq 0$ for infinitely many k , the intermediate value theorem (with $H_{J_k}(0) = 1$) gives real zeros $s_k \in [-T, T]$ of H_{J_k} , and a limit point of (s_k) contradicts (i). Otherwise $H_{J_k}(t_k) \rightarrow 0$ along a subsequence; passing to a further subsequence, $t_k \rightarrow t^*$ and $H_{J_k} \rightarrow H_\infty$ locally uniformly, so $H_\infty(t^*) = 0$, and by Hurwitz's theorem [26, VII.2.5] t^* is a limit of zeros of H_{J_k} , again contradicting (i). (iii) $\Rightarrow$ (ii): if H_∞ is the limit along $J_k \rightarrow \infty$, then for $|t| \leq T$, $H_\infty(t) = \lim_k H_{J_k}(t) \geq \liminf_{\mathcal{J}} \min_{|s| \leq T} H_J(s) > 0$. (ii) $\Rightarrow$ (i): if zeros z_k of H_{J_k} , $J_k \rightarrow \infty$, converge to $t^* \in \mathbb{R}$, take a convergent subsequence $H_{J_k} \rightarrow H_\infty$; then $|H_\infty(t^*)| \leq |H_\infty(t^*) - H_\infty(z_k)| + |H_\infty(z_k) - H_{J_k}(z_k)| \rightarrow 0$. Finally, a uniform lower bound $|\operatorname{Im} t| \geq \eta$ for the zeros of H_J ($J \geq J_0$) implies (i), since then no zero lies in Σ_η .

Norm convergence. Along a convergent subsequence, $\|F_J - F_\infty\|_{\delta'} \rightarrow 0$ for every $\delta' < \delta$. Indeed, with m as in the proof of Lemma 4.7, on $\{|\operatorname{Im} z| \leq \frac{1}{2} + \delta', |\operatorname{Re} z| \geq R\}$ we have $(1 + |z|)^2 |F_J - F_\infty| \leq 2(2 + |\operatorname{Re} z|)^2 m(\operatorname{Re} z)$, which is small for large R uniformly in J , while on the compact part $|\operatorname{Re} z| \leq R$ of the strip the convergence is uniform. Since $\mathcal{A}$ is $\|\cdot\|_{\delta'}$ -continuous on $\mathcal{T}_{\delta'}$ (Lemma 2.2), $\mathcal{A}(F_J) \rightarrow \mathcal{A}(F_\infty)$ along the subsequence.

What (N) and (G_a) give alone. The proof of Lemma 4.7 without hypotheses (iii) and (iv) shows: every limit $F_\infty = \Xi^2 H_\infty$ lies in $\mathcal{T}$, has $H_\infty \in \mathcal{H}$, $H_\infty(0) = 1$, $\hat{F}_\infty(\xi_n) = \hat{F}'_\infty(\xi_n) = 0$ for every $n \in \text{PP}$, and $\mathcal{A}(F_\infty) = 0$ (Corollary 4.4). So a proof of (G_a) would produce an exact solution of the infinite interpolation problem; what Theorem 4.8 adds is the positivity of this particular limit, which is the content of (Z_∞) and (Mg).

Further remarks. (a) In (N) it suffices that the linear system $M_J p = -b_J$ be solvable for infinitely many J , choosing any solution; invertibility is used nowhere else, although the family is then not canonical. (b) Different subsequences may have different limits; under (Z_∞) and (Mg) all of them have the zero sets Z_ζ and ξ_{PP} , and convex combinations of limits are again exact magic functions. (c) The strict inequality $a < \pi/2$ in (G_a) is essential: for $a = \pi/2$ the majorant m is not integrable, F_∞ need not lie in $\mathcal{T}$, and neither the explicit formula nor complementary slackness is available. Uniformity of C and a in J is equally essential, since every single polynomial satisfies the bound with some constant.

B.2.4. Coefficient bounds.

Proof of Proposition 4.9. (1) All terms of $H_J(t) = \sum_k p_k t^{2k}$ are ≥ 0 , and $p_0 = 1$. (2) $|H_J(t)| \leq \sum_k p_k |t|^{2k} \leq C \cosh(a|t|)$, and $|t| \leq |\operatorname{Re} t| + b$ on $|\operatorname{Im} t| \leq b$. (3) Montel on $\mathbb{C}$, and Cauchy's formula for the coefficients. (4) The first claim is clear. For the second, if $p_{k-1} = 0 < p_k$ for some k the bound is trivial; otherwise $p_0, \dots, p_d > 0$ and $p_k = 0$ for $k > d$. Put $r := \min_{k \leq d} p_{k-1}/p_k$ and $c_k := p_k r^k$ ($k \leq d$), a non-increasing positive sequence. For $|z| < 1$, $(1 - z) \sum_k c_k z^k = c_0 - \sum_{k \geq 1} (c_{k-1} - c_k) z^k - c_d z^{d+1}$, and the subtracted terms have modulus $< c_0$; so $P_J(rz) \neq 0$ (this is the Eneström–Kakeya bound). $\square$

APPENDIX C. ANCILLARY FILES

The ancillary files contain, for every computer-assisted statement, the exact input, a standalone verifier and the log of a fresh run. Their file `README.md` describes each script in detail; this appendix is an index (Table 4), with the hashes of the headline inputs in Table 5.

Requirements. Python ≥ 3.10 with python-flint 0.9.0 (FLINT 3.6.0, which contains Arb) and mpmath 1.3.0; nothing else. All commands are run from the top directory of the ancillary files, with PY set to such an interpreter. The wrapper `runlog.sh LOG "$PY" SCRIPT ARGS` writes a log whose header records the command, the date and the software versions, and whose footer

records the exit code and the wall time. A run succeeds if it exits with code 0 and its final output line reports success (the exact line for each script is listed in `README.md`). The ancillary files are licensed under the Apache License 2.0 (copyright 2026 Nic Johns); the paper itself is licensed under CC BY 4.0.

TABLE 4. Index of the certificates. Directories: `kappa/` (the slack), `exact/` (exact members P_J), `general/` (Gaussian–Laguerre functions and pairs of Herglotz form). Runtimes are wall times from the footers of the shipped logs.

Statement	Script and input	Arithmetic	Runtime
Thm. 5.1, Table 1	<code>kappa/verify_kappa.py</code> with <code>kappa/params/JJ.json</code> , $J = 20, 23, 24, 30, 35, 40, 50, 60, 100$	Arb, 500–3300 bits	7 s–1560 s
Lemma A.4 (constants)	<code>kappa/cushion_A.py</code> , <code>lib/cushion.py</code>	Arb integration	0.5 s
Lemma A.3 (validation only)	<code>kappa/selftest.py</code>	Arb vs. mpmath	244 s
Prop. 5.5	<code>exact/verify_exact_member.py</code> , $J = 10, 60, 61, 110, 111$	Arb, 800–3000 bits	up to 620 s
Prop. 5.3	<code>general/verify_general.py</code> with <code>zeta_K128_s13.json</code>	exact Sturm; Arb	467 s
Prop. 5.4	<code>general/verify_general.py</code> with <code>zeta_K80_s12_F80.json</code> ; <code>general/verify_windows.py</code> with <code>F80_windows.json</code>	exact Sturm; Arb; exact	289 s; 2 s
Prop. 5.8	<code>general/verify_pair.py</code> with <code>pair_zeta_logq002_X10.json</code>	Arb	12 s
Prop. 5.9	<code>general/verify_general.py</code> with <code>gammaR2_K80_s7.json</code> , <code>gammaC_K64_s9.json</code> ; <code>general/verify_pair.py</code> with <code>pair_gammaR2_X240.json</code> , <code>pair_gammaC_X40.json</code> ; <code>general/corollary_slack.py</code> with <code>corollary_qsqrt5.json</code> , <code>corollary_qsqrtm3.json</code>	exact Sturm; Arb; exact	173 s, 187 s; 36 s, 19 s; < 1 s
Lemma 5.7 (consistency check only)	<code>general/check_pair_identity.py</code> on the three pairs above, two lines of integration each	mpmath, 30 digits	63 s–88 s

TABLE 5. SHA-256 of the input files of the headline certificates (all others: SHA256SUMS).

File	SHA-256
<code>kappa/params/J100.json</code>	1186be2c9c2e575810781c02fc8a22ee060ffa4c61be4e8eab21b63a90719662
<code>kappa/params/J60.json</code>	e22afd8721e40313f47f4ee5ab3415dcc4d5c24502c17c4994499a9971e94a7e
<code>kappa/params/J20.json</code>	d686e47b0412358c8ad519370655c6f8600c2d44d25ee8da8e39ab66551c5ec8
<code>general/params/zeta_K128_s13.json</code>	0976a6711193adc0f02a532931974b8ef4912337f79e39f59aa21445879a4df2

Details of the headline runs. Theorem 5.1 with $J = 100$ uses 3300 bits, Taylor order 28 (64 for $x \geq 121$) and 5848 pieces, and its decisive output line is (4) `kappa* <= A_upper / Fhat_H(0)_lower <= 1.1508391e-906`. Proposition 5.3 uses one Arb integral on $[0, 220]$ at 512 bits and prints `kappa*_OPS` (and `kappa*`) of the data `<= 9.9461827001452550725e-41`; the run for Proposition 5.4 prints the bound $7.7252542459488902828 \cdot 10^{-25}$ for $\mathcal{A}(F_{80})/\int F_{80}$, and the enclosures of $\mathcal{A}(F_{80})$ and $\int F_{80}$ in its log imply the two bounds stated there. The exact members use 800 bits ($J = 10$), 1800 (60), 2000 (61) and 3000 (110, 111); the maximal relative radii of the enclosures are of order 10^{-222} , 10^{-474} , 10^{-534} , 10^{-789} and 10^{-788} . The pair for Proposition 5.8 uses 8437 mean-value steps on $[0, 2000]$ and prints `MU FLOOR : mu(t) >= 0.000471949` for all real t ; the pairs for Proposition 5.9 use 17214 and

9501 steps on $[0, 800]$, and the combinations print $1.6420748e-5 \leq \kappa \cdot (Q(\sqrt{5})) \leq 0.00048294147$ and $0.00015006672 \leq \kappa \cdot (Q(\sqrt{-3})) \leq 0.00099854968$. The pairs also print `supp nu~` in $[x_0, \infty)$ with $x_0 \geq 4.049150$ (resp. 3.011664); run with `--gap-x 4` (resp. `3`), they print `admissible at gap xi_4` (prime measure on $[4, \infty)$): `True` (resp. `xi_3`), and the combinations print that the bracket and the slack interval hold at the natural gap ξ_4 (resp. ξ_3). The consistency check of Lemma 5.7 compares both sides of the explicit formula for the three pairs; its residuals are at most $3.2 \cdot 10^{-30}$, and it is not used in any proof.

Re-running. Each directory has a script `run_all.sh` that repeats its certificates (39.0, 44.4 and 27.5 CPU-minutes for `kappa/`, `exact/` and `general/`, summed from the log footers), after which `sha256sum -c SHA256SUMS` checks that inputs and shipped outputs are unchanged. A single certificate is repeated with the command recorded in the first line of its log, for example

```
./runlog.sh kappa/logs/J100.log "$PY" kappa/verify_kappa.py \
  kappa/params/J100.json --prec 3300 --order 28 --order-far 64 \
  --order-switch 121 --workers 1
```

The precision, Taylor orders and number of worker processes affect only the running time and the width of the enclosures, not the validity of a successful run. The floating-point table `computed/family_J1_110.tsv`, used only in Numerical observation 5.6, is listed separately in `README.md`.

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